

Supplemental Material

Captured weight and boundary leakage bound the error of sample-based spectral functions

S1. THE PRIMITIVE: PROVENANCE DECLARATIONS IN FULL

Section II states the measurement primitive, fixes the one definition the rest of the paper turns on, and then says — channel by channel, figure by figure — which objects here were reconstructed from a finite number of measurements and which are full-sector references computed classically. That line is routinely drawn nowhere in the text, and a single clause of an abstract is enough to draw it in the wrong place. It is drawn here, explicitly, before any result is quoted.

Two consequences of Rayleigh–Ritz. Two corollaries of Eq. (3) are used later. A determinant acts on A_S through H_S even carrying no seed amplitude: at sector fraction 0.85, 30.2% of the selected determinants at $L = 6$ and 42.0% at $L = 8$ lie outside $\text{supp}(\phi)$, add exactly nothing to the captured weight, and still move the Ritz spectrum. And on the smallest subspace of *unit* captured weight, $S^\star = \text{supp}(\phi)$, of size $(\frac{1}{2} + \frac{1}{L})$ dim at $L = 6$ and 0.571 dim at $L = 8$, the relative L_1 error of Eq. (3) does not vanish either; it is measured there at both resolutions, on both boundary conditions and against its ceiling, in Sec. IIID.

Table I gives, for every dynamical object in this paper, the subspace it was built on and the measurements that built it. Three readings the word *sampled* invites are closed off by it.

The neutral channels are references, not reconstructions. $S(q, \omega)$ and $S^{zz}(q, \omega)$ are computed here by a continued fraction over the *entire* N -particle sector of 853 776 determinants at $L = 12$, at recursion depth $n_\ell = 200$: no subspace is selected and no shot is drawn, and the generating scripts say so in their own headers. The primitive *targets* these two channels — that is what Fig. 1 asserts, and what the seed algebra supports — and targeting a channel is not reconstructing it from measurements; they were not, nor were the $A(k, \omega)$ of Fig. 4 or the N_2 series of Fig. S6. A reference at finite recursion depth is not the exact spectral function either, so its depth is quoted with it, and we quote no error that lies within a factor 20 of the drift of its own reference.

What the Born ranking is, and what it is not. The two objects it would be easiest to call sampled without a budget — Fig. 3(a,b) and the dashed curve of Fig. 5(b) — select S by sorting the accumulated Born weight $\sum_k |\langle d | e^{-iHt_k} \phi \rangle|^2$ of the evolved state over the $K+1$ snapshots and keeping the top fraction. That weight is the distribution the device draws from, not the ground-state amplitudes and not the exact Lehmann residues, so the ranking is *not* a clairvoyant oracle: the repository contains one of those, in `amortized_recovery.py`, and labels it as such there. It is the infinite-shot limit $T \rightarrow \infty$ of the declared protocol, approached from above in error by any finite budget — at matched subspace size finite sampling is worse by 1.50 ± 0.25 at $L = 6$ and 1.96 ± 0.12 at $L = 8$ (Table S6) — and in that limit it selects nothing the device could never return: the only determinants of exactly zero accumulated weight are the reflection-forbidden ones (4, 18, 48, 200 at $L = 6$ –12), which rank last and never enter S . What it is not is a finite-shot experiment, and it must not be called one. The replacement is measured and quoted with its budget: at $L = 8$, $T = 2.60 \times 10^6$ shots over $K+1 = 19$ snapshots reach sector fraction 0.835 ± 0.0025 at relative L_1 error $(4.81 \pm 0.27) \times 10^{-3}$ over eight independent shot streams (Fig. 3(c)), and the measured ladder from 2.4×10^4 to 1.9×10^7 shots for $L = 6$ to $L = 12$ is Fig. 5(c) and Table IV. The literal configuration of Fig. 3(a,b)— $\eta = 0.18t$, $K = 16$, all sixteen channels—has also been run at finite shots, at two budgets a factor two apart: 5.2×10^6 per channel suffice for panel (b)’s accuracy, but on a *larger* subspace, and at half that budget, at a fraction just under the announced one (0.839 against 0.850), the penalty is a measured 2.61 and not the 2.05 Table S6 would transfer (numbers in the caption of Fig. 3).

What the a posteriori checks establish. The branch sum rules $\int A_p^\pm d\omega$ are identities of the ground state, not measurements of a reconstruction. Integrated on the grid the figures use, at $L = 6$ and $\eta = 0.15t$, the exact and reconstructed addition branches carry 0.48659 and 0.48647 — agreement to 2.4×10^{-4} relative, both falling 2.7% short of the analytic $1 - \langle \hat{n}_p \rangle = 0.5$ because Lorentzian tails of width η leave the finite frequency window. That is a genuine consistency check on a zeroth moment, and not an error bound. The certificate is the two-sided bound of Theorem 1, and where it is non-vacuous is measured, not asserted, in Sec. V — an essential property given that the classical hardness of a specific instance is empirical rather than a theorem (Sec. IX). What is guaranteed in advance is narrower: a subspace containing the order- K Krylov space reproduces that branch’s exact spectral moments through order $2K+1$ (Theorem 2), and $T \geq \varepsilon^{-1} \log(1/(\varepsilon\delta))$ shots capture every configuration of Born probability at least ε (Prop. 1); the second does not imply the first (Sec. IV).

S1.1. The two budgets

The two halves of this paper are the two terms of one inequality. With Q_S the projector complementary to P_S and $w_S = \|P_S \phi\|^2 / \|\phi\|^2$, the upper branch of Eq. (7) reads

$$\|A - A_S\|_1 \leq (\|\phi\| + \|P_S\phi\|) \left(\underbrace{\|Q_S\phi\|}_{\text{missed weight}} + \underbrace{\Lambda_S(\eta)/\eta}_{\text{boundary leakage}} \right).$$

The captured-weight term $\|Q_S\phi\| = \|\phi\|\sqrt{1-w_S}$ vanishes once the sampler has exhausted the Born support of the probe, and Table I prices that exhaustion in measurements. The leakage term $\Lambda_S(\eta)/\eta$ falls as S grows along the sampler’s ranking (Sec. V), but the captured weight does not control it, and it does not vanish with it. On the subspace that *is* that support — $S^* = \text{supp}(\phi)$, of size $(\frac{1}{2} + \frac{1}{L}) \dim$ at $L = 6$ and $0.571 \dim$ at $L = 8$, fixed by the probe and not by any ranking, so that $w_S = 1$ exactly by construction, with no tie-breaking and no zero-amplitude padding — the weight term is identically zero and the relative error is still 0.43 at $L = 6$ and 0.35 at $L = 8$ at $\eta = 0.18t$, rising to 1.15 and 0.73 at $\eta = 0.05t$. $\Lambda_S(\eta)$ is defined by the resolvent at resolution η and exists only inside the spectral half of this work: the obstruction theorem of Sec. VI takes it as its subject and would have no subject without Sec. III. Conversely, Sec. III leaves open precisely the question Sec. VI answers — whether the a posteriori certificate can be replaced by an a priori predictor computed from the state. Neither half can be removed without leaving the other incomplete in a way we can name.

S2. THE LEAKAGE CERTIFICATE AND THE WEIGHT-ONLY BOUND: HYPOTHESES AND DETAILS

The hypotheses of Theorem 1, and its proof. Four hypotheses carry the theorem, each a property of the pipeline rather than of the inequality: (H1) A and A_P are referred to the same E_0 ; (H2) the L_1 norm is taken over the whole line and normalized by $\|\phi\|^2$; (H3) A is the exact spectral function of the full sector; (H4) the probe is built from the exact ground state. They are stated with the theorem in Sec. III, and each is checked against this work’s own pipeline, with numbers and with the measured consequence of its failure, in Sec. S11. The proof of the theorem and of the closed form, Eq. (6), is Appendix A of the main text; its steps are numbered (1) to (5) there.

What object is certified. If the resolvent inside $\text{ran } P$ is itself evaluated by a Krylov recursion of depth n_ℓ , then Λ computed on that Krylov space bounds the error of the depth- n_ℓ reconstruction *and of nothing else*. The convenient inequality $\Lambda_K \geq \Lambda_S$ that would transfer the statement to the dense A_S is false: 1169 of 6000 random triples violate it, at a minimum ratio of 0.5258. Every table of Sec. V carries the depth of the object it certifies for that reason. Full reorthogonalization is mandatory throughout: without it P_K is not a projector—the orthogonality defect reaches 0.238 at $L = 8$, where the captured weight then reads 1.4868, an impossible value—and Λ is inflated thirtyfold.

Relation to the classical error theory, and the scope of the novelty claim. Nothing in the proof of Appendix A is new as analysis. The split, Eq. (A1), is Löwdin partitioning [1], the same algebra as the Feshbach projection of scattering theory [2]; and sixty years of Rayleigh–Ritz and Krylov error analysis for resolvents and matrix functions [3, 4] contains every rung of steps (2) and (3), along with far sharper residual, Temple and Kato–Temple estimates for individual eigenpairs [5–7], their extension to the noisy, ill-conditioned eigenvalue problem of quantum subspace diagonalization [8], and the loss-of-orthogonality analysis of Paige [9] that makes step (5) conditional on reorthogonalization, and the polynomial-restart convergence theory of Beattie, Embree and Sorensen [10]. Step (3) in particular is the reduced-basis bound on a Galerkin projection onto an arbitrary subspace—the dual norm of the residual over an inf-sup constant [11, 12]—with that constant equal to η .

Nor is “a posteriori” ours to claim, and we name the prior art rather than wait to be shown it. Stopping criteria for rational matrix functions of Hermitian matrices, computed from the residual alone, are Frommer and Simoncini’s [13]; Jia and Lv’s title is this section’s claim word for word—*a posteriori error estimates of Krylov subspace approximations to matrix functions* [14]—and Frommer and Schweitzer carry that line to Stieltjes functions [15]; Lanczos approximations to matrix functions carry a posteriori bounds from the same Cauchy-integral argument as step (5) [16, 17], and stochastic Lanczos quadrature carries them on an approximated spectral measure itself, in Wasserstein and Kolmogorov–Smirnov distance, through Gauss-quadrature brackets that hold for each seeded (weighted) spectral measure and for the global density of states that averages them [18]—bounds on a full-space Lanczos quadrature, not on a projection onto a sampled determinant subspace with boundary leakage as bounded here; and in April 2026, before this work, on *quantum* Krylov, Oliveira, Ziemis and Glaser introduce “two new metrics, the imaginary and unitary filters, that successfully assess the reliability of the obtained solutions without any knowledge of the true eigenspectrum” [19]. That is the framing of this section, established earlier and by others, and the priority is theirs. On the selection side, that a subspace chosen for the energy misses configurations a response needs, and a selection criterion built for the response instead, are Reinholdt, Kjellgren and Kongsted’s, for classical selected configuration interaction [20].

What we did not find in that literature is the conjunction of three things, and it is the whole of the novelty claimed here. First, a closed form for the frequency-integrated residual $\Lambda_P(\eta)$, assembled from the Ritz pairs, on a subspace of *coordinates* carrying no Krylov structure, no invariance and no spectral-gap assumption—so that the estimate attaches

to the set of determinants a sampler returned, which is not the span of the iterates of any recursion, and Eq. (A1) is evaluated on that set directly. Second, that the bound is two-sided on the same object: the lower branch is fixed by the captured weight and the upper branch adds what the weight does not control, and Sec. V uses the distance between them rather than either alone. Third, a *measured* calibration of where that bound crosses the trivial one, over sixteen (L, η) cells whose true errors span five decades (Sec. VD), in place of an asymptotic statement about the crossing. Whether such a bound is *useful* at the fractions sampling reaches is a separate question; it is not settled by the proof, and Sec. V answers it—in the negative—by measurement.

The pooled sweep of Fig. S5, and its limits. The pooled sweep behind Fig. S5 contributes 621 subspaces from five Hamiltonian families and eight construction rules, written by two independent implementations; 15 reproduce their target to double-precision round-off and define no ratio. On the 606 that remain the weight-only bound fails on 469, and restricted to the regime this work operates in, $w_S \geq 0.99$, it fails on 363 of 363, the mildest by a factor 2.00 (Sec. S9). The compliant cases are subspaces so poor that the relative error is already of order unity, where the bound asserts nothing. In the configuration of Fig. 3 itself— $L = 8$, $\eta = 0.18t$, the 3920-dimensional $(N+1)$ sector, momentum seeds $\hat{c}_{k\uparrow}^\dagger |\Psi_0\rangle$, both branches, eight momenta, subspaces grown in order of accumulated Born weight—the median violation factor at sector fractions 0.50, 0.70, 0.85, 0.95 is 18.5, 23.6, 26.5, 29.9, the 64 rows spanning 15.1 to 57.3 overall and 15.1 to 33.7 at the fraction 0.85 of Fig. 3. Those medians *rise* monotonically, by a factor 1.6, while the measured error over the same range falls by three orders of magnitude, from the 10^{-1} level at the lowest fraction to 1.0×10^{-4} at 0.95: the weight-only bound fails worst exactly where the reconstruction is best, which is the opposite of what an error bound is for. That is a defect of the functional form and not a pathology at an edge of parameter space, and it is therefore measurable: Fig. S5 plots it against sector fraction and against η , beside the replacement bound on the same subspaces.

Three properties of that sweep are declared with it, because each of them limits how it may be read. Its ranking uses $K = 16$ slices, one protocol step short of the $K = 18$ of the resource scan of Sec. VII. Its momentum seeds have support on 3848 to 3920 of the 3920 determinants and captured weight below one at every fraction, so the support floor derived below for the local seed does not apply to it; the members of a translation orbit carry equal Born weight, so the cut can fall inside a tied orbit. How far a tie-breaking can move an error was measured on a local-seed subspace at $L = 6$, fraction 0.85, whose cut fell among zero-amplitude determinants: ten tie-breakings moved the measured error by $\pm 20\%$, which is why the violation factors are quoted to three figures at most and why the rise across fractions is read as a direction and not as a rate. (That $\pm 20\%$ is the spread of the *measured error* over tie-breakings of a zero-amplitude cut. It is a different quantity from the 6.6×10^{-4} by which ties move the *bound* in the sweep of Sec. V, where the cut falls below the support floor and only two determinants are tied.) And the error at fraction 0.85 has two independent determinations that differ by a factor 2.3, so only its order of magnitude is quoted. None of this touches the conclusion: the two-level counterexample is analytic, and the counting argument of the next paragraph but one uses no ranking, no protocol and no numerics at all. The sweep is corroboration of a conclusion that does not depend on it.

Why the weight-only bound fails, in full. The reconstruction is Rayleigh–Ritz on $H_S = P_S H P_S$ (Sec. II), not the restriction of A to a subset of its own eigenpairs: the Ritz energies $\tilde{E}_m$ are not eigenvalues of H , so weight the subspace *keeps* is nonetheless moved in ω . The obvious sketch elides exactly this. Having correctly noted $\int A = \|\phi\|^2$ and $\int A_S = \|P_S \phi\|^2$, it bounds the L_1 discrepancy by twice the missing weight—a step legitimate when $[P_S, H] = 0$, where A_S is A with the outside contributions deleted and $\|A - A_S\|_1 = \|\phi\|^2(1 - w_S)$ identically, and controlling nothing otherwise. The two-level example is the extreme: nothing is discarded there, and every unit of retained weight is displaced by g . What the integrals do give, with no commutator hypothesis, is $|\int A - \int A_S| \leq \|A - A_S\|_1$, that is $\|\phi\|^2(1 - w_S) \leq \|A - A_S\|_1$ —a *lower* bound, with 0 violations across the 606 subspaces of the pooled sweep. It survives intact as the left-hand half of Theorem 1: the captured weight does say how wrong the reconstruction must be, it never said how wrong it can be.

The normalization of the witness errors. (Relative here and throughout this subsection means divided by $\|\phi\|^2$, the normalization of Theorem 1; dividing instead by the in-window integral of the exact A , as the repository’s own relative L_1 does, moves them by about 1% at $\eta = 0.05t$ and by 2.6 to 3.3% at $\eta = 0.18t$, measured in Sec. S11, and the two conventions are not interchanged anywhere in this paper without saying so.)

S3. MOMENT EXACTNESS: THE PROOF ROUTE, THE CONJECTURE AND THE SHOT BOUND

That proof is three lines of Rayleigh–Ritz, and it is not the proof this statement is usually given. The usual route argues that an order- $(K+1)$ quadrature “matches the first $2K + 1$ moments exactly”, and $2K + 1$ is exactly what Table II measures: the *reach* is right, the route is not. What does not hold here is the identification of A_S with a Gaussian-quadrature rule built by a nonorthogonal Lanczos/Krylov construction. No such object is built anywhere in this work: every reconstruction diagonalizes $P_S H P_S$ on a coordinate subspace, so $d\mu_S$ carries $|S|$ nodes, not $K + 1$, and the orders that separate the two objects measure the distance between them, not an error in the published reach.

The conclusion is right, the route is not, and the correct route is also the more general one (Sec. S10.1).

Equation (11) invites a corollary: “consequently, at fixed broadening $\eta > 0$, the smoothed reconstruction converges geometrically, $\|A - A_{\mathcal{K}_K}\|_1 \leq C(\eta) \rho(\eta)^{-K}$.” It does not follow, and it is not a corollary here. Moment exactness is a list of $2K + 2$ linear identities, and the L_1 distance between two Lorentzian-broadened measures is not a finite linear functional of them; moment exactness supplies neither of the two ingredients a conversion would need (Sec. S10.3), and neither $C(\eta)$ nor $\rho(\eta)$ is derived or exhibited anywhere in this work or in the deposit. We therefore state the claim as what it is.

One constraint on any answer is already available. The two-level family of Sec. III D has $\mathcal{K}_0 = \text{span}\{\phi\}$, and by direct integration at $\eta = 0.15$ its $\|A - A_{\mathcal{K}_0}\|_1$ is 1.983 at $g/\eta = 2 \times 10^2$ and 1.998 at $g/\eta = 2 \times 10^3$, rising to $2\|\phi\|^2$. Evaluating Conjecture 1 at $K = 0$ therefore forces $C(\eta) \geq 2\|\phi\|^2$: the same family that forces the trivial constant on the weight-only bound forces it here on the prefactor. Whatever content the conjecture has must be carried by $\rho(\eta)^{-K}$ alone, and in particular it cannot be non-vacuous at the small K at which a shot-limited subspace operates.

One figure of this work is read against that conjecture and not against a theorem. A geometric collapse $\text{rel-}L_1 \sim \rho^{-K}$ with $\rho \approx 1.7$, fitted to panel (b) of Fig. 2, is easily reported as “precisely the geometric convergence guaranteed by the moment-exactness theorem”. It is guaranteed by no such thing—that is the corollary ruled out above—and the fit does not carry the claim on its own either. The panel is one of the two places in this work where the sampling is genuine and its budget is declared—multinomial draws at 4000 shots per time step, eight seeds, $\eta = 0.15 t$, so $T = 4000(K+1)$ and $T = 5.2 \times 10^4$ at the last point—and the fit uses six of its thirteen measured points, inside a window selected after the fact ($4 \leq K \leq 9$, set as a literal in `spectral_validation_max.py`). Across that window the mean subspace the sampler returns grows from 179 to 236 determinants of the 300-dimensional sector, so decay in the Krylov order is confounded with growth in $|\mathcal{S}|$; and the K of the abscissa is the number of measured time slices, not the K_S of the weight-only bound, which is -1 at all thirteen points by Eq. (9). No rate is fitted to that window here, and the caption claims none. What the panel does show, over all thirteen points, with its eight-seed band and with T attached, is that the error falls by a factor 32 as the measured Krylov order is climbed at fixed η , from 0.59 at $K = 0$ to 0.0185 at $K = 12$, with a single non-monotone step at $K = 3$ inside the discovery phase; that is the statement its caption makes.

There is a sharper reason not to read Eq. (11) as an error bound, and Table II contains it. For a coordinate subspace P_S is diagonal, so $\text{span}(\mathcal{S}) \supseteq \mathcal{K}_K$ at any $K \geq 0$ already forces $\text{supp}(\phi) \subseteq \mathcal{S}$, that is $w_S = 1$ exactly—as the fourth column records for all three rows. Moment exactness here is strictly stronger than unit captured weight, and unit captured weight is precisely the condition under which Sec. III measures the error to be $O(1)$. The two meet in the first row: the $K = 0$ member of the ladder is the witness $\mathcal{S}^* = \text{supp}(\phi)$ of Sec. III, of size $(\frac{1}{2} + \frac{1}{L})D$ at $L = 6$ and fixed by the probe rather than by any ranking. It is moment-exact through order 1, it holds the entire Born weight of the probe, and at $\eta = 0.18 t$ its relative L_1 error is 0.28 on the open chain of Table II at $L = 6$, and 0.43 at $L = 6$ and 0.35 at $L = 8$ on the periodic rings on which Sec. III reports it: a measured range of 0.28–0.43 at unit captured weight, whose lower end is the very system on which the method validation of Sec. II is performed. (All three are $\|A - A_S\|_1$ divided by $\|\phi\|^2$, the normalization of Theorem 1; the size of the conversion to the repository’s own in-window relative L_1 is measured in Sec. S11, and the two conventions are never interchanged silently.) Exactness of $2K + 2$ numbers is what the structure certifies; the curve is certified, at fixed η and on the subspace in hand, by Theorem 1, whose own non-vacuity is measured rather than assumed (Sec. V).

What Eq. (13) does *not* say belongs in the same breath, and in the main text rather than in an appendix. The parameter ε thresholds a Born probability; it is not a spectral accuracy, and no step of the argument connects it to $\|A - A_S\|_1$. Arbitrarily many configurations may each carry $p(x) < \varepsilon$ while jointly carrying most of the mixture; and, decisively, even at $w_S = 1$ exactly—nothing left to capture at any ε —the relative error is $O(1)$, as Sec. III measures and the first row of Table II exhibits. Capturing the high-probability configurations is a statement about the sampler, not about the spectrum.

Two compositions of the bound follow easily from it, and neither is made here. “A sampling-capture bound polynomial in $|\mathcal{S}|$ and independent of the Hilbert-space dimension” is a pair of clauses that are literally true and jointly empty, because this same work measures $|\mathcal{S}| \sim e^{1.05L}$ for these lattices, Eq. (20), so a budget polynomial in $|\mathcal{S}|$ is exponential in L , and independence of the Hilbert-space dimension is no saving when the quantity the budget is polynomial in grows exponentially itself. And the fall of the required sector fraction with L is not “the favourable face of the polynomial-shot bound”: it is not a face of that bound at all. Equation (13) relates a shot budget to a Born-probability threshold and contains no statement about $|\mathcal{S}|/D$, about L , or about any trend in either.

What stands in their place is the narrower statement Eq. (13) does support, and it is the stronger one for being narrow: that many shots buy, at confidence $1 - \delta$, every configuration above Born weight ε , and nothing further. What the resulting subspace costs, and what it is worth, are separate measurements, reported as three independent prices—support, resolution and shots—in Sec. VII.

S4. THE NON-VACUITY THRESHOLD: PROTOCOL, TABLES AND LIMITS

Throughout this section we call FR^* the *non-vacuity threshold of the certificate of Theorem 1*: the sector fraction above which the leakage branch of the bound first improves on the trivial bound $\|\phi\|^2(1 + w_S)$. We do not call it a frontier of the method, and it is not one. It is a property of the inequality chain that proves Theorem 1, measured here for the first time, together with one structural fact about the determinant basis. Two measurements separate the two contributions. First, at fixed system and fixed accuracy target the threshold sits far above the fraction the reconstruction actually needs: at $L = 8$, $\eta = 0.18t$, a relative L_1 error of 0.05—the acceptance criterion used throughout this paper—is reached at $\text{FR} \approx 0.52$, while the certificate does not reach 0.05 until $\text{FR} \approx 0.98$, and the intervening shift of ≈ 0.45 in fraction is the looseness of the proof. Second, at full captured weight $\Lambda_S = 0$ requires $\text{span}(S) \supseteq \mathcal{K}(H, \phi)$, whose coordinate support is every determinant the reflection about the seed site allows, at every size measured—a statement that the leakage is nonzero, not about its size, and not the source of the vacuity measured below, which is at $w_S < 1$ —and a Krylov-diagonal oracle ranking gives a *worse* certificate than the Born ranking the device realizes at $L = 6$, $\eta = 0.18t$, at every fraction checked from 0.50 on. Section V E measures both. What is a property of the method, and is new, is the calibration of Sec. V D: the crossing occurs at an essentially fixed level of true error, so the crossing itself is worth computing even where the bound around it is not.

What is evaluated, in full. We evaluate Theorem 1 on the subspaces of Fig. 5 at $L = 6, 8, 10$ and 12—sector dimensions 300, 3920, 52 920, 731 808—with the ranking protocol of Sec. VII ($K = 18$, $\delta t = 0.5$, 16 substeps, 19 snapshots), over fractions 0.20–0.95 and $\eta = 0.10$ –0.25 t . Those subspaces are the $T \rightarrow \infty$ limit of that protocol and not finite-shot draws, and every statement below inherits that limit: the marginal determinant at the $L = 12$ threshold carries Born probability $1\text{--}2 \times 10^{-9}$, so at the budgets of Table IV that subspace barely exists as a sampled object. Of the 204 evaluations, 187 are converged in Krylov depth, 6 marginal and 11 not converged, at a total cost of 2874 s with a peak of 1.64 GiB. *No row flagged not converged is quoted anywhere in this section*; the tables mark the ones that are marginal.

To that grid we add twenty evaluations that are not on it. A grid point near a published fraction is not the published subspace, and the difference is not cosmetic: this sweep reads 0.80, 0.56, 0.40 and 0.18 against published fractions of 0.8200, 0.5561, 0.3625 and 0.1700. The rows of Table III are placed instead at the integer determinant count $|S| = \text{round}(\text{frac} \times \text{dim})$ that Fig. 5 reports—246, 2180, 19 183, 124 407 and 824 504 determinants at $L = 6, 8, 10, 12$ and 14—at the same four resolutions and under the same protocol, at depths $n_\ell = 245$ (exhausting the subspace), 900, 700, 300 and 250, with orthogonality defect at most 4.9×10^{-15} and $|w_K - w_S|$ at most 6.4×10^{-15} . The sixteen up to $L = 12$ pass the depth test defined below; at $L = 14$, where Λ_{out} was not recorded, only its indicator half can be applied, and it passes at all four resolutions. All twenty clear the reference-drift rule of Sec. S11, the weakest— $L = 14$ at $\eta = 0.10t$, where the reference itself drifts by 1.9×10^{-3} on halving its depth—by a factor 51. They cost 129 s of core time up to $L = 12$ and 2238 s of wall time at $L = 14$, and they add the size the sweep never reached.

Three conventions are fixed before any number is read. Full reorthogonalization is used throughout, without which the projector is not a projector (Sec. S11 gives the measured defect). What is certified is the object the code forms— Λ on $\mathcal{K} = \text{Krylov}(H_S, \phi_S, n_\ell)$ bounds the error of A_K at that depth and of nothing else, the convenient inequality $\Lambda_K \geq \Lambda_S$ that would extend it to the dense A_S being false (1169 of 6000 random triples violate it, minimum ratio 0.5258), and A_S is never formed. And depth is certified per point by the internal indicator $\Lambda_{\text{in}}/\Lambda_{\text{out}}$, the truncation part of Λ over its boundary part, rather than assumed from a fixed recursion depth [21] or from a priori Krylov convergence bounds [8]: a point passes the depth test, and is counted converged, when that indicator is below 0.05 and Λ_{out} is constant to 10^{-3} relative over the last three depths of its ladder, and is marginal when the indicator is below 0.30 otherwise (`fsum.py`); Sec. S11 bounds what the depth cap does to FR^* .

Four hypotheses carry the bound, each a statement about the pipeline rather than about the theorem; they are stated in full with the theorem in Sec. III, and all four are checked against this pipeline in Sec. S11, along with the ranking, tie and support declarations of the cut. It is (H3), the exactness of the reference, that enforces the rule that no measured error is quoted within a factor 20 of its own reference drift. One asymmetry runs through every table, and it is narrower than it looks: neither term of Eq. (7) is free of Ψ_0 . The leakage needs the retained probe ϕ_S , the amplitudes of Ψ_0 on the determinants the probe operator maps into S , which the reconstruction itself uses and which is exact here by (H4); the weight term needs in addition the full norm $\|\phi\|^2$, and estimating w_S against an already-truncated Ψ_0 is biased optimistically, while Λ_S evaluated with such a Ψ_0 belongs to another probe (Sec. III). What the tables report is accordingly one object and not a family of them: a bound on $\|A - A_K\|_1/\|\phi\|^2$ for the reorthogonalized Krylov reconstruction at a declared depth, against a reference whose drift is declared and enforced.

The two largest sizes. The largest classically converged case carries the evidence. At $L = 12$ and the published subspace itself— $|S| = 124\,407$ of 731 808 determinants—the captured weight is $w_S = 0.998622$, so Eq. (14) asks for $\hat{\Lambda}_S/\eta < \tau(w_S) = 0.9625$ and the measured value is 4.236 at $\eta = 0.18t$, putting the leakage branch at 8.544 against a trivial branch of 1.9986. That is converged in depth: the depth indicator is $\Lambda_{\text{in}}/\Lambda_{\text{out}} = 2.8 \times 10^{-3}$ and between depths 250 and 300 Λ_{out} moves by 3.7×10^{-7} relative and the bound by 1.4×10^{-5} , while at the neighboring grid

TABLE S1. **Where the certificate stops being vacuous, and what the reconstruction actually needs, measured on the same sweep.** Protocol of Sec. VII, full reorthogonalization, depth certified per point. Column 3 is the fraction Fig. 5 reports; column 4 the control, where this sweep—independent code, the normalization of Theorem 1, the whole line, a deep reference—first reaches relative $L_1 < 0.05$, the parenthesized value being the log-linear interpolation inside the bracket and not a measurement at three digits; column 5 is FR*; columns 6–7 are where the certified value itself first falls below 0.50 and 0.20. Every bracket is delimited by two converged evaluations unless marked: * means the upper end is marginal in depth, *n.c.* that the bracket rests only on evaluations failing the depth test and is therefore not quoted. Columns 5–7 are statements about A_K at the declared depth, not about the dense A_S . Columns 4 and 5 must be read together: FR* alone falls with size and would suggest the certificate improves as the problem grows, whereas what the reconstruction needs falls with it and the gap widens—a gap we do not quote as a ratio, both columns being interpolations inside brackets 0.02–0.10 wide.

L	$\dim(N+1)$	published	rel- $L_1 < 0.05$	FR*	cert < 0.50	cert < 0.20
<i>$\eta = 0.18t$, the resolution of the momentum-resolved channel</i>						
6	300	0.82	[0.70, 0.80]	[0.90, 0.92]	> 0.97	> 0.97
8	3920	0.56	[0.50, 0.56]	[0.75, 0.80]	[0.90, 0.92]	> 0.95
10	52920	0.36	[0.30, 0.40]	[0.60, 0.70]	[0.80, 0.85]	[0.85, 0.90]
12	731808	0.18	[0.14, 0.18]	[0.40, 0.45]	[0.60, 0.70]*	<i>n.c.</i>
<i>$\eta = 0.15t$, the resolution at which Fig. 5 is drawn</i>						
6	300	0.82	[0.80, 0.85] (0.816)	[0.92, 0.95]	> 0.97	> 0.97
8	3920	0.56	[0.50, 0.56] (0.553)	[0.80, 0.82]	[0.92, 0.95]	> 0.95
10	52920	0.36	[0.30, 0.40] (0.356)	[0.60, 0.70]	[0.80, 0.85]	[0.85, 0.90]
12	731808	0.17–0.18	[0.14, 0.18] (0.168)	[0.45, 0.50]	[0.60, 0.70]*	<i>n.c.</i>

point FR = 0.18 the same quantity was checked twice more— Λ_{out} changing by 7.5×10^{-7} between depths 250 and 300, and an independently written re-implementation moving it by 8.6×10^{-5} between depths 100 and 300. In all sixteen deposited depth series Λ at every depth from 100 on lies at or above its deepest value—it is not monotone in depth in general, and rises by up to 2.2% from one depth to the next below depth 100 at $L = 12$ —so the bound at any depth from 100 on, including the depth 150 at which Fig. 5 is drawn, is no smaller, so the vacuity is not an artifact of Lanczos truncation in either direction. Nor is it peculiar to these subspaces: over the 606 live subspaces of the pooled sweep of Sec. S9, Fig. S5, the leakage branch is the smaller of the two on only 61. The reconstructions at the fractions this paper publishes—Fig. 5 and the operating points of Secs. VII and VIII—are therefore uncertified. Figure 3, drawn at 85% of its sector, is the exception: there the certificate of the depth-500 Krylov reconstruction is non-vacuous in all sixteen channels, at a relative bound of 0.77–1.75 against the trivial 2, and loose, 350–900 times the true error of the channel integrated over the whole frequency line (`cert_akw.json`).

One row is quoted with a caveat attached. $L = 14$ is the size at which Fig. 5 reports its headline fraction of 0.08, and the certificate is evaluated there at $|S| = 824\,504$ of 10 306 296 determinants—one determinant above the round($\text{frac} \times \text{dim}$) = 824 503 of the deposited grid—with $w_S = 0.998388$ and a leakage branch of 17.21, 11.18, 9.21 and 6.50 against a trivial branch of 1.9984. Ground state, seed and order there are the repository’s stored checkpoint—residual 2.7×10^{-9} , the order verified to be a permutation of the sector—and the $K = 18$ ranking protocol, re-run at that size, reproduces the stored selection as a set to 1.000000 at $|S| = 824\,504$ and at the fractions 0.04, 0.16 and 0.32, as it does at $L = 10$. The caveat is the depth indicator $\Lambda_{\text{in}}/\Lambda_{\text{out}} = 0.0453$ at $\eta = 0.10t$, the loosest value quoted anywhere in this section, comparable to the 4.7×10^{-2} of the last row of Table S3. It is not repaired, and the row is reported because it makes the verdict of this section worse rather than better.

One column of Table S1 is a positive control and has to be read before the rest. Applying this paper’s own acceptance criterion to the same sweep—independent code, the theorem’s normalization, the whole line rather than a window, a deep reference rather than one recomputed at the subspaces’ own depth—returns 0.816, 0.553, 0.356 and 0.168 at $\eta = 0.15t$ for $L = 6$ –12, against the published 0.82, 0.56, 0.36 and 0.18: agreement to 7% at worst, and to under 2% at every size once each value is compared with the fraction the scan itself reports rather than with the two-figure entry of the replication grid (0.5561, 0.3625, 0.1700 at $L = 8, 10, 12$). None of the three ingredients is shared. Figure 5 is thereby reproduced independently.

The last column of Table III is that control taken one step further, and it is the sharper of the two: instead of re-deriving the fraction it evaluates the relative L_1 error at the published subspace, at the resolution and against the threshold that define it ($\eta = 0.15t$, $\text{rel-}L_1 < 0.05$). The five values—0.0477, 0.0478, 0.0484, 0.0485 and 0.0463—all sit just below the threshold and are flat to within 5% of one another across a sector dimension running from 300 to 10 306 296, $L = 14$ included. The criterion that defines the published fractions is therefore reproduced at the subspaces themselves, at every size the figure plots above $L = 4$, again with nothing shared. What fails in this section is not the reconstruction and not that curve; it is the word *certificate* attached to the fractions at which the curve is read.

In shots the distance between those two columns is a budget and not a rhetorical gap (Table S2): reaching the

TABLE S2. **The threshold in the only currency a sampler has.** T_{marg} is the shots per snapshot at which the marginal determinant of the cut appears once in expectation—the reciprocal of its Born probability in the accumulated distribution of the 19 snapshots, from the protocol’s own cut weights; covering all of S costs a further factor $\log|S|$. *It is a different quantity from the measured budgets T of Table IV*, which are totals over the 19 snapshots at which the finite-shot reconstruction reaches relative $L_1 < 0.05$; the two carry different names for that reason, they are not interchangeable, and only ratios within this table are used. Column 2 uses the grid point nearest the published fraction: 0.80 against a published 0.82 at $L = 6$, and 0.40 against 0.36 at $L = 10$, where the published fraction is the worse of the two. Factors are from unrounded budgets and need not reproduce from the two-figure entries. The marginal Born probability at the $L = 12$ threshold is 2.2×10^{-9} at $\text{FR} = 0.40$, 1.2×10^{-9} at 0.45.

L	published fraction $\rightarrow T_{\text{marg}}$	$\text{FR}^* \rightarrow T_{\text{marg}}$	factor
6	$0.80 \rightarrow 2.2 \times 10^4$	$[0.90, 0.92] \rightarrow 3.1\text{--}3.6 \times 10^4$	1.4–1.6
8	$0.56 \rightarrow 2.1 \times 10^5$	$[0.75, 0.80] \rightarrow 1.6\text{--}2.1 \times 10^6$	7.5–10
10	$0.40 \rightarrow 5.2 \times 10^6$	$[0.60, 0.70] \rightarrow 3.4 \times 10^7\text{--}1.1 \times 10^8$	6.7–21
12	$0.18 \rightarrow 2.3 \times 10^7$	$[0.40, 0.45] \rightarrow 4.7\text{--}8.1 \times 10^8$	20–35

TABLE S3. **The strongest certified statements available from converged data.** Each row is the tightest certified value at that size whose bracket is delimited by evaluations passing the depth test; rows that would be stronger but rest on evaluations failing it are not listed. “True error” is the measured relative L_1 at the same point, in the normalization of Theorem 1, given only where it exceeds its own reference drift by more than a factor 20 (Sec. S11, H3); at $L = 10$ it does not, and is withheld rather than rounded. At $L = 6$ the recursion runs to $n_\ell = |S|$, so the evaluation is exact and the depth indicator is not defined. The gap between the certified value and the true error is three to four decades: the object is a certificate and not an error estimate, and Sec. VD states what it is good for.

L	η/t	fraction	certified rel- $L_1 \leq$	true rel- L_1	$\Lambda_{\text{in}}/\Lambda_{\text{out}}$
6	0.18	291/300 = 0.97	0.502	3.94×10^{-4}	exact ($n_\ell = S $)
8	0.18	3 724/3 920 = 0.95	0.219	4.33×10^{-5}	8×10^{-4}
10	0.18	47 628/52 920 = 0.90	0.136	withheld (H3)	1.6×10^{-3}
12	0.18	439 085/731 808 = 0.60	0.717	1.89×10^{-4}	7.0×10^{-3}
12	0.15	0.60	0.878	2.66×10^{-4}	1.4×10^{-2}
12	0.25	512 266/731 808 = 0.70	0.218	1.99×10^{-5}	4.7×10^{-2}

non-vacuous band multiplies the shots per snapshot needed for the marginal determinant by 7.5–10 at $L = 8$ and by 20–35 at $L = 12$, and only by 1.4–1.6 at $L = 6$, where the sector is small. Driving the certified value down to 0.10 costs more again—a further factor 241 at $L = 8$ (fraction 0.95) and 634 at $L = 10$ (fraction 0.90), the corresponding $L = 12$ fraction resting only on evaluations that failed the depth test, so that no budget is quoted for it. This section reports the measured regime of validity: four sizes, four resolutions, 204 evaluations, every bracket delimited by converged rows, and a shot budget attached to each.

The strongest certified statements. Table S3 collects what the sweep does certify, taking at each size the strongest statement whose bracket is converged—at $L = 12$, that 60% of a 731 808-dimensional sector gives relative $L_1 \leq 0.717$, tightening to ≤ 0.218 at 70% and $\eta = 0.25 t$. Each row is a statement about A_K at the declared depth, and none of them uses the exact spectrum.

One restriction belongs with that table. At the acceptance criterion this paper uses everywhere else—relative $L_1 < 0.05$ —the certificate requires at least 0.98 of the sector at $L = 8$, measured at integer subspace size rather than interpolated: the certified value is 0.084 at 3802 of 3920 determinants, 0.060 at 3830 and 0.039 at 3855, so the crossing lies in $[0.977, 0.983]$; on our grid it is not reached at all at $L = 6$, $L = 10$ or $L = 12$. At such fractions diagonalizing the whole sector is already what one would do instead, and a certificate informative only there does not certify the regime in which sampling is of interest. We state it without qualification: at the accuracy this paper accepts elsewhere, Theorem 1 does not certify the operating points of Fig. 5, and the shot cost of the regime where it would exceeds the saving that figure reports.

What the table does certify at a fraction a sampler can plausibly reach is weaker, and it is not empty. The trivial bound caps the relative error at $1 + w_S \leq 2$ for *any* subspace, a reconstruction of pure noise included; at $L = 12$ and 60% of a 731 808-dimensional sector the certified 0.717 sits a factor 2.8 below that cap, and at 70% and $\eta = 0.25 t$ a factor 9.2 below it. Those are exclusions of gross failure made from the subspace alone, at a dimension where the exact spectrum is expensive and, at the sizes this method is aimed at, unavailable.

The calibration and its limits, in full. The calibration is the reason the sweep is reported. Across all sixteen (L, η) cells the crossing of the trivial bound occurs at a true relative L_1 error of 1.2×10^{-3} to 7.3×10^{-3} , median 2.8×10^{-3} : a spread of 5.9 over a table whose true errors span five decades, and in every cell at least 47 times the drift of our

own reference. $\hat{\Lambda}_S(\eta)/\eta$ is therefore a calibrated a posteriori risk tracker—its crossing of the trivial bound locates a true error of $\approx 3 \times 10^{-3}$ to within a factor of three over $L = 6$ –12 and $\eta = 0.10$ –0.25 t , and it is computed from H , S , the retained probe ϕ_S , the probe norm $\|\phi\|$ and η , without A , given the exact probe (H4). That factor of three is a drift and not scatter, and the deposited grid settles it in a few lines: the sixteen crossing errors are ordered without a single exception, falling with L at every resolution—5.65, 3.50, 2.52 and 1.60×10^{-3} at $\eta = 0.18 t$ for $L = 6$ –12, a factor 2.8–3.7—and rising with η at every size by 1.6–2.1, the product of the two being the observed 5.9. The drift runs the conservative way in the variable one would extrapolate: the larger the system, the smaller the true error at which the certificate first says anything, so reading the crossing as $\approx 3 \times 10^{-3}$ beyond the sizes measured overstates the error rather than understating it. In η , which the user chooses and which lies inside the measured range, it does not. The objection to anticipate is that this is Fig. 5 relabeled. It is not, and the difference is the content: that figure is built knowing the exact spectrum and this quantity is computed without it. That the two track each other to within a factor 5.9 across five decades of error is not a redundancy—it is the calibration, and it is what makes the quantity usable where the second curve cannot be drawn.

Four limits come with it. The crossing is one point per cell, so the tracker locates one error level and does not estimate the error away from it: at a *fixed* subspace the looseness of the bound against the measured error is constant to within $\pm 17\%$ ($L = 6$) and $\pm 25\%$ ($L = 8$) over a decade in η , but across subspaces it varies by a factor 51, and at the threshold itself it is 3×10^2 to 2.4×10^3 —the object is a certificate, not an error estimate, and calling it the latter would be refutable in one line. Neither of its two terms is free of the ground state (Sec. III), and the comparison needs more than the leakage: it is $\hat{\Lambda}_S/\eta$ against $\tau(w_S)$, and τ needs w_S , which a user does not have. That dependence can be discharged conservatively, and doing so is what makes the decision itself a posteriori. Equation (14) is decreasing in $1 - w_S$ for $w_S \geq 0.066$, with $\tau(1) = 1$, and positive for $w_S > 0.296$, so any assumed floor on the captured weight above 0.296 yields a usable threshold: at the $L = 12$ published operating point $w_S = 0.998622$ and $\tau = 0.9625$, and every subspace with $w_S \geq 0.995$ has $\tau \geq 0.9280$. A user willing to assume only that the sampler captured 99.5% of the probe’s Born weight can therefore run the comparison as $\hat{\Lambda}_S(\eta)/\eta < 0.928$ and use no other ground-state information, at the cost of at most 7% in the threshold. Where not even that floor can be assumed, or where w_S is estimated against an already-truncated Ψ_0 —an estimate biased optimistically—the test is not conservative and the bound is not a posteriori. All of this presupposes the exact probe of (H4): with a Ψ_0 estimated on a sampled subspace, $\hat{\Lambda}_S$ itself is evaluated for another probe. And sixteen cells over four sizes and four resolutions is the whole of the evidence: no exponent is fitted to it and no extrapolation to $L = 14$ is made, the local log–log slopes of $\text{FR}^*(L)$ already varying by a factor 4.1 across the four sizes measured.

The fourth limit is the one a reader holding the deposited grid will test first, and it is a real restriction on what the crossing is worth. A sampler already holds the captured Born weight, so the question is whether a stopping rule indexed on $1 - w_S$ alone locates the same error level, in which case the crossing would be that weight relabeled. It is not: over the sixteen cells, $1 - w_S$ at the crossing ranges over a factor 45 against the true error’s 5.9, and still over 9.4 against 3.8 once the extreme cell at each end is dropped. A rule with constants fitted to the grid, however, comes close enough. Take $1 - w_S < c\eta^p$, fit c and p on some of the four sizes with its operating point matched to the certificate’s, evaluate it on the sizes held out, and repeat over all fourteen train/test splits: out of sample it localizes the crossing error about as tightly as the parameter-free certificate does, and more tightly on eleven of the fourteen (`null_ws7.py`). Out of sample the certificate has no edge in sharpness over the captured weight as a tracker of the error; its edge is provability (Sec. VD). The difference lies elsewhere. The fitted rule fires at $1 - w_S \sim 10^{-5}$ – 10^{-4} , one to two orders of magnitude finer than the floor $1 - w_S \leq 5 \times 10^{-3}$ at which the threshold above still works, and it asks that precision of the one quantity this section has already said a user does not have and, estimated against an already-truncated Ψ_0 , has optimistically; strip its second parameter and an η -blind $1 - w_S < c$ is the looser of the two on thirteen of the fourteen splits. Theorem 1, for its part, is an inequality with a proof: it holds at a size no grid has seen, and its decision survives replacing w_S by a floor. The fitted rule is a two-constant regularity with no guarantee outside its fitted range—the only place either would be used.

Within those limits the procedure is operational and identical at every size we measured: assemble $\hat{\Lambda}_S(\eta)/\eta$ from the Ritz vectors already in hand and compare it with $\tau(w_S)$. Above τ the certificate says nothing, which is the regime Fig. 5 operates in; at τ the true relative error is $\approx 3 \times 10^{-3}$ within a factor of three; below τ it is smaller than that, by an amount the bound states loosely and the calibration does not sharpen. That is the whole content of the certificate, and it is a number a user can compute before knowing whether the reconstruction is right.

Where the threshold comes from, in full. Where does the ≈ 0.45 of sector fraction between what the reconstruction needs and what the certificate certifies go? We measured the displacement rung by rung along the chain of inequalities that proves Theorem 1, at $L = 8$, and the answer removes the one repair that looked available: the pointwise step from a scalar distance to a distance in the Hilbert space accounts for 44% of it, the operator-norm step $\|R(z)\| \leq 1/\eta$ applied to a source that depends on ω for 38%, the Feshbach/Minkowski split for 11%, and the Cauchy–Schwarz step in ω for 4.6% (medians over 15 of 18 cells; Sec. S11). The working hypothesis that Cauchy–Schwarz is what places the

threshold is therefore refuted by its own measurement: its total margin for tightening is a factor 1.05 typically and 3.29 at the worst point measured, which moves the threshold by a fraction of one grid interval.

The second measurement concerns the subspace itself, and it is the one that turns part of the result into a statement about the problem. $\Lambda_S = 0$ holds if and only if $\text{span}(S)$ contains $\mathcal{K}(H, \phi_S)$, the Krylov space of the retained part of the probe (Sec. III); for a subspace that holds the whole probe, $w_S = 1$, that space is $\mathcal{K}(H, \phi)$ itself, so the coordinate support of $\mathcal{K}(H, \phi)$ is a floor under every coordinate subspace that holds the whole probe without leaking. Measured, it is 296 of 300 determinants at $L = 6$, where $\dim \mathcal{K} = 65$, and 3902 of 3920 at $L = 8$, where $\dim \mathcal{K} = 596\text{--}663$. (It is the *coordinate support* of $\mathcal{K}$ that is measured, not $\mathcal{K}$ itself, whose dimension is three orders of magnitude smaller; containment, not spanning, is what $\Lambda_S = 0$ requires.) The determinants missing are not an accident of threshold. The reflection $j \rightarrow -j$ about the seed site commutes with H , and ϕ is an eigenvector of it, so every vector of $\mathcal{K}$ vanishes exactly on the reflection-fixed determinants of the opposite parity: 4, 18, 48, 200 and 600 at $L = 6, 8, 10, 12$ and 14. Every other determinant is in the support. At $L = 10$ the same construction—a Lanczos recursion on the whole sector to depth 700 with full reorthogonalization, orthogonality defect 1.6×10^{-15} —leaves no row exactly zero in floating point, although 48 of them vanish in exact arithmetic: round-off is not a support.

Beyond $L = 10$ that basis cannot be stored, and we use a lower bound instead: every $H^k \phi$ lies in $\mathcal{K}$, so the union over k of the coordinate support of $H^k \phi / \|H^k \phi\|$ is contained in the support of $\mathcal{K}$. A count of this kind is meaningless without its threshold, so both are given. Projected onto the reflection sector of ϕ at every power (`z_suppK_sym.py`), so that round-off cannot populate the forbidden sector, the union over $k \leq 36$ is every symmetry-allowed determinant at every size: 296, 3902, 52 872, 731 608 and 10 305 696 at $L = 6$ to 14, at an amplitude threshold of 10^{-12} , and the same at 10^{-10} except at $L = 14$, where 232 genuine amplitudes lie between the two thresholds. Unprojected, the power iteration amplifies round-off in the forbidden sector until it crosses either threshold and reports the entire sector from $L = 8$ on (`z_suppK_power.py`); that count is not a support.

At every size measured, therefore, a coordinate subspace that holds the whole probe leaks unless it contains every symmetry-allowed determinant—the sector minus its forbidden determinants contains $\mathcal{K}$ and does not leak—so at full captured weight no choice of determinants short of all of them makes Λ_S vanish, and that is a property of the basis in which the device measures, not of the bound and not of the ranking. It is a statement that $\Lambda_S \neq 0$, not about its size, and the vacuity at the published fractions is not deduced from it: that is the measurement of Table III. Below full captured weight the space that decides is $\mathcal{K}(H, \phi_S)$, which changes with S , and nothing structural is claimed there. Nor is the ranking the weak link: replacing the Born ordering by a Krylov-diagonal oracle—determinants ordered by the diagonal of the projector onto $\mathcal{K}(H, \phi)$, an ordering no sampler could produce—gives a *worse* certificate at $L = 6$, $\eta = 0.18t$, at every fraction checked from 0.50 on, 12.97 against 4.27 at 0.85 and 5.82 against 0.911 at 0.95; only at 0.30, where both are far above the trivial value, is the oracle marginally better, 21.76 against 24.52.

At fixed size the displacement is the looseness of the proof, and part of the looseness has a structural reason: the leakage branch is of first order in the boundary coupling, the error it bounds of second. For a subspace that holds the whole probe, eliminating Q exactly replaces H_P by $H_P + PHQ(z - QHQ)^{-1}QHP$, and the two sum rules then give $\|A - A_P\|_1 \leq \|\phi\|^2 \|QHP\|^2 / \eta^2$. Sharper, with $\phi_Q = 0$ Eq. (A1) and Cauchy–Schwarz in ω give $\|A - A_P\|_1 \leq \Lambda_P(\eta) \tilde{\Lambda}_P(\eta) / \eta^2$, where $\tilde{\Lambda}_P$ is Eq. (6) with the energy difference in its kernel reversed in sign, equal to Λ_P for a real Hamiltonian and probe. The ratio of the leakage branch, $2\|\phi\|\Lambda_P/\eta$, to the error is therefore at least $2\eta\|\phi\|/\tilde{\Lambda}_P$, and grows without limit as the subspace approaches invariance; at full captured weight the product $\Lambda_P \tilde{\Lambda}_P / \eta^2$ is itself an a posteriori certificate of second order. Every published fraction lies below full captured weight, and whether a certificate of second order exists there is open; we have not constructed one. Section VI closes the picture from the other side: nothing cheaper than the crossing of Sec. VD—no invariant of the state, computable before the subspace exists—can be put in its place.

One route out of the determinant basis has been measured, and it closes this section as an outlook and not as a result, because the same measurement shows where it stops. The qualifier *in the determinant basis* above is load-bearing, and it is measurable. An orbital rotation is a Givens network and the circuit of Sec. II already contains them, so the measurement basis is a choice; for the momentum-resolved probe of Fig. 3 it is a forced one, since H commutes with lattice translation and ϕ_k is an eigenvector of it, so that $\mathcal{K}(H, \phi_k)$ lies inside a single momentum block—rigorously, at every L —and that block is exactly $\dim(N+1)/L$ plane-wave determinants. At $L = 6$ and $L = 8$ the Krylov support fills nearly all of the block, and the certificate becomes non-vacuous only close to the full block: at 48–50 of its 50 determinants at $L = 6$ and at 455–483 of its 490 at $L = 8$, over the momenta of the addition branch (`z_kblock.py`), a factor 5–7 fewer than the site determinant basis requires; on the block itself Λ_S vanishes identically and the bound is zero rather than vacuous. The gain, though, is $1/L$ against an exponential: as a fraction of the fraction Fig. 5 publishes the block is 0.20 at $L = 6$ and then 0.22, 0.28, 0.49 and 0.89 at $L = 14$, the two crossing just past $L = 14$. It is also a laptop Lanczos—60 984 determinants at $L = 12$, 736 164 at $L = 14$ —so the regime it certifies is one that momentum-resolved exact diagonalization has enumerated since the 1970s. Three things are unmeasured here and would have to be measured before more is said: whether the support fills the block at $L \geq 10$, where the threshold then sits, and what the $\approx L(L-1)$ extra two-qubit gates do to the sampled distribution.

S5. ONE-BODY MAGIC, THE GEMINAL WITNESS AND THE DETERMINANT SUPPORT: FULL ANALYSIS

The candidate measures. The candidates have a sharp algebraic origin. Matchgate (free-fermion) circuits are classically simulable [22–25], and what leaves that manifold is fermionic non-Gaussianity: any pure non-Gaussian state injected into a matchgate circuit promotes it to universality [26], the fermionic counterpart of qubit magic [27, 28]. A ladder of measures quantifies it—Gaussian rank and extent [29], non-Gaussian gate counts [30, 31], including the count that makes a state efficiently learnable [32], covariance-matrix measures and monotones [33, 34], and convolution- and entropy-based measures built on the violation of Wick’s theorem [35]; distinct from these is the qubit non-stabilizerness that Gaussian states themselves carry [36].

The collapse, in detail. Equation (17) is exact on any number-conserving state and verified state by state to 10^{-13} across the dissociation of N_2 and the suite of Table S12, that is on the 38 molecular ground states of this work. For spin-restricted natural orbitals ($g_{a\uparrow} = g_{a\downarrow} = n_a/2$) it equals $2N_u$, the quadratic effectively-unpaired-electron index $N_u = \sum_a n_a(2 - n_a)$ of Takatsuka–Fueno–Yamaguchi and of Staroverov–Davidson [37, 38], and $\text{tr}[\gamma(1 - \gamma)]$ is one quarter of the Gigena–Rossignoli one-body quadratic (linear) entropy, $S_2 = 4 \text{tr}[\gamma(1 - \gamma)]$ on a number-conserving state [39], a trace-form one-body entanglement entropy in the sense of Gigena, Di Tullio and Rossignoli [40]. Here $2N_u$ is the quadratic index, distinct from the linear $\sum_a \min(n_a, 2 - n_a)$ of Head-Gordon [41]. A cleanly singly-occupied spin orbital contributes 0 to $\mathcal{F}_1$ but 2 to $2N_u$, so on the five spin-polarized members of Table S12 (O_2, NO, CN, OH, BeH) the two differ and we report $\mathcal{F}_1 = 4 \text{tr}[\gamma(1 - \gamma)]$ directly. Outside number conservation the collapse fails: a paired Gaussian state hides fractional occupations in anomalous correlators and keeps $\mathcal{F}_1 = 0$.

The algebra is not new. Equation (17) is an immediate number-conserving specialization of the antiflatness of Ref. [33]; N_u has been the standard diagnostic of unpaired density since 1978 [37]; and the relation was noted independently and contemporaneously by Świątek *et al.* [42], who state it for fixed-particle-number states and write its ensemble-averaged form for random superpositions of Gaussian states, their model of interacting-integrable eigenstates. What we add is the exact, state-by-state *dictionary*: three literatures—fermionic magic, effectively-unpaired electrons, one-body entanglement—name one quantity. The verification is the 38 molecular states just quoted; the further equality to $2N_u$ is checked on the closed-shell members, the five spin-polarized ones being excluded for the reason given above. The 36 lattice ground states of this work are number conserving, so Eq. (17) applies to them by the same algebra, but they were not part of that state-by-state check and we do not count them in it. The consequence we carry forward is structural and equally exact: being a function of the occupation spectrum alone, $\mathcal{F}_1$ inherits the orbital-rotation invariance of the eigenvalues of γ . It is a genuine, cheap, exactly computed invariant of the state, and it is that invariance the two following subsections use.

The witness and the fibers. The proof is four short block-diagonal computations (Sec. S12); the step worth flagging here is that at $\theta = \pi/4$ a single maximally-paired geminal is the two-determinant cat, whose Majorana covariance *vanishes identically*, so *every* order of antiflatness saturates its maximum at once. The values of $\mathcal{F}_1$, $\mathcal{F}_2$ and $|\mathcal{S}|$ are reproduced by exact diagonalization for $K = 1, \dots, 4$, and so is $\chi = 2$, the largest Schmidt rank over all cuts of the pairing ordering (Sec. S12).

The orbit is the argument. The witness is not the trivial “a product state is easy” (Fig. S1). Because $\mathcal{F}_1$ and $\mathcal{F}_2$ are invariant under number-conserving orbital rotations [33], the whole orbit $\{U|\Psi_K\rangle\}$ carries the *same* $(\mathcal{F}_1, \mathcal{F}_2) = (4K, 4K)$ while the cost in the fixed computational basis—Born-sampling time, through the bond dimension—is $O(K)$ at the localized representative and, at generic rotated points, not established here. Two conclusions follow, each in the only direction the construction supports: a Gaussian-invariant fermionic magic is not a sufficient certificate of hardness and does not lower-bound the bond dimension, nor the cost beyond linear order in K (there $\chi = 2$ while $\mathcal{F}_k = 4K \rightarrow \infty$); and $\mathcal{F}_k$ determines the cost only if the rotated points are in fact easy, being constant on an orbit that contains an $O(K)$ -cost point and points at which sampling is plausibly hard. In a generic rotated basis the Born sampling of $|\Psi_K\rangle$ is plausibly classically hard—the free-fermionic-linear-optics-with-magic-input regime of the Fermion Sampling advantage scheme [43]—while $\mathcal{F}_k$ is unchanged: any hardness that appears belongs to the sampler’s fixed frame. We are silent on whether magic is *necessary* for hardness, and we anchor the easy end to an external result rather than our own, since for this paired class the additive-error estimation of amplitudes, overlaps and number correlators under free-fermionic dynamics is classically efficient [44].

The same orbit reaches the leakage. The orbit argument was written for $|\mathcal{S}|$; it extends to the second budget of Eq. (7). An invariant upper bound on $\Lambda_S(\eta)/\eta$ *does* exist—it is exhibited inside the proposition below, and we tested it on 1600 hostile cases (GOE/GUE, near-degenerate, banded, $\|H\| \sim 300$) with zero violations and with its first link attained, minimum ratio 1.0000000000. The correct statement is therefore one of non-vacuity, not of non-existence: the same logical form the $|\mathcal{S}|$ no-go always had, where the sector dimension is likewise an invariant upper bound that says nothing.

Table S4 is the exhibit. The two endpoints of each fiber are the *same* rotation angle applied within one geminal block and across two blocks, which is why the captured weight agrees bit for bit. An exhaustive search over all 276

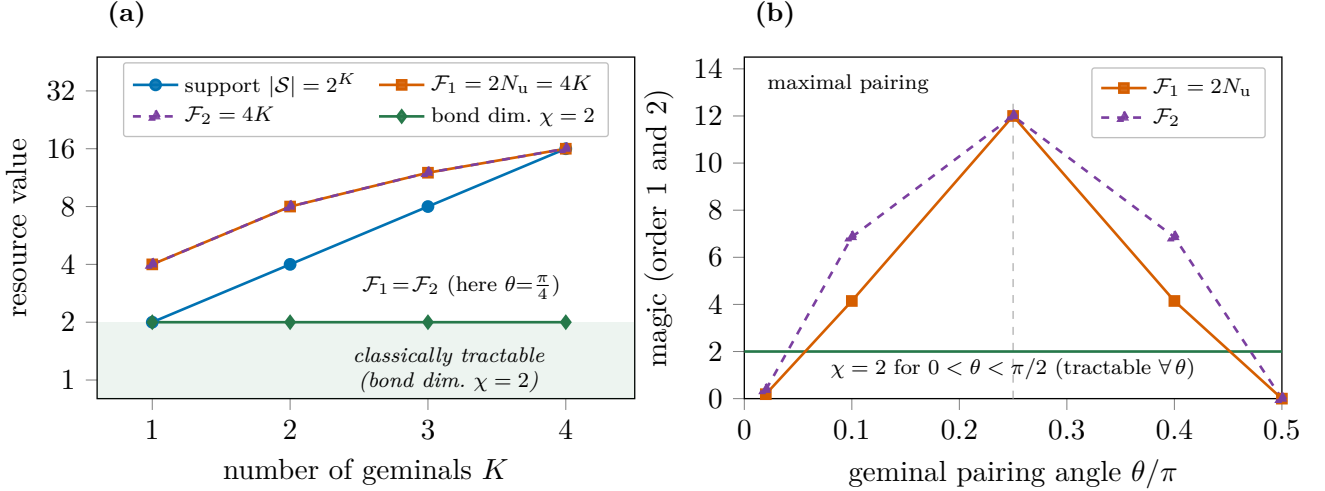


FIG. S1. **The geminal orbit witness (Theorem 3).** Antisymmetrized products of strongly orthogonal geminals in their localized (pairing) basis, where $\chi = 2$ and the state is a classically tractable tensor network. **(a)** As K grows, $\mathcal{F}_1 = 2N_u = 4K$, $\mathcal{F}_2 = 4K$ and $|\mathcal{S}| = 2^K$ all diverge while the bond dimension is 2; the χ markers are the largest Schmidt rank over all cuts of the pairing ordering, computed by the deposited diagonalization (Theorem 3); at the central cut alone it is 2, 1, 2, 1 for $K = 1, \dots, 4$, because that cut falls between two geminals when K is even. **(b)** At $K = 3$, sweeping the pairing angle drives both $\mathcal{F}_1$ and $\mathcal{F}_2$ to their maximum at $\theta = \pi/4$ —where the covariance vanishes and all orders saturate—with $\mathcal{F}_2 > \mathcal{F}_1$ off the peak, and $\chi = 2$ at every interior angle, dropping to 1 at $\theta = 0$ and $\theta = \pi/2$, where the state is a single determinant (the line drawn at $\chi = 2$ is the analytic interior value). Because $\mathcal{F}_k$ is orbital-rotation invariant the same values persist on the whole orbit, which contains the $O(K)$ -cost pairing point and rotated points at which Born sampling is plausibly hard (not established here).

TABLE S4. **Fibers of fixed invariants with moving leakage**, at $s = \pi/16$, $\delta = 0.01$, $\eta = 0.10$, $|\mathcal{S}| = 2^{K-1}$, $w_S = 0.961940$, $\tau(w_S) = 0.795$. Along each row the spectrum, every Lehmann weight, $\|\phi\|^2$, $|\mathcal{S}|$ and all $\mathcal{F}_k$ are identical between the two endpoints (w_S to 2.2×10^{-16}), so the first term of Eq. (7) is fixed and only the second moves. Quantities are relative ($\hat{\Lambda}_S = \Lambda_S/\|\phi\|$) and “certificate” is $\min\{(1 + \sqrt{w_S})(\sqrt{1 - w_S} + \hat{\Lambda}_S/\eta), 1 + w_S\}$, the relative form of Theorem 1; “al.” and “rot.” label the aligned and rotated endpoints of the fiber. Bold marks a non-vacuous certificate, below the trivial bound $1 + w_S$. A continuous angle sweep is in Sec. S12.

K (dim($N+1$))	$\mathcal{F}_1=\mathcal{F}_2=\mathcal{F}_3$	$\hat{\Lambda}/\eta$ al.	$\hat{\Lambda}/\eta$ rot.	cert. al.	cert. rot.
2 (24)	8	0.237	2.326	0.856	1.962 (trivial)
3 (300)	12	0.323	2.336	1.026	1.962 (trivial)
4 (3920)	16	0.391	2.346	1.161	1.962 (trivial)

coordinate subspaces of that size at $K = 2$ settles both ends. At the aligned endpoint the global minimum of the certificate over the 276 is 0.856 and it is attained by the protocol’s own subspace: the subspace the sampler would return is the best available one. At the rotated endpoint the minimum over the same 276 is exactly the trivial bound, attained only by subspaces of vanishing captured weight: *no* determinant subspace of that size certifies anything there.

What this exhibit is not. Four limits are declared with it, and they are why we call it a fiber and not a sweep of the orbit. *(i)* The evidence is a matched pair at $s = \pi/16$ for $K = 2, 3, 4$. At the maximal-pairing angle $s = \pi/4$ the Born distribution of ϕ is exactly flat over 2^K determinants, so “the top $|\mathcal{S}|$ of the Born distribution” does not define a subspace at all (6, 70 and 12870 equally valid outputs at $K = 2, 3, 4$); any contrast quoted at that angle is a tie-breaking artifact and not a property of the fiber, and the sweep of Sec. S12 stops at $s = 0.654$ for that reason. *(ii)* Nothing here grows with the system, and two separate quantities shrink. The ratio of the two leakages falls, $9.80 \rightarrow 7.23 \rightarrow 6.01$ from $K = 2$ to $K = 4$, and it dies as the perturbation is turned up, from 28.7 at $\delta = 0.0025$ to 1.03 at $\delta = 0.3$. More to the point—because it is the quantity the proposition actually rests on—so does the margin of the certificate over the trivial bound: $1.962/0.856 = 2.29$, $1.962/1.026 = 1.91$, $1.962/1.161 = 1.69$, a fall of 26% across the only three sizes available. Where that fall would reach unity we do not say: a linear extrapolation in K , a log-linear one and the last step alone place it between $K = 6$ and $K = 8$, the decrements are themselves decelerating, and three points do not support a number—which is exactly the objection this paper makes to its own fraction headline in Sec. VII. A statement of non-vacuity needs one fiber and not a magnitude, and the fiber at $K = 4$ is still non-vacuous by 69%; but “the effect grows with K ” is false here. *(iii)* The small-leakage endpoint is realized only by the aligned basis, which

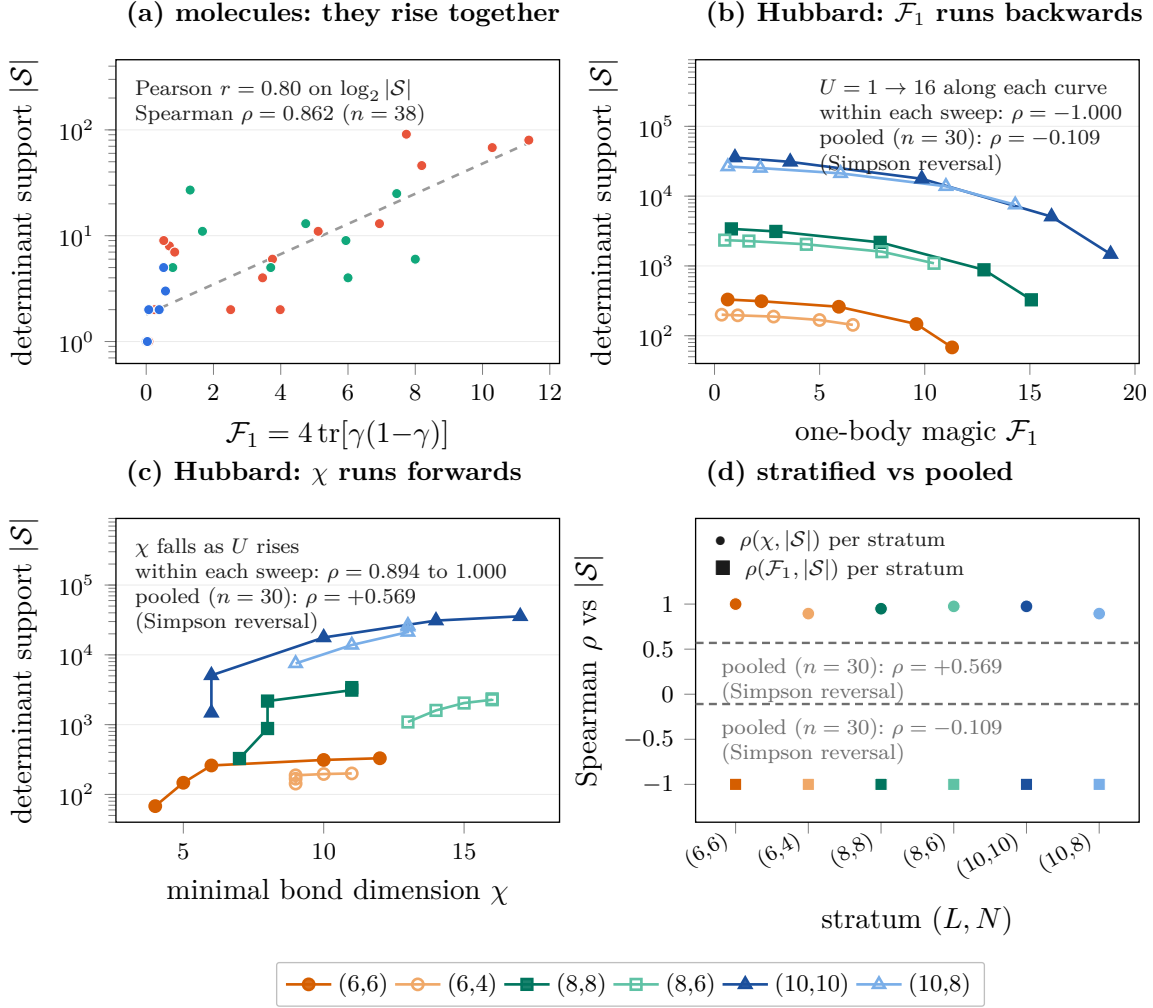
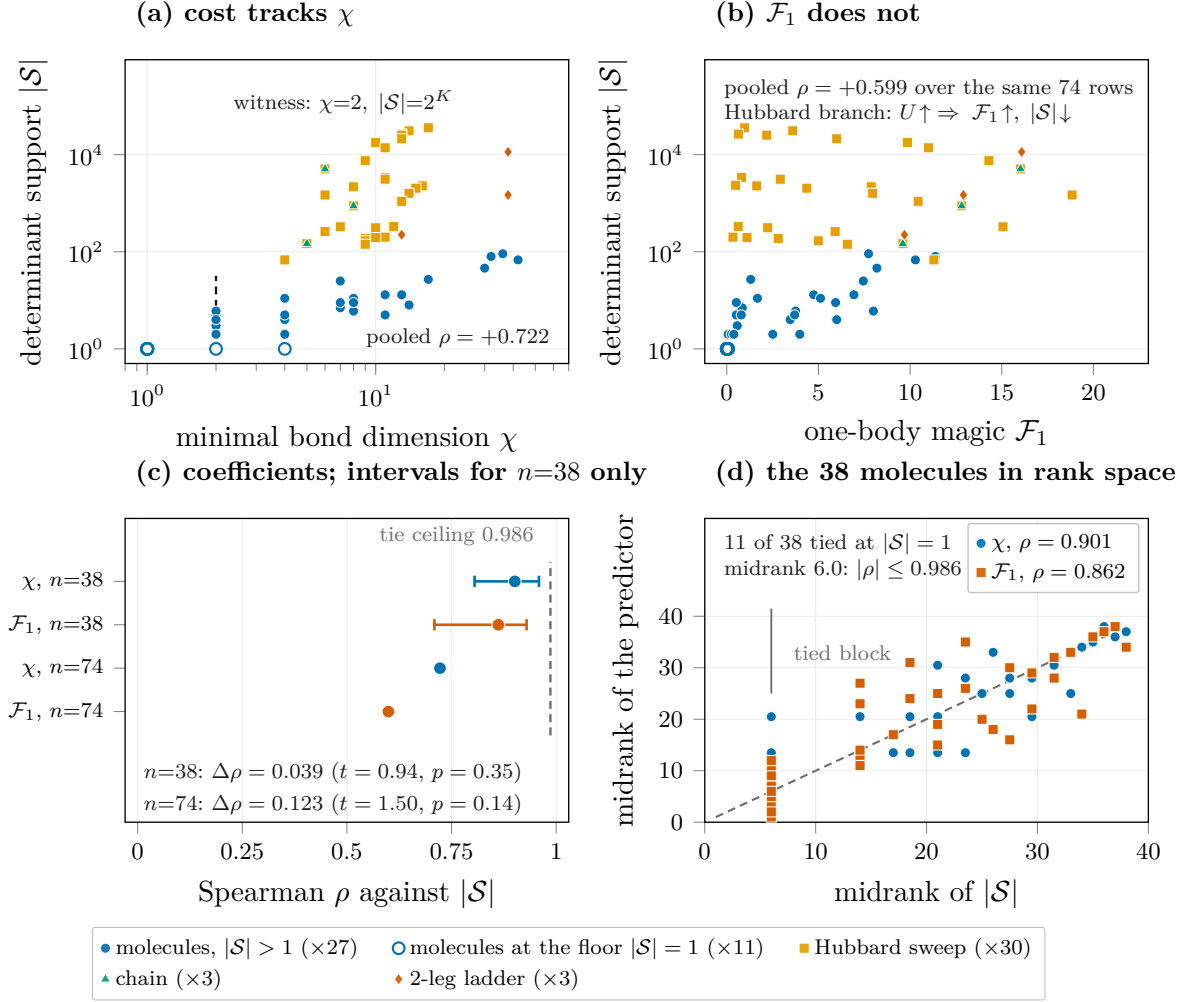


FIG. S2. **Inside every stratum one-body magic $\mathcal{F}_1$ orders the static determinant support $|\mathcal{S}|$ perfectly and with the sign inverted; what fails is the *pooled* coefficient, not $\mathcal{F}_1$.** (a) Nineteen molecular active spaces at equilibrium and at dissociation (38 points): in the static-correlation regime $\mathcal{F}_1$ and $|\mathcal{S}|$ rise together (Pearson $r = 0.80$ on $\log_2 |\mathcal{S}|$; Spearman $\rho = 0.862$). The grey dashed line is the least-squares fit $\log_2 |\mathcal{S}| = 0.474 \mathcal{F}_1 + 0.850$ over those same points, a guide to that correlation and not a model of the cost. Color: covalent bond-breaking (red, 20 points), multireference/open-shell (green, 10) and ionic/weakly correlated (blue, 8), the groups of Table S12. (b) The Hubbard interaction sweep is not one cloud of 30 points but six strata (L, N) of five on-site repulsions $U = 1, 2, 4, 8, 16$; one curve per stratum. Inside every one of the six, raising U raises $\mathcal{F}_1$ and lowers $|\mathcal{S}|$, and does so with a perfect rank reversal: $\rho(\mathcal{F}_1, |\mathcal{S}|) = -1.000$ in all six. Pooling across strata collapses the coefficient to $\rho = -0.109$; that collapse is a Simpson reversal, not an absence of signal. (c) The same six strata against the minimal bond dimension: $\rho(\chi, |\mathcal{S}|)$ runs from $+0.894$ to $+1.000$ within strata, with a mean of 0.948 , and does not change sign between them. Its pooled coefficient, like $\mathcal{F}_1$'s, is a Simpson reversal and is not used: pooling six deterministic five-point sweeps destroys within-stratum structure whichever way the result falls, and it fell in our favour on one of the two. (d) The six within-stratum coefficients against the two pooled ones. Stratified, the $\mathcal{F}_1$ evidence is one-sided and exact: Kendall $\tau = -1.000$ with 0 concordant of 60 within-stratum pairs and no ties, one-sided exact permutation $p = 3.35 \times 10^{-13}$ over the $(5!)^6 = 2.99 \times 10^{12}$ within-stratum relabellings. The pooled reading, $\rho = -0.11$ ($n = 30$, $p \approx 0.56$), is not used. Two limits of this evidence, which the stratified test does not remove. First, the five points of a stratum are a deterministic monotone sweep in U , so the effective number of independent observations is 6, not 30; the exact p counts relabellings, not independent samples. Second, this is a statement about rank direction only: inside a fixed sweep $\mathcal{F}_1$ orders the five costs *perfectly*, so the claim is not that $\mathcal{F}_1$ is uninformative about $|\mathcal{S}|$ but that its sign is not stable—inverted here, positive over the molecular set of panel (a)—which is exactly what an orbital invariant compared against a basis-dependent cost must do.



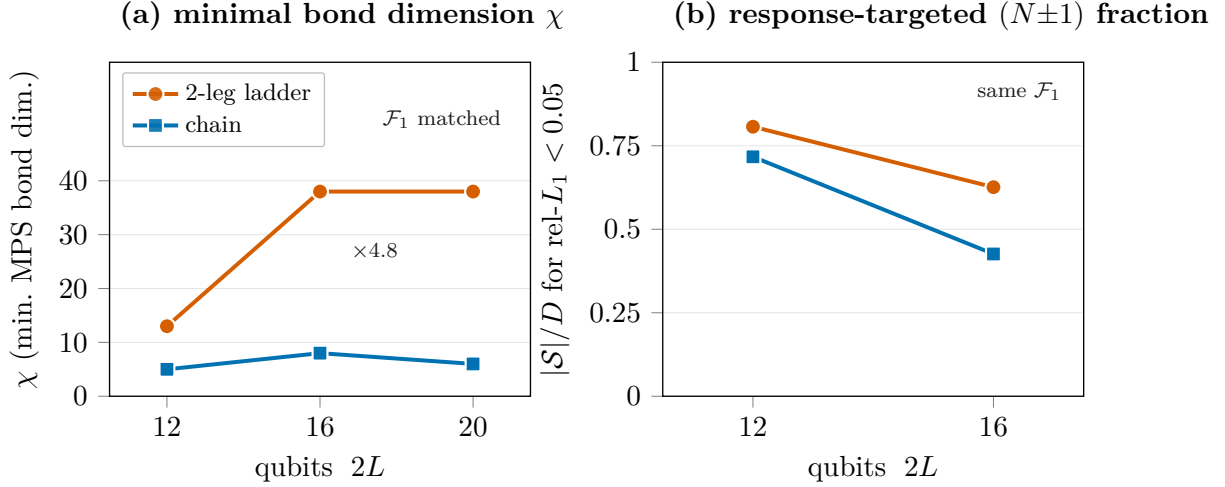


FIG. S4. **The cost is tracked by entanglement, not one-body magic—a controlled chain-versus-ladder contrast.** A two-leg Hubbard ladder versus a chain of the *same* site count (half filling, $U/t = 8$; exact diagonalization). At matched fermionic magic $\mathcal{F}_1$ (open chain vs. ladder: 9.60/9.69, 12.82/12.91, 16.03/16.09 at 12/16/20 qubits— $\mathcal{F}_1$ is a geometry-blind one-body invariant) the ladder’s perpendicular cut has an area-law boundary of two rather than one. **(a)** Its minimal matrix-product-state bond dimension χ is far larger; the chain-to-ladder ratio grows over $12 \rightarrow 20$ qubits ($\times 2.6$, $\times 4.8$, $\times 6.3$), the ladder χ saturating at 38 while the chain χ stays small and does *not* grow (5, 8, 6 at 12/16/20 qubits—non-monotone, with no systematic increase over this range, consistent with the at most logarithmic entanglement growth of the spin sector of the one-dimensional chain)—the reported χ are threshold-truncated Schmidt ranks. **(b)** The response-targeted $(N\pm 1)$ fraction needed for spectral relative- $L_1 < 0.05$ is correspondingly higher for the ladder (these union $(N\pm 1)$ response fractions differ from the single-sector $(N+1)$ $A(\omega)$ fractions of Fig. 5, a different observable). Same one-body magic, higher cost: the determinant support is lower-bounded in the raw count and empirically tracked by the basis-dependent bond dimension χ (Sec. S5.3), not by $\mathcal{F}_1$. A ladder of this width ($\chi \approx 38$) remains classically near-exact for dynamical DMRG; it is a resource contrast and no claim of advantage is attached to it.

is the stabilizer of the state ($|\langle \Psi(0) | \Psi(U_{\text{in}}) \rangle| = 1.000000000000$); at the fiber angle $s = \pi/16$, every generator that actually moves the state gives $\hat{\Lambda}/\eta \in [2.34, 5.61]$. That range is a statement about generators at one angle and not about the angle sweep of Sec. S12, whose rotated endpoint descends to 1.61 as $s \rightarrow 0$ because the rotation itself is being switched off. The rhetorical parallel with the cost witness—where the orbit contains an $O(K)$ -cost point and points of plausibly exponential cost—is therefore lost; Sec. S5.2 states that asymmetry. (iv) The scale is a toy: $2K \leq 8$ orbitals, sector dimension ≤ 3920 , no instance at the $L = 12$ Hubbard scale of Sec. V.

What the exhibit delivers survives all four limits and is measured: on fibers where every invariant is fixed to 10^{-14} and the captured Born weight to 2.2×10^{-16} , the certificate of Theorem 1 separates a non-vacuous value from the trivial bound in every row of Table S4, and a quantity constant on those fibers cannot make that distinction. The a priori predictor asked for at the head of this section does not exist in the invariant class; what stands in its place is the subspace-aware certificate, whose non-vacuous region is mapped, not asserted, in Sec. V.

S5.1. The obstruction is bilateral, and its sign is not stable

The same symmetry closes the cost side, in both directions (Fig. S2). The cost statistic of this section is the *static* determinant support $|\mathcal{S}|_{\varepsilon_w}$: the number of fixed-basis Slater determinants of the *ground state* carrying all but ε_w of its weight. Three tolerances called 0.05 or 0.01 appear in this paper and they are three different objects, so we subscript all of them. ε_w cuts the support: $\varepsilon_w = 10^{-2}$ for the molecular suite (`n19_suite.py`, 99% of the weight) and $\varepsilon_w = 2.5 \times 10^{-3}$ for the lattice families (`cost_vs_entanglement.py`, 99.75%, written there as $1 - \varepsilon^2$ with $\varepsilon = 0.05$). $\varepsilon_\chi = 0.05$ cuts the Schmidt spectrum for χ , by discarded tail $\leq \varepsilon_\chi^2$, in both families. Neither is the third 0.05 of this work, which is a threshold on a relative L_1 spectral error at a stated resolution and defines the *dynamical* support $|\mathcal{S}|_\eta$ of Sec. VII; and neither is the S of Theorem 1, which is whatever the sampler returned. This subsection is about $|\mathcal{S}|_{\varepsilon_w}$ and about nothing else. Where the *raw* support—determinants of nonzero amplitude—is meant we say so, because the two do not obey the same bounds. Unlike $\mathcal{F}_1$, $|\mathcal{S}|$ is basis dependent: a single Slater determinant has $|\mathcal{S}| = 1$ in its own natural orbitals and exponentially many terms in a generic basis. An orbital rotation is a free operation that leaves the occupation spectrum, hence $\mathcal{F}_1$, exactly fixed while changing $|\mathcal{S}|$; a quantity invariant under operations that change $|\mathcal{S}|$ at fixed value cannot determine it.

Both directions are realized and neither is a limiting case. A free-fermion ground state at $U = 0$ —one Slater determinant, $\mathcal{F}_1 = 0$, the minimum—has site-basis support growing exponentially, $|\mathcal{S}|_{\varepsilon_w} = 35, 336, 3496, 37361$ for $L = 4, 6, 8, 10$ on the open chain at the lattice threshold $\varepsilon_w = 2.5 \times 10^{-3}$, the discarded weight being at most ε_w (`free_fermion_support.py`; at $L = 4$ the four smallest determinant weights equal ε_w exactly, and a strict cut gives 36): minimal invariant, divergent cost. Along a Hubbard chain at fixed filling, raising U raises $\mathcal{F}_1$ while lowering $|\mathcal{S}|$ as the ground state localizes: rising invariant, falling cost. Across the nineteen molecules of Table S12 the two rise together (Pearson $r = 0.80$ on $\log_2 |\mathcal{S}|$), both merely tracking static correlation. The relationship is two-sided and its sign is not stable—the signature of comparing an orbital invariant to a basis variant.

We state what this is and what it is not. It is not a refutation of anyone’s claim: Ref. [33] proposes $\mathcal{F}_k$ as a measure of fermionic magic (non-Gaussianity) for matchgate circuits—faithful, Gaussian invariant and (sub)additive—which is what it is, and no one proposed it as a predictor of the determinant support. It is our own no-go, about our own cost, and a corollary consistent with—not a competitor to—the two families of Tarabunga *et al.* [34], who in July 2026, before this work, showed that occupation entropies depend only on the covariance matrix, two members of which (orders 1 and 2) are strong monotones that lower-bound the non-Gaussian gate count, while natural-orbital participation entropies depend on the full state and upper-bound the classical simulation cost in an orthonormal Gaussian basis. Both of their families are Gaussian invariant; the basis dependence at issue here enters only because $|\mathcal{S}|$ is counted in a fixed determinant basis rather than in the state’s own natural orbitals. That is the quantity on the useful side of the division, and the priority is theirs. On the invariant side the same acknowledgement is owed to Leone and Bittel [45], who in July 2026, before this work, proved that the fermionic entropy $M_f = n(1 - \|\Gamma\|_2^2/2n)$ on n modes—an affine function of the squared Frobenius norm of the correlation matrix—is a strong pure-state Gaussian monotone (its convex roof one for mixed states)—on $n = 2n_o$ modes M_f is $\mathcal{F}_1$ itself and Tarabunga *et al.*’s quadratic occupation entropy, whose strong monotonicity the revised version of Ref. [34] proves independently—and that its fermionic purity $\|\Gamma\|_2^2/2n$ has an unbiased estimator using $O(\varepsilon^{-2})$ two-copy measurements. What our construction adds to that division is the constructive instance it does not contain: a family (Theorem 3) on which every order of the invariant is maximal while the cost is $O(K)$, and fibers (Proposition 2) on which the invariants are frozen to 10^{-14} while the certificate of Theorem 1 still separates.

S5.2. Two blindnesses, and why we do not sell them as one symmetry

The two halves of this section are not one statement wearing two hats. On the cost side the invariance is a genuine physical group: the Gaussian orbital rotations $U(n_o)$ are free operations, they act on the whole problem, and along one orbit the cost is $O(K)$ at one point and, at others, plausibly exponential but not established here, at fixed invariant—a symmetry argument in the ordinary sense. On the precision side it is weaker and of a different kind: $\Lambda_S(\eta)$ reads the off-diagonal block $P_S H Q_S$ in the working basis, and an invariant is blind to that block—informational blindness, not a gauge. The orbit there does not sweep an extensive range; as declared above, the small-leakage endpoint is realized only by the stabilizer of the state, and the contrast shrinks with K rather than growing.

What joins the two is therefore not a symmetry at all, and nothing stronger is claimed for it than what it is: the “+” of Eq. (7), one inequality whose two summands are the two halves of this paper, one controlled by the captured weight and one not, each carrying its own obstruction. What makes that join a measurement rather than an analogy is that both summands have been evaluated on the same object, $\mathcal{S}^* = \text{supp}(\phi)$, where the first summand is identically zero while the second still leaves a relative L_1 error of 0.43 at $L = 6$ and 0.35 at $L = 8$ on the periodic rings at $\eta = 0.18t$, and 0.28 on the open chain (Secs. III and IV).

S5.3. What does track the cost: χ , and the statistics the right way up

The computable lower bounds on $|\mathcal{S}|$ are themselves basis dependent. The orbital-Schmidt rank across a bipartition—the tensor-network bond dimension χ [46]—is a rigorous floor on the raw support,¹ and, empirically, a tracker of the measured $|\mathcal{S}|_{\varepsilon_w}$. It is not the winner of a contest: χ is measured rather than predicted, and it has no purchase on the

¹ For any state and any bipartition into (L, R) , $\chi \leq |\mathcal{S}|$ with $|\mathcal{S}|$ the number of determinants of nonzero amplitude: each determinant factorizes across the cut, so the matricization $A_{D_L, D_R} = c_D$ has at most $|\mathcal{S}|$ nonzero entries and rank at most $|\mathcal{S}|$. It is elementary, which is why we record it as an observation, and it is proved for the raw support only: once amplitudes are thresholded the inequality need not survive—in the molecular suite $|\mathcal{S}|$ is cut at 99% of the weight while χ is cut at a discarded tail of $\varepsilon_\chi^2 = 2.5 \times 10^{-3}$, i.e. at 99.75%, so the two are not truncated at the same place—and the measured $|\mathcal{S}|_{\varepsilon_w}$ falls below χ in 7 of the 38 molecular points. It is also one-sided and can be exponentially loose: Theorem 3 has $\chi = 2$ at $|\mathcal{S}|_{\text{pair}} = 2^K$.

leakage at all. On the fibers of Table S4 it is *identical* at the two endpoints—2 and 2 at $K = 3$, 1 and 1 at $K = 4$, at equal raw support—while $\hat{\Lambda}/\eta$ moves by a factor 6–7; in a control at $K = 3$ it jumps from 2 to 16 with $\hat{\Lambda}/\eta = 2.336$ and relative $L_1 = 0.238$ unchanged to three figures; and across a subfamily of 58 points on which χ and the raw support are *exactly* equal, $\hat{\Lambda}/\eta$ still varies by 2.8 and the measured error by 8.3. The only tracker of the leakage is the boundary coupling $\|P_S H Q_S\|$ of Sec. III. What follows is a statement about $|\mathcal{S}|_{\varepsilon_w}$ and about nothing else.

The Hubbard sweep, stratified. The 30 lattice points are not 30 independent observations. They are six (L, N) strata—(6, 6), (6, 4), (8, 8), (8, 6), (10, 10), (10, 8)—each swept over the same five values $U = 1, 2, 4, 8, 16$. Within every stratum the rank correlation of $\mathcal{F}_1$ with $|\mathcal{S}|_{\varepsilon_w}$ is -1.0000 , exactly, in all six; the corresponding $\rho(\chi, |\mathcal{S}|_{\varepsilon_w})$ are 1.000, 0.894, 0.949, 0.975, 0.975, 0.894, mean 0.948. The stratified concordance statistics $(n_C - n_D)/(n_C + n_D)$ are -1.000 for $\mathcal{F}_1$, with 0 concordant pairs out of 60 and no ties, where the statistic is Kendall’s τ , and $+1.000$ for χ , with 50 concordant, 0 discordant and 10 tied, where it is the Goodman–Kruskal γ , which drops the tied pairs; Kendall’s τ_b , which charges them, is below 1. Because each $\mathcal{F}_1$ stratum is a perfect reversal of five points, the exact one-sided stratified permutation p -value is $(1/120)^6 = 3.35 \times 10^{-13}$, against 1.93×10^{-10} for χ , which has ties. Both take a product across strata and therefore assume the six independent. They are not six independent experiments: they are three lengths crossed with two fillings, on one code path and one U grid. The per-stratum coefficients, Fisher’s more conservative combination and the tie structure are in Sec. S14.

Two pooled coefficients are available over these same 30 points: $\rho = 0.57$ for χ , which invites the reading that the bond dimension tracks the cost, and $\rho = -0.11$ for $\mathcal{F}_1$, which invites the reading that it carries no evidence. Neither reading is used here, and for one reason. The 30 points are six deterministic five-point sweeps; the effective n is ≈ 6 , not 30; and pooling strata with different offsets destroys within-stratum structure. Here it destroys it in our favour on one coefficient and against us on the other, which is precisely why neither may be kept. What is used instead is the within-stratum evidence above, which is stronger on both sides and symmetric between them: $\mathcal{F}_1$ anti-orders the support perfectly in all six strata, and χ orders it at a mean ρ of 0.948.

That the evidence is stronger does not make it what we were about to call it. A rank correlation of exactly -1 in all six strata is the strongest evidence a rank statistic can give of a *deterministic monotone* relation between $\mathcal{F}_1$ and the support inside a stratum—with the wrong sign. In this family $\mathcal{F}_1$ is not decoupled from the cost; it is an excellent rank predictor of it whose *sign* reverses from family to family: $+1.000$ in the chain and ladder sweeps below, $+0.86$ across the molecules, -1.000 within every Hubbard stratum. That is the failure mode the no-go of Secs. VIB and S5.1 predicts for an orbital invariant compared against a basis variant, and “unstable in sign”, not “decoupled”, is the phrase the measurement supports. The physical content is one mechanism observed six times—raising U localizes the ground state, so $|\mathcal{S}|_{\varepsilon_w}$ falls and χ falls while $\mathcal{F}_1$ rises—and a p -value of 3.35×10^{-13} is the probability of six perfect reversals under independent random orderings within each stratum—a null that six strata sharing one mechanism do not satisfy—and not evidence that it generalizes beyond the one-dimensional Hubbard family in which it was measured.

The molecular set does not discriminate. The $n = 38$ molecular points are 19 molecules at two geometries—dependent pairs, not 38 draws—and 76% of them lie in some tie in $|\mathcal{S}|_{\varepsilon_w}$, 11 pinned at the floor $|\mathcal{S}| = 1$. The comparison the molecular set is asked to settle is not resolvable by any of three procedures: $\rho = 0.90$ for χ against 0.86 for $\mathcal{F}_1$ differ by 0.039, a molecule-level cluster bootstrap puts a 95% interval of $[-0.070, +0.189]$ on that difference, and Williams’s and Steiger’s tests for dependent correlations give $p = 0.35$. The pooled pair over all 74 states, 0.72 against 0.60, is likewise unresolved. The molecular set therefore does not separate the two candidate resources (Sec. S14), and the whole weight of the separation rests on the stratified Hubbard sweep above.

The two families where $\mathcal{F}_1$ wins, written out. In the matched chain and two-leg ladder families of Fig. S4 ($n = 3$ each), $\rho(\mathcal{F}_1, |\mathcal{S}|) = +1.000$ exactly, against $\rho(\chi, |\mathcal{S}|) = 0.50$ and 0.87. We report it and then dismantle it. The parameter swept there is L , not U : for the open chains $\mathcal{F}_1/L = 1.6008, 1.6020, 1.6031$ at $L = 6, 8, 10$, that is $\mathcal{F}_1 = 1.60 L$ to three figures, so $\rho = +1$ restates “both quantities grow with system size” and nothing else, and conditioning on L leaves zero degrees of freedom. The three chain rows are moreover literal duplicates of three of the 30 Hubbard rows—the $U = 8$, half-filled entries at $L = 6, 8, 10$, identical to the last digit in $\mathcal{F}_1$, $|\mathcal{S}|$ and χ —so they are not independent evidence. And with $n = 3$ the smallest attainable permutation p -value is $1/6$: less evidence than three coin flips. What the chain/ladder pair does establish is the controlled contrast it was built for, which needs no correlation coefficient to read: at $\mathcal{F}_1$ equal to within 1%, the ladder’s support exceeds the chain’s by up to $\sim 2.2\times$, while χ moves from 5 to 13 at $L = 6$ and from 8 to 38 at $L = 8$.

What stands, and where the tracker fails too. Figure S3 consolidates the picture over all 74 exact ground states: the determinant support is bounded below by the bond dimension in the raw count, and the bond dimension is the better empirical tracker of the thresholded one. It is not a monotone tracker, and the exception is in the same file as the rule. Within every Hubbard stratum χ orders the support at $\rho = 0.89$ –1.00. Along the chain family, where the swept parameter is L and not U , it does not: $\chi = 5, 8, 6$ at $L = 6, 8, 10$ while $|\mathcal{S}|_{\varepsilon_w} = 147, 880, 5084$, so χ *falls* across the last step while the cost grows by 5.8, and $\rho(\chi, |\mathcal{S}|_{\varepsilon_w})$ there is 0.50; in the ladder family χ is flat at 38 over that same step while the cost grows by 7.7. That non-monotonicity of the chain χ is stated here explicitly, and it is a limit on χ itself. What $\mathcal{F}_1$ does instead is fail in two *additional* and visibly distinct ways—the Hubbard

branch runs the wrong way at a stratified $\tau = -1.000$, and the chain/ladder points sit at identical $\mathcal{F}_1$ with $|\mathcal{S}|_{\varepsilon_w}$ differing by up to $\sim 2.2\times$. Neither statement supplies the predictor this section set out to find, χ being itself basis dependent, measured rather than predicted, and blind to the leakage. What $\mathcal{F}_1$ is remains intact and useful: a faithful, efficiently measurable detector of static (multireference) correlation, generalizing to fermionic non-Gaussianity, and to the nineteen molecules of Table S12, the qubit non-stabilizerness peak of molecular bonding [47] and its reported proportionality to Hartree–Fock weight, itself reported to break down beyond the Coulson–Fischer point [48]; that a single-particle invariant can be maximal while bond entanglement stays $O(1)$ is the fermionic counterpart of the separation between magic and entanglement reported for qubit matrix-product states [49]. The cost is a property of the sampler’s frame, read after the fact by χ and priced in Sec. VII; the error is a property of the subspace’s boundary, read after the fact by $\Lambda_S(\eta)$ and mapped in Sec. V. Per-species $|\mathcal{S}|_{\varepsilon_w}$, D and χ are in Sec. S15; the full stratified analysis—the six per-stratum coefficients, the exact one-sided permutation values, the tie structure, the cluster bootstrap and the dependent-correlation tests—is in Sec. S14.

S6. THE THREE PRICES: THE RESOLUTION GRID AND FINITE SHOTS

The support exponent and its fit range. Restricting the fit of Eq. (20) to $L = 4\text{--}12$, the five sizes Table S5 covers, gives 1.082 at $R^2 = 0.998$; we quote the fit range with the exponent throughout, because it matters at the second decimal. The same caution applies to the object fitted: 1.082 is the fit to the published points, while the 1.090 in the bottom block of Table S5 is the fit to that table’s own coarser replication grid at the same resolution. They are two fits to two datasets and the difference is theirs, not a rounding.

The two exponents are one exponent. It would be convenient to report that the support exponent, unlike the fraction exponent, is resolution-independent. It is not, and the arithmetic is immediate: the fraction *is* the support divided by the sector dimension—exactly, to the last digit, at all six sizes—and the sector dimension does not depend on η . The two exponents therefore differ by a constant, $b_{|\mathcal{S}|}(\eta) - b_{\text{frac}}(\eta) = b_{\text{dim}} = 1.291$ on $L = 4\text{--}12$, and they move together: across the decade $\eta = 0.05 \rightarrow 0.50 t$ both shift by 0.148 in absolute value (Table S5). There is one resolution-dependent exponent in this analysis, and quoting the fraction and the support separately reports it twice.

What differs is the base it is measured against, and that is not cosmetic. On the fraction exponent 0.148 is three quarters of the quantity itself: $e^{-0.13 L}$ becomes $e^{-0.27 L}$, and the headline changes qualitatively with a choice the model does not make. On the support exponent the same 0.148 carries it from 1.165 to 1.017—a seventh of its own value, 13% measured against the larger end and 15% against the smaller—and “exponential in L with an exponent near one” survives at every resolution in the decade. We therefore report the support, and we report it for two reasons, neither of which is stability: it is the absolute resource a device has to populate, and it is not divided by a sector dimension that grows for a reason—the combinatorics of the Hilbert space—that has nothing to do with the method. The fraction is reported as the derived, resolution-dependent quantity that it is, with its full grid attached.

The headline these numbers offer—that the fraction needed for a fixed spectral accuracy falls from 0.92 at $L = 4$ to 0.08 at $L = 14$, as $e^{-0.25 L}$ —is tunable, and no headline of that form is claimed here. Over the decade of η the fraction exponent moves by a factor 2.2, from -0.1266 at $\eta = 0.05 t$ to -0.2743 at $\eta = 0.50 t$. At $\eta = 0.50 t$ the fraction at $L = 12$ is already 0.10, a quarter above the $L = 14$ value at the resolution used throughout this work and reached two sizes earlier: any headline of this form is reachable by choosing the broadening.

Nor is the fraction exponential even at fixed η . The local logarithmic slopes between consecutive published points are $-0.056, -0.194, -0.214, -0.379, -0.377$ —a factor 6.8 across five intervals, steepening monotonically over the first four. A single-exponential fit over $L = 4\text{--}14$, with $R^2 = 0.94$, is a straight line drawn through a curve of increasing slope, and the fitted exponent depends on where the fit stops: $L = 4\text{--}10$ gives -0.159 , $L = 4\text{--}12$ gives -0.209 , and the -0.2477 of the fit over $L = 4\text{--}14$ requires reaching $L = 14$.

The mechanism that makes the fraction move with η is available without derivation, and Sec. IV states it as Conjecture 1 rather than as a theorem: the polynomial-approximation route to a geometric rate in the Krylov order places the kernel poles at $\omega \pm i\eta$, so any such rate approaches 1 as $\eta \rightarrow 0^+$ and any prefactor diverges (Sec. S10.3). At finer resolution more Krylov order—hence more of the sector—is needed for the same L_1 error. We use that as a mechanism and not as a bound: no $C(\eta)$ or $\rho(\eta)$ is exhibited anywhere in this work, and none is needed to read Table S5, which is a measurement.

One further constraint has to be declared rather than inherited. The reference against which the threshold is applied is itself a finite-depth Lanczos computation, recomputed at the same depth as the subspaces so that the comparison is internally consistent ($n_\ell = 150$; `scaling_lanczos_mf.py`). Two consequences follow, and both belong in the open. First, the published relative L_1 is a distance between two depth-150 objects, not between the reconstruction and the exact A . Second, that reference drifts with depth, and the drift is resolution-dependent: between $n_\ell = 100$ and $n_\ell = 200$ at $L = 12$ the relative L_1 change of the reference *alone* is 11.1% at $\eta = 0.05 t$, 3.0% at $\eta = 0.10 t$, 0.93% at $\eta = 0.15 t$ and 0.05% at $\eta = 0.25 t$. At the depth used here the protocol is self-consistent only for $\eta \gtrsim 0.15 t$ —which is

TABLE S5. **Sector fraction required for relative $L_1 < 0.05$, versus system size and resolution.** Same ground state, same time-evolution ranking (protocol $K = 18$, $dt = 0.5$, 16 substeps, 19 snapshots; `scaling_lanczos_mf.py`), same threshold, same depth $n_\ell = 150$; only η changes. The scan over subspace size in this replication is coarser than the bisection that produced the published points, and returns the smallest scanned fraction meeting the threshold, so the $\eta = 0.15t$ column is an upper bound on them: against the published 0.9167, 0.8200, 0.5561, 0.3625, 0.1700 it reads 0.917, 0.820, 0.560, 0.380, 0.180, agreeing at the two smallest sizes and standing 0.7%, 4.8% and 5.9% high at the three largest. The same granularity is why the fraction exponent of this column, -0.2012 , is not identical to the -0.2093 obtained by fitting the published points themselves over $L = 4$ –12. Bottom block: least-squares exponents. *Both* fit windows are given for the support, because the choice moves the third digit and this work criticizes exactly that elsewhere in this section. The fraction exponent moves by a factor 2.2 over the decade, from -0.1266 to -0.2743 , while the support exponent moves from 1.165 to 1.017; in absolute value both move by 0.148, which is the whole content of the comparison.

L	η/t							
	0.05	0.075	0.10	0.15	0.18	0.25	0.35	0.50
4	0.917	0.917	0.917	0.917	0.917	0.917	0.917	0.875
6	0.900	0.890	0.870	0.820	0.800	0.730	0.680	0.630
8	0.740	0.680	0.640	0.560	0.540	0.480	0.420	0.380
10	0.520	0.480	0.440	0.380	0.340	0.260	0.220	0.200
12	0.340	0.280	0.240	0.180	0.160	0.140	0.120	0.100
fraction exponent, $L=4$ –12	-0.1266	—	-0.1681	-0.2012	—	-0.2395	—	-0.2743
R^2	0.904	—	0.918	0.929	—	0.970	—	0.982
support exponent, $L=4$ –12	1.165	—	1.123	1.090	—	1.052	—	1.017
R^2	0.999	—	0.999	0.998	—	0.999	—	0.999
support exponent, $L=6$ –12	1.136	—	1.088	1.053	—	1.022	—	0.992

exactly where the published column sits, at the margin. Fig. 8, a different channel drawn at $n_\ell = 200$, sits *below* that margin, at $\eta = 0.10t$, which is why its caption quotes its lineshape with its depth and not as exact.

The fraction is therefore reported as a declared window rather than as a number: only for $\eta \geq 0.15t$ at $n_\ell = 150$, with the full $\text{frac}(L, \eta)$ grid of Table S5 published beside it.

The shot budget, in full. Populating a subspace by measurement is a coupon-collector problem on the Born distribution of the time-evolved seed, so the budget exceeds the support it buys. Table IV reports that budget *measured*, by multinomial sampling of the declared protocol, at the five sizes where the measurement is affordable; only $L = 14$ carries no measurement. Two independent runs contribute, and we name them as two rather than merging them: run A draws eight seeds at $L = 4, 6$ and 8, run B draws four seeds at $L = 4, 6, 8$ and 12 and six at $L = 10$, with different seeds and a different budget grid. Where they overlap they agree—2.43 against 2.44×10^4 at $L = 6$ and 2.70 against 2.61×10^5 at $L = 8$, that is to 0.2% and 3.2%—which is why the two larger rows are reported as measurements and not as extrapolations of the smaller ones. Every budget is for *one* orbital seed and *one* branch; the momentum-resolved observable consumes the full set of orbital seeds in both $(N \pm 1)$ sectors, sixteen channels at $L = 8$.

The published curve and finite shots, in full. Precision about the label matters more here than anything else. The ranking that produced the published fractions is `argsort` applied to the Born distribution of the time-evolved state accumulated over the 19 snapshots of the protocol (`scaling_lanczos_mf.py`, `stage_rank`)—that is, to the distribution the device would itself measure, in the limit of infinitely many shots. It is *not* a clairvoyant ordering by exact eigenvector amplitudes; the repository contains such an oracle elsewhere (`amortized_recovery.py`) and labels it as one. But it is also not a finite-shot reconstruction, and to describe it as *sampled* without a shot count is to mislabel it. The published curve is the $T \rightarrow \infty$ limit of the declared protocol, and it is labelled that way from here on.

The published curve is therefore a strict lower bound on the finite-shot cost. At $L = 4$ the two agree because the sampler exhausts the sector; at $L = 6$ and $L = 8$ the finite-shot fraction exceeds the published one by 0.037 and 0.071. That is a direction across three sizes, one of which is degenerate, and we state it as a direction and do not fit an exponent to it—which is the same objection this section raises against the fraction headline, applied to our own finite-shot curve. At the published subspace itself, $L = 8$ and $|\mathcal{S}|_\eta = 2180$, the infinite-shot ordering gives relative $L_1 = 0.0490$, just inside the threshold, against 0.1006 ± 0.0036 for finite shots at $T = 1.47 \times 10^5$: a factor 2.05 ± 0.07 , and the reason the finite-shot fraction has to rise to 0.63.

The finding that runs the other way is the more useful one, and it is what this subsection closes on: the penalty is budgetary, not structural, and it extinguishes with a budget that is affordable at this size. On the same pair as the last row of Table S6— $L = 8$, $|\mathcal{S}|_\eta = 2331$ —raising the budget from $T = 2.04 \times 10^5$ to $T = 2.60 \times 10^6$, a factor 12.7, takes the ratio of the finite-shot error to the infinite-shot-ordering error from 1.96 ± 0.12 to 1.06 ± 0.01 , through 1.51, 1.18 and 1.09 at the intermediate budgets. No sampling penalty in this family can be quoted without its budget, because the penalty is a statement about T and not about the method. We therefore report finite-shot fractions only

TABLE S6. **Finite shots, with the budget attached.** Multinomial sampling of the declared protocol, eight independent seeds, $\eta = 0.15t$; $\pm$ is the sample standard deviation across seeds, not the standard error of the mean. Top block: the sector fraction the finite-shot reconstruction needs to reach relative $L_1 < 0.05$, and the budget at which it reaches it. Bottom block: at *matched* subspace size, the error of the infinite-shot ordering against the error of finite shots at the stated budget, and the ratio between them.

	$L = 4$	$L = 6$	$L = 8$
fraction, $T \rightarrow \infty$ (published)	0.9167	0.8200	0.5561
fraction, finite shots	0.9167 ± 0.0000	0.8571 ± 0.0135	0.6275 ± 0.0031
budget T at which it is reached	$(7.8 \pm 2.0) \times 10^2$	$(2.43 \pm 0.34) \times 10^4$	$(2.70 \pm 0.13) \times 10^5$
matched $ \mathcal{S} $ (size the sampler returned)	22	247	2331
budget T at that size	7.8×10^2	1.74×10^4	2.04×10^5
relative L_1 , $T \rightarrow \infty$ ordering	1.4×10^{-14}	0.0469	0.0362
relative L_1 , finite shots	1.4×10^{-14}	0.0702 ± 0.0115	0.0709 ± 0.0042
ratio	1.00	1.50 ± 0.25	1.96 ± 0.12

with T and the seed count attached, and we report the infinite-shot ordering for what it is—the floor that any budget approaches from above, at the price Table IV now states.

S7. PHYSICAL VALIDATION: SPIN SCALING, POLE CENSUS, SUM RULE AND SIZE

The charge scale, in full. $\Delta(L)$ decreases monotonically and stays above the Lieb–Wu thermodynamic value $4.67952t$ at every size, the residual falling from $0.679t$ at $L = 6$ to $0.289t$ at $L = 12$ —that is, to 6.2% of the thermodynamic value (Table V). $\Delta(L = 12) = 4.96876t$ reproduces the $4.97t$ this work has quoted throughout, and is the same number that appears as the Hubbard-band separation of Fig. 4 and as the low-frequency scale of Fig. 7. We fit no extrapolation exponent: a straight line in $1/L$ through the four points extrapolates to $4.561 \pm 0.044t$, which undershoots Lieb–Wu by $0.12t$ and is therefore evidence that the functional form is not $1/L$, not evidence about the limit. The statement supported is that the charge scale converges to a finite thermodynamic value and is already within 6.2% of it at the largest size reached here.

Spin. The lowest pole of $S^{zz}(q = \pi, \omega)$ gives $L\Delta_s = 2.091, 1.917, 1.996, 1.996$: a $1/L$ closing with a coefficient of about 2 (mean 2.000, range 1.917–2.091). Two fits make that precise and bound it. Fitting $\Delta_s = c_0 + c_1/L$ over these four sizes gives an intercept $c_0 = -0.020 \pm 0.020$ —consistent with zero at one standard error—against $c_0 = 4.561 \pm 0.044$ for the charge gap: one intercept is consistent with zero and the other is not, and that separation, not either intercept on its own, is the result. It has to be read with the control that limits it, which we state here rather than leave to the figure: the pure Heisenberg ring, whose spin gap is known to vanish in the thermodynamic limit, gives a linear-in- $1/L$ intercept of $0.0142 \pm 0.0014t$ —above zero, not consistent with it—so at these sizes a vanishing intercept is not by itself evidence that a gap closes. The statement that survives without a fitted model is the flatness of $L\Delta_s$ below. Fitting instead $L\Delta_s = d_0 + d_1L$ gives $d_0 = 2.09 \pm 0.17$ with $d_1 = -0.010 \pm 0.018$, a slope consistent with zero: over $L = 6$ – 12 the product $L\Delta_s$ is flat to the resolution of four points. What is *not* supported is a third significant figure on the coefficient. The four values span 8% and are not monotone, and we quote $L\Delta_s \approx 2$ with the spread attached, not 2.00; the intercept $d_0 = 2.09$ is the fit’s value at $L = 0$, not the level over $L = 6$ – 12 .

The independent control is the effective spin model. The pure Heisenberg ring at $J = 4t^2/U = 0.5t$ —no charge degree of freedom, no double occupancy, a different Hamiltonian on a different Hilbert space, solved by dense diagonalization on a different code path—gives $\Delta_{\text{ST}}(N = 12) = 0.35585J = 0.17792t$ and reproduces the same $1/L$ law across the same four sizes, with $L\Delta_{\text{ST}} = 2.054, 2.091, 2.116, 2.135$. Two things have to be said about that agreement. First, the two numbers at $L = 12$ differ by 6.5% of the Heisenberg value, equivalently 7.0% of the Hubbard one; we name the denominator because the two differ in the third figure. Second, the residual is not monotone and changes sign—the Hubbard gap is 1.8% *above* the Heisenberg one at $L = 6$ and 8.3%, 5.7% and 6.5% *below* it at $L = 8, 10, 12$ —so it cannot be attributed to a smooth $O(t^4/U^3)$ correction to the effective exchange, which would not change sign. A correction of that order is the expected *scale* of the difference at $U/t = 8$, and it is the right scale; its size dependence is not resolved here and we claim nothing about it. The control confirms the $1/L$ scaling and the magnitude of the coefficient to within 10%; it does not pin the coefficient, and its own $L\Delta_{\text{ST}}$ drifts upward by 0.0134 ± 0.0014 per site over the same range.

What establishes the physics instead is measurement rather than illustration: the four-size scaling of Table V, plotted in Fig. 6, in which the spin scale is a converged pole position at every size and closes as $\approx 2/L$ with an independent effective-Heisenberg control, and the pole census of Table S7, which states what is actually inside the window at each

TABLE S7. **Pole census, across three sizes.** Poles carrying more than 1% of the channel maximum, counted per momentum; each entry is the range over the momenta plotted (7 values of k for $A(k, \omega)$, 6 values of q for the collective channels). The $L = 6$ and $L = 8$ columns are dense diagonalizations, where the count needs no recursion, on the subset of momenta available at those sizes; the $L = 12$ column is the reorthogonalized recursion at $n_\ell = 120$, which reproduces the dense counts pole by pole at the two smaller sizes. “Leading weight” is the fraction carried by the single largest pole at $L = 12$. The addition branch of $A(k, \omega)$ and the charge channel are at $\eta = 0.18t$, the spin channel at $\eta = 0.10t$. The last column is the smallest separation between retained poles in units of the full width at half maximum, which is why the figures show bands: at the tightest momenta the lines are genuinely merged, and the census says how many of them there are.

channel	poles above 1%			dim ratio	leading	$\delta\omega_{\min}/$
	$L=6$	$L=8$	$L=12$		weight	FWHM
$A(k, \omega)$, addition	2–3	2–4	3–7	2440	0.32–0.69	0.12–2.04
$S(q, \omega)$, charge	3–4	3–6	6–9	2440	0.38–0.68	0.24–2.20
$S^{zz}(q, \omega)$, spin	1–2	1–3	1–3	2134	0.80–0.99	1.08–2.37

size instead of naming it.

S7.1. A pole census

At $L = 12$ and the broadenings used here, none of the three momentum-resolved channels is a dense spectrum inside the plotted window; each momentum carries a small number of discrete poles that the broadening merges into a band. Counting poles that carry more than 1% of the channel maximum—Lanczos with full reorthogonalization at $n_\ell = 120$, checked for depth stability at $n_\ell = 60, 120$ and 240 at $q = \pi$ in every channel, and validated pole by pole against dense diagonalization at $L = 6$ and $L = 8$ —gives Table S7; the recursion is the Lanczos one of Refs. [4, 21].

The census is the deposited `pole_structure.json` and is taken on the momentum grid the figures of this section plot, seven values of k and six of q ; a count over a coarser grid is a count of different momenta and is not comparable entry by entry. The census covers the three channels the figures of this section plot; the k -integrated $A(\omega)$, which is a sum over the same branch, and the removal branch are not counted, and no statement is made about them here.

The count does grow with size, and the size of the growth is the point. Between $L = 6$ and $L = 12$ the $(N+1)$ sector grows by a factor 2440, while the largest per-momentum pole count grows from 3 to 7 in the addition branch, from 4 to 9 in the charge channel and from 2 to 3 in the spin channel. A line count that roughly doubles while the Hilbert space grows by three orders of magnitude is a discrete spectrum whose resolvable lines are being broadened together, not a dense one forming. The spin channel is the extreme case: at every momentum a single pole carries between 80% and 99% of the weight, at separations of at least 1.08 full widths, and at $q = \pi$ that single pole is the $0.16635t$ scale of Table V. What Fig. 8 shows at $L = 12$ is one dominant antiferromagnetic mode plus at most two satellites.

This has a consequence for how the reference itself must be described: $A(k, \omega)$ is not computed exactly, as the Lehmann sum would be. With $n_\ell = 150$ and no reorthogonalization in a sector of dimension 731 808, what is computed is a 150-node Gauss quadrature of the spectral measure [4, 21]. It is accurate enough here—and it is accurate *precisely because* the measure it integrates is a handful of poles; a dense spectrum at this dimension would not be reproduced by 150 nodes. It is not, however, converged: the same depth moves the reference by 0.93% at $\eta = 0.15t$ and by 3.0% at the $\eta = 0.10t$ of Fig. 8 when it is taken from $n_\ell = 100$ to $n_\ell = 200$ (Sec. VII). The reference is therefore labelled for what it is: a finite-depth continued fraction, checked against dense diagonalization at $L \leq 8$ and quoted with its depth. The pole positions of Table V are a separate and stronger object—they are computed at full reorthogonalization and are stable across $n_\ell = 60, 120, 240$ —which is why the physical claim of Sec. VIII A rests on them and not on a lineshape. The census counts poles, and the count does not depend on the broadening either; only the picture does.

S7.2. The sum rule, as measured

The analytic identity $\int A^+ d\omega = 1 - \langle \hat{n}_p \rangle = 0.5$ is exact and grid-independent, but it is an identity of the *ground state*: evaluating it costs one ground-state expectation value, $\langle \Psi_0 | \hat{c}_p \hat{c}_p^\dagger | \Psi_0 \rangle$, and never touches the reconstruction. It is therefore not an a posteriori check of A_S , and the caption of Fig. 2 does not present it as one.

Integrating the plotted curves instead, over the 600-point window at $\eta = 0.15t$:

$$\int_{\text{window}} A_{\text{exact}} d\omega = 0.486588, \quad \int_{\text{window}} A_S d\omega = 0.486471. \quad (\text{S1})$$

Both fall 2.7% below the analytic 0.5 because Lorentzians of width η leak outside any finite window: the analytic in-window mass is 0.486589, the tail outside it is 0.013411, and the residue of the trapezoidal quadrature itself is 3×10^{-6} relative—so the deficit is entirely the window and not the reconstruction. The a posteriori statement the reconstruction does support is the comparison of the two integrated curves, which differ by 1.16×10^{-4} absolute, that is 2.4×10^{-4} relative. That is the number we report, and it is consistent with the $\sim 2\%$ window effect that Fig. 4 states.

S7.3. Size: the convention, and what it does not license

The system size is not hidden and is not the weak point. The exact-diagonalization convention for this observable is $L = 12$ –16 at $\eta \approx 0.1$ – $0.2t$: Xu, Lu, Tohyama and Shao [50] compute this spectral function by Lanczos at $L = 12$ and $\eta = 0.2t$ at half filling, and at $L = 16$ at quarter filling. Their model is the extended Peierls–Hubbard chain, and their $L = 16$ run is a dimerized lattice away from half filling; it is their half-filled $L = 12$, and the plain-Hubbard $L = 12, 8, 6$ of their Appendix A, that sit alongside ours. Our $L = 12$ at $\eta = 0.10$ – $0.18t$ is that convention, not an outlier within it.

What the convention does not license is the vocabulary of the thermodynamic limit, and the criterion that says so is in the dynamical-DMRG literature this paper already cites. Jeckelmann’s rule is explicit [51]: “ $\eta_0(N)$ must be larger than the distance $\delta\omega$ between two consecutive excited states.” At $L = 12$ the largest spacing between consecutive excited-state energies in the window is $\approx 0.48t$, a factor 2.7 above the $\eta = 0.18t$ used for $A(k, \omega)$ and 4.8 above the $\eta = 0.10t$ used for the spin channel. The rule is not satisfied here. Satisfying it would demand $\eta > 0.48t$. That is only a tenth of the Mott scale, so the Hubbard bands would survive; but it is $0.96J$ and 2.9 times the $L = 12$ spin gap itself, so the entire low-frequency structure of Fig. 8 would merge into one feature. The number the rule demands is set by the spacing, $0.48t$; we take it from the spacing and not from a rounded-up interval. The correct response is not a larger η : it is to stop using words that require the rule to hold. The dynamical-DMRG standard fixes the product ηL and lets η fall with the size: Benthien and Jeckelmann use $\eta = 5t/L$ in one place, the $L = 100$ spin calculation, and declare $\eta L = 12t$ for the finite-size scaling that carries the zone-boundary charge peak to $L = 120$ [52]; Jeckelmann’s own prescription is $\eta(N) = c/N$ with $c = 6.4$ and $12.8t$, on chains of up to $N = 256$ [51]. This work is at $L = 12$ with $\eta L = 1.2$ – $2.2t$, a factor of two to ten below every constant in that range, and it is the words, not the size, that we do not use. Section IX C records the same restriction: the spin-charge statement rests on pole positions at four sizes.

S8. HARDWARE AND SCOPE: FULL ACCOUNT

Three claims one could make for this method are larger than the quantity that fixes them: the hardware claim is fixed by a subspace size, the chemistry claim by a set of sector dimensions, and the advantage claim by the same bond dimension this paper puts forward as its own cost tracker. This section states each with its fixing quantity attached, and derives the region in which this method has no advantage over a classical one.

The two hardware runs. Both device runs are reported in full and they are the entirety of the quantum hardware used in this work. The first was executed on the IBM Heron processor `ibm_fez` (job `d9s16avpemt73ct6g8g`): the $L = 6$ periodic Hubbard ring at $U/t = 4$, $\eta = 0.15t$, 12 qubits, 3.5×10^5 computational-basis shots— 5×10^4 on each of seven circuits $k = 0, \dots, 6$, of which $k = 0$ is the $t = 0$ reference—pooled into one subspace, under measurement twirling, Pauli twirling and dynamical decoupling, with no readout-error mitigation applied to the counts and no zero-noise extrapolation (Fig. 9(a) is the result; the circuits prepared the computational-basis $(N+1)$ determinant with orbitals $0\text{--}3\uparrow$ and $0\text{--}2\downarrow$ occupied rather than $\tilde{c}_p^\dagger|\Psi_0\rangle$, and their hopping layer applies $R_{xx}R_{yy}$ rotations without Jordan–Wigner strings, so they are not the primitive of Fig. 1; because the retained subspace is the whole sector, neither affects the reported A_{hw}). No configuration recovery was applied. The notebook post-selected on particle number and S_z after reading each bitstring in reversed qubit order (`int(bs[::-1], 2)`), which maps qubit q to $11 - q$: site i to $5 - i$ and spin $\uparrow$ to $\downarrow$. The circuits conserve both quantities and start from four $\uparrow$ and three $\downarrow$ electrons, so under that reading their own output falls in the $S_z = -\frac{1}{2}$ sector and is discarded: in a noiseless statevector simulation of the same seven circuits, 0 of 3.5×10^5 shots are retained, against all 3.5×10^5 and the full sector of 300 under the correct reading (`src/hw_bitorder_check.py`, `data/hw_bitorder_check.json`). The $|\mathcal{S}| = 300$ determinants retained from the device therefore all came from device errors that changed the spin balance; the device counts needed to redo the post-selection are not part of the deposit (Sec. IX D). The recorded relative L_1 is 0.0: the reconstructed A_{hw} agrees with the exact A to 7.9×10^{-15} at every frequency, not byte for byte, because the retained subspace spans the whole sector.

That number is exact *by coverage*, and the distinction is not rhetorical: at $|\mathcal{S}| = \dim$, Rayleigh–Ritz on $\text{span}(\mathcal{S})$ is diagonalization of the full sector, $P_{\mathcal{S}} H P_{\mathcal{S}} = H$, and a vanishing error is an identity rather than a measurement of device fidelity. The run establishes only that the steps of the pipeline execute end to end on a processor—transpiled circuits, device counts, post-selection to a particle-number and S_z sector, and the classical Lehmann assembly of Sec. II.

The post-selection itself was wrong: it read every bitstring in reversed bit order, so every determinant it kept came from a device error, and the run is not evidence that the device sampled the intended distribution. It establishes nothing about accuracy, neither at the complete coverage it reached, where agreement is an identity, nor at incomplete coverage, the regime the rest of this paper is about. The runs are therefore described as *executed on* the named processors, not as demonstrating the method on them, and the coverage qualification applies wherever they are mentioned.

The second run, and a comparison the record cannot decide. Stretched N_2 ($R = 2.0$ Å, CAS(10e,12o)/cc-pVDZ, 24 qubits) was executed on `ibm_marrakesh` (job `da125f2ein7c73bcsqs0`, 10^4 shots, CCSD-seeded local unitary cluster-Jastrow (LUCJ) circuit). Averaged over 8 recovery seeds on the cached device counts it reaches 0.59 ± 0.14 mHa of the exact active-space value (best seed 0.40 mHa), while the noiseless statevector simulation of the same circuit stops at 29.5 mHa (Fig. 9(b)). That contrast invites the reading that device noise *helps*, and the record cannot support it, because the mechanism named for it is a statement about a quantity the record does not contain: recovery is said to turn corrupted bitstrings into valid configurations and thereby *enlarge* the variational subspace—a valid-shot economy, not a fidelity gain—so the comparison is decided by the two recovered subspace sizes, and neither is stored. The same gap runs through the controlled sweeps, whose scripts save at most the size of the naively post-selected subspace and not of the recovered one (the spectral script of Fig. S7 saves neither), so the places where an error *below* the noiseless value is reported beyond the seed bands— NH_3 from 2.53 to 1.24 mHa at $\varepsilon = 10\%$, CO from 1.73 to 0.63 mHa at 20% and C_2 from 0.184 to 0.026 mHa at 30% in Fig. S8(a), and the spectral relative L_1 of Fig. S7, from 0.105 at $\varepsilon = 0$ to 0.045 at 13%—rest on the same unrecorded variable, at the price of one line in each script, both of which already build the subspace.

The mechanism itself is reported elsewhere. On the dissociation of N_2 in a large active space, with the same LUCJ ansatz used for the run above, Vaquero-Sabater, Carreras, Broers, Shirakawa, Yunoki and Casanova report that “sampling noise can enhance Hilbert-space exploration by generating additional configurations beyond those supported by the ideal ansatz,” and that “when combined with configuration recovery, this leads to systematically improved energies” [53]: the same molecule, the same ansatz and the same mechanism, and there it is carried by a controlled comparison under a simple noise model that the work introduces. The same work also reports that “recovery alone (starting from randomly generated configurations) can efficiently construct accurate CI spaces” [53]. Our record contains no control of that kind (uniformly random bitstrings passed through the same recovery with the same eight seeds), so it cannot show that the device counts contributed anything to the 0.59 mHa beyond what recovery supplies on its own. What this work cannot claim is its own measurement of the effect, not the effect: our two arms differ in the one quantity neither of them recorded, so they cannot be evidence either way, whatever the literature independently establishes.

What *is* controlled is the comparison in which both arms consume the same corrupted bitstrings at the same rate with the same seeds and differ only in the recovery step, and there the gain from recovery is measured and grows with the noise: at $\varepsilon = 2\%$ eighteen of nineteen molecules reach chemical accuracy with recovery against seventeen without it, at $\varepsilon = 30\%$ recovery roughly halves the error on N_2 ($31.9 \rightarrow 13.2$ mHa, Fig. S8), and on the periodic Hubbard ring it keeps the spectral relative L_1 near 0.05 for $4\% \leq \varepsilon \leq 16\%$ while naive post-selection on the same bitstrings degrades steadily (Fig. S7)—seed-averaged, with ± 1 s.d. bands, at matched noise. That is the noise result of this work.

Scale. These runs are 12 and 24 qubits, and the same class of observable has been executed elsewhere at larger scale—retarded spin Green’s functions of the one-dimensional Fermi–Hubbard model on IBM Heron r2 at up to 30 qubits [54], an ARPES-type $A(k, \omega)$ of a free-fermion 27-site chain at 54 qubits [55]—and, at a comparable 20 sites on trapped ions, dynamical structure factors against scattering data [56].

In those works the device data determine the reported spectrum, as they do in the four-qubit trapped-ion Hubbard-dimer reconstruction of Ref. [57]; in our spectral run they do not: the retained subspace is the whole 300 -dimensional $(N+1)$ sector and every retained determinant came from a device error, so the spectral function it returns is the exact-diagonalization one, the run is not an entry in that class, and it is not offered as a scale result. The run is also at a size where the coverage question is degenerate, and Sec. VII says how degenerate in shots: the 3.5×10^5 spent—seven circuits at 5×10^4 shots each, pooled into one subspace—are 14.4 times the $(2.43 \pm 0.34) \times 10^4$ that Table IV measures for one orbital seed and one branch at $L = 6$, so complete coverage of a 300-dimensional sector is what the budget predicts for a correctly read run, as the noiseless simulation above confirms; the coverage this run recorded came from device errors and does not test that prediction.² The first size at which a device run would test the reconstruction rather than only the execution of the pipeline is $L = 8$: sector 3920, at $(2.70 \pm 0.13) \times 10^5$ shots per orbital seed and branch—a factor 1.3 *below* what was already spent here, so what confined this run to $L = 6$ was not the shot budget. The momentum-resolved observable at that size is a different matter: sixteen channels, 8.3×10^7 shots as measured for Fig. 3 to reach its relative L_1 of 2.16×10^{-3} , over two orders of magnitude above this job. That is not its cost at the 0.05 threshold used above, which was not measured: half that budget already gives 6.16×10^{-3} .

² Table IV replicates the repository protocol, $K = 18$ with 19 snapshots, at $U/t = 8$ rather than the $U/t = 4$ of the device run; this is a consistency statement between two budgets, not an identity.

The region of no advantage, in full. The resource statement of Sec. VI has a consequence for the standing of this method. What tracks the determinant support is χ , the minimal bond dimension across a bipartition in the working basis (Sec. S5.3). But χ is, by definition, the cost parameter of a matrix-product representation of the same state. A cost claim built on a basis-dependent boundary quantity therefore cannot select for the quantum side of the comparison: wherever our own diagnostic says the determinant support is small it says a matrix-product state is small too, and both methods are charged the same number. The thesis of Sec. VI implies its own region of no advantage, and implies it more sharply than an external objection would.

The instance is in the literature and it is not marginal. Ouyang, Chi and Chan reproduce the 7 260 trajectories of the Fermi–Hubbard dynamics experiment of Hartnett *et al.*—120 qubits, up to 90 Trotter steps [58]—by tensor contraction at bond dimension 32, faster and to higher accuracy than the device run [59]: the same pattern in which tensor-network and Pauli-propagation methods have overtaken earlier hardware demonstrations, the first “quantum utility” claim [60] among them [61, 62], with Pauli propagation since extended to real-time two-point correlators [63], and in which unitary cluster-Jastrow circuits are themselves now simulated classically: the energy of any single-layer circuit in polynomial time, reproducing the largest (77-qubit) experiment of Ref. [64] on a laptop [65], and, to additive approximation, the Born-rule probabilities of hardware-sampled bitstrings by an extended matchgate simulator, at a cost exponential only in the magnitudes of the controlled-phase angles [66]—against which the standing worst-case hardness result for that circuit family, its embedding of instantaneous-quantum-polynomial sampling [67], hard in the worst case [68] and, under average-case conjectures, to additive error [69], is not a contradiction but a statement about a different regime. Two architecture-independent results point the same way: polynomial-time classical estimation of expectation values in noisy circuits [70], and classical estimation in noiseless chaotic circuits across geometries [71]. On the ground-state energy the strong classical selectors [72–74] already win on cost [75, 76], even optimally ordered SQD subspaces need a number of configurations that grows exponentially with the size of Heisenberg and Hubbard lattices [77], and the case for a generic exponential advantage in ground-state chemistry was found wanting before that [78].

One framing does not survive. It is tempting to argue that the energy is where classical methods have caught up and to move the target to dynamics; in one dimension that inverts, because dynamical DMRG computed all three of the momentum-resolved channels reported here nineteen years ago [52] (Sec. I, Sec. VIII). Dynamics in one dimension is where classical methods arrived first, and no framing of this work says otherwise. Within the region this work measures—one-dimensional Hubbard rings at $L \leq 14$, the nineteen active spaces, and a largest minimal bond dimension of 42 over the 74 deposited exact ground states, attained on a molecular active space, the largest on the lattice being 38 in the two-leg ladder—there is no computation performed here that a classical method cannot perform, and we make no advantage claim at any size reached. Deriving one’s own no-advantage region is a stronger position than conceding it; the credit does not change the fact, and the fact is the one just stated.

The same derivation says where the boomerang stops, and that is the measurement this subsection closes on. The controlled contrast of Sec. S5.3 is a search for a target on which the two costs come apart: at one-body magic matched to within 1%, the two-leg ladder carries $\chi = 13$ against the chain’s 5 at $L = 6$ and $\chi = 38$ against 8 at $L = 8$, while its determinant support exceeds the chain’s by 1.52, 1.68 and 2.24 $\times$. The quantity to look for is a target whose minimal bond dimension is large in *every* ordering while the sampled support stays compact—a property of geometry and orbital connectivity rather than of dimensionality, so two-dimensional lattices and molecular orbital graphs are where it would be measured, and nothing here measures it. Section V supplies the second coordinate of that search: in the determinant basis the Krylov space of the probe already has coordinate support on every symmetry-allowed determinant at $L = 8$ (3902 of 3920), so zero leakage at full captured weight requires a different working basis; whether a non-vacuous certificate at small sector fraction does is open (Sec. VE).

S9. COUNTEREXAMPLES TO THE WEIGHT-ONLY BOUND, AND THE POOLED SWEEP

The three-level family. Section IIID exhibits a two-level instance forcing the constant of the weight-only bound (8) to its trivial value at $w_S = 1$. With $w_S < 1$ strictly, the weight term alone fails without bound. Take $H = \text{diag}(\begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix}, \delta)$, $\phi = e_0 + \varepsilon e_2$ and $S = \{0\}$, so that $1 - w_S = \varepsilon^2/(1 + \varepsilon^2)$ may be made as small as wished while the weight displaced to $\pm g$ stays $O(1)$. The ratio of the measured error to the weight term then diverges as ε^{-2} at fixed g/η : the error stays of order $\|\phi\|^2$ while the weight term is $2\|\phi\|^2(1 - w_S) = 2\varepsilon^2$, and δ , the energy of the discarded component, does not enter. Any $C > 0$ in the second term caps the ratio.

The pooled sweep behind Fig. S5. Five machine-readable scans, written by two independent implementations, contribute 621 subspaces: 243 from the original sweep, and 162 + 112 + 104 from a re-implementation deposited as `cert_stress.json`, `cert_akw.json` and `cert_teqsci.json`. Fifteen reproduce the target to double-precision round-off (relative $L_1 \leq 10^{-10}$) and define no ratio, leaving 606 live subspaces over five Hamiltonian families — Hubbard chains, XXZ, t -V, Anderson and sparse random matrices — and eight construction rules, of which 156 are finite-shot

realizations at 4000 shots per snapshot and the remainder deterministic weight-ordered or structured selections. On those 606: the weight-only bound fails on 469; among the 363 with $w_S \geq 0.99$ — the regime a sampling experiment aims for — it fails on all 363, the mildest by a factor 2.00, with 50 of them carrying $w_S = 1$ exactly, so that the bound is identically zero and the ratio infinite; below $w_S = 0.99$ it survives on 137 of 243, in every case where the relative error is already of order unity and the bound excludes nothing. Theorem 1 is violated by none of the 606; the largest ratio of error to bound is 0.9999999997, attained on four subspaces with $w_S = 0$ exactly, where $A_S \equiv 0$ and the trivial branch is an identity rather than an inequality. The leakage branch is the smaller of the two on only 61 of the 606: the sweep shows that the bound *holds*, not that it is tight.

Independent re-verification. The 378 rows of the three deposited scans were re-audited by a script sharing no code with the one that produced them. It confirms 378 of 378 for each side of Eq. (7); a largest error-to-bound ratio of 0.9999999997 over the 376 rows that define one; the exclusion lemma (9) on 378 of 378, with 357 rows at $K_S = -1$ and 21 at $w_S = 1$ exactly; and violations of the weight-only bound on 219 of the 219 finite-ratio rows with $w_S \geq 0.99$, the mildest by a factor 2.00, on the same subspace (XXZ, $\Delta = 2$, $\eta = 0.9$, fraction 0.75) that is the mildest violation of the pooled sweep: every row with $w_S \geq 0.99$ lies at least a factor two from compliance.

S10. MOMENT EXACTNESS: STATEMENT, PROOF, SHARPNESS, AND THE SHOT COUNT

We make precise the two things a Krylov-adequate subspace does deliver. Fix the seed $|\phi\rangle = \hat{c}_p^\dagger|0\rangle$ (the electron-addition branch; removal is identical), with weight $\|\phi\|^2 = \langle\phi|\phi\rangle$, and let $A(\omega) = \sum_n |\langle n|\phi\rangle|^2 \mathcal{L}_\eta(\omega - (E_n - E_0))$ be the exact spectral function, where $\{|n\rangle, E_n\}$ diagonalize H in the $(N+1)$ -particle sector and $\mathcal{L}_\eta(x) = \frac{\eta/\pi}{x^2 + \eta^2}$. For a configuration set $\mathcal{S}$ let $P = P_S$ be the orthogonal projector onto $\text{span}(\mathcal{S})$, let $H_S = PHP$ with eigenpairs $\{|\tilde{m}\rangle, \tilde{E}_m\}$, and let $A_S(\omega) = \sum_m |\langle \tilde{m}|\phi\rangle|^2 \mathcal{L}_\eta(\omega - (\tilde{E}_m - E_0))$ be the reconstruction. Write $\mathcal{K}_K = \text{span}\{\phi, H\phi, \dots, H^K\phi\}$ and $w_S = \|P\phi\|^2/\|\phi\|^2$. Moments are taken on the point measures $d\mu = \sum_n |\langle n|\phi\rangle|^2 \delta(\omega - (E_n - E_0))$ and $d\mu_S = \sum_m |\langle \tilde{m}|\phi\rangle|^2 \delta(\omega - (\tilde{E}_m - E_0))$, not on the broadened A , whose moments of order ≥ 2 diverge.

S10.1. What the proof uses

Theorem 2 and its proof are stated in Sec. IV A. The proof uses $P = P^\dagger = P^2$, $H = H^\dagger$ and $\mathcal{K}_K \subseteq \text{span}(\mathcal{S})$, and nothing else. In particular it does not use that $\mathcal{S}$ is a coordinate subspace, that it was obtained by drawing bitstrings, or that any Lanczos basis is well conditioned [8]; it holds verbatim for the orthogonal projector onto a Krylov space, which is the form evaluated in Sec. V. The single asymmetry in Eq. (12)— a runs to K while b runs to $K+1$ —is what produces the odd reach $2K+1$ rather than $2K$. Two structural remarks follow at no cost.

(a) *The node count.* $d\mu_S$ has $|\mathcal{S}|$ atoms, one per Ritz pair, not $K+1$. Equation (11) is therefore a weak quadrature statement for a rule of that size: a Gauss rule of n nodes for $d\mu$ is exact through order $2n-1$ [4], so the $K=1$ row of Table II, with 280 nodes, would reach order 559 if its nodes were the Gauss nodes of $d\mu$. They are not, and it reaches order 3; those 555 orders measure the distance between the two objects. This is the measurable content of the distinction: identifying A_S with an order- $(K+1)$ Gauss rule misdescribes the object and the reach such an object would have; the reach $2K+1$ of Theorem 2 itself is unaffected.

(b) *Unit captured weight is implied, not assumed.* For a coordinate subspace P is diagonal, so $P\phi = \phi \iff \text{supp}(\phi) \subseteq \mathcal{S} \iff w_S = 1$. Since $\mathcal{K}_0 = \text{span}\{\phi\}$, the hypothesis of Theorem 2 at any $K \geq 0$ already implies $w_S = 1$. Every subspace admissible for moment exactness therefore sits at the extreme of the weight budget of Sec. III, where the shot term vanishes identically and the whole residual error is leakage—which is why the two statements are complementary rather than competing.

S10.2. Sharpness, control, and the limits of the validation system

Table II is computed by exact diagonalization of the $(N+1, S_z = +\frac{1}{2})$ sector of the open $L=6$, $U/t=8$ Hubbard chain ($D=300$, $\|\phi\|^2 = 1/2$), with $\mathcal{S} = \bigcup_{a \leq K} \text{supp}(H^a\phi)$. Four measurements support it.

Reach. Through order $2K+1$ the relative discrepancy is $\leq 3.2 \times 10^{-15}$ for $K=0, 1$ and $\leq 7.5 \times 10^{-15}$ for $K=2$. At order $2K+2$ it is 9.5×10^{-4} ($K=0$) and 6.0×10^{-8} ($K=1$), eleven and seven decades above the arithmetic floor. The absolute discrepancies at that order, 3.104583×10^{-2} and 1.340167×10^{-4} , agree to six significant figures across all six implementations listed below and are properties of the construction, not of the arithmetic. The $K=2$ row carries no entry there, for a structural reason and not a numerical one: the ladder has filled the sector, $A_S = A$ identically, and their relative L_1 distance is 0 to machine arithmetic. One run of one of the six routes does report a nonzero

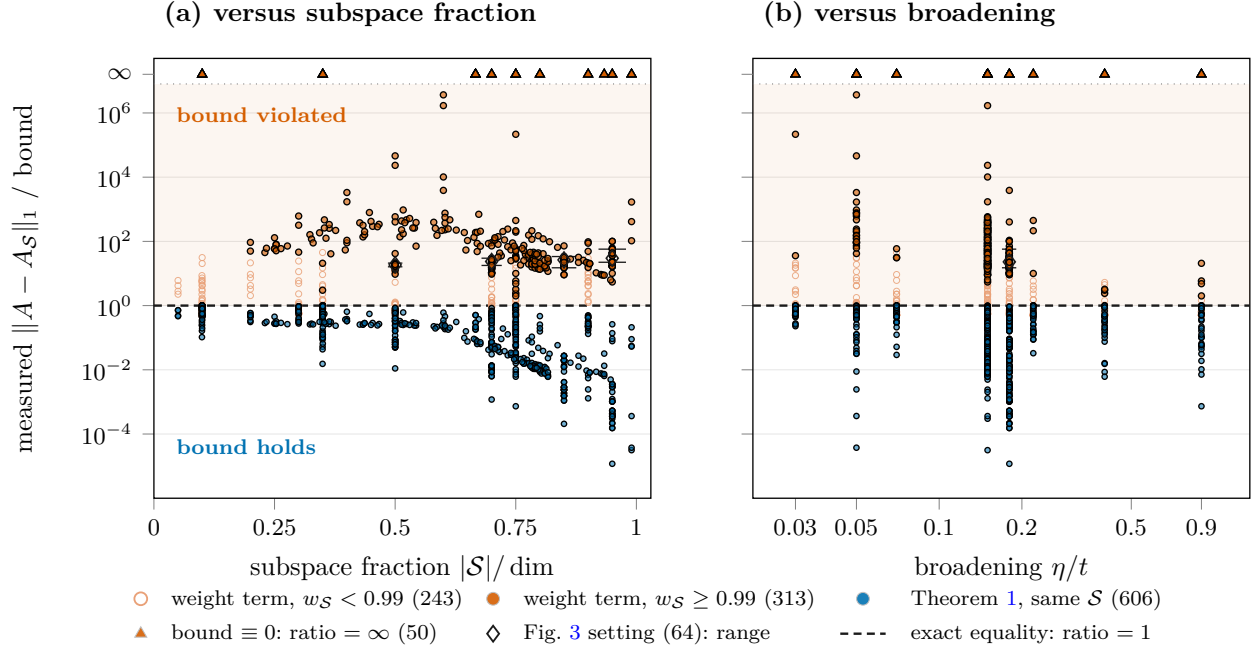


FIG. S5. **A bound indexed on the captured weight alone fails on exactly the subspaces this work relies on; Theorem 1 does not.** For each subspace $\mathcal{S}$ we plot the measured error $\|A - A_S\|_1$ divided by the bound, so every point above the dashed line at 1 is a violation. Orange: the weight term of Eq. (8), $2\|\phi\|^2(1 - w_S)$, split by captured weight (filled, $w_S \geq 0.99$; open, $w_S < 0.99$). Blue: Theorem 1 as stated, $\min\{\|\phi\|^2(1 + w_S), (\|\phi\| + \|\phi_S\|)(\|\phi_Q\| + \Lambda_S(\eta)/\eta)\}$, on the same subspaces. We pooled 621 subspaces from five machine-readable sweeps written by two independent implementations (239 + 4 from the original sweep; 162 + 112 + 104 from an independent re-implementation); 15 of them reproduce the target to within double-precision round-off (relative $L_1 \leq 10^{-10}$) and define no ratio, leaving 606 live subspaces, of which 156 are finite-shot realizations at 4000 shots per snapshot and the remainder are deterministic weight-ordered or structured selections. *The weight term alone is violated on 469 of the 606.* On 50 of them it is identically zero: those subspaces carry the whole weight to double precision ($w_S = 1$ exactly, and $K_S \in \{0, 1, 2\}$, so the remaining term is the undefined constant $C\rho^{-K_S}$); their ratio is infinite and they are drawn on the row marked ∞ , above the dotted rule. Among the 313 finite-ratio points with $w_S \geq 0.99$ —the regime the paper invokes—*every one* is a violation, the mildest by a factor 2.00; with the 50 infinite ones that is 363 of 363, and the finite factors reach 3.66×10^6 . Below $w_S = 0.99$ the bound survives on 137 of 243 points, but only where the relative error is already of order unity, so it excludes nothing there. Diamonds mark the configuration of Fig. 3 ($L = 8$, $U/t = 8$, $\eta = 0.18t$, momentum seeds $c_{k\uparrow}^\dagger|\Psi_0\rangle$, subspace fractions 0.50/0.70/0.85/0.95, both branches, eight momenta): that reconstruction reproduces the deposited `sampld_akw_L8.json` to 3.1×10^{-8} per momentum, and its violation factor is 15.1–57.3, of which 15.1–33.7 at the fraction 0.85 of Fig. 3. Those 64 rows sit at only four abscissae, so they are drawn as a min–median–max whisker at each one rather than as 64 overlapping markers (medians 18.5, 23.6, 26.5, 29.9 at fractions 0.50, 0.70, 0.85, 0.95); each of them is also plotted individually inside the filled orange series, which is where their scatter can be read. Theorem 1 is violated on none of the 606; its ratios span 1.18×10^{-5} to 0.9999999997. That maximum is reached on four subspaces with $w_S = 0$ exactly, where $A_S \equiv 0$ and $\|A - A_S\|_1 = \|\phi\|^2$ equals the trivial cap by algebra—the branch that carries no information—and the leakage branch $\Lambda_S(\eta)/\eta$ is the smaller of the two on only 61 of the 606 rows: the figure shows that Theorem 1 *holds*, not that it is tight. In (b) the η axis is not a sweep within one model: the Hubbard-chain rows exist only at $\eta/t = 0.05, 0.15, 0.18$ and the stress families (XXZ, t -V, Anderson, sparse random) only at 0.03, 0.07, 0.22, 0.40, 0.90, so no trend across the full axis may be read from it.

residue in that row; the same run reports a cyclic-subspace dimension of 301 inside a sector of dimension 300 and emits floating-point overflow while forming the high powers. It is discarded, and the corrected run is the one quoted.

Reproducibility of the residues. The relative discrepancies are quoted, and the absolute ones below order $2K+2$ are not, for a reason worth recording. The exact moments $\mu_j = \langle \phi | (H - E_0)^j | \phi \rangle$ run from 5.0×10^{-1} at $j = 0$ to 1.6×10^5 at $j = 6$, so an absolute residue at the top of the ladder is a roundoff quantity whose size depends on the route taken to the same identity. Across six equivalent routes—Rayleigh–Ritz on the shifted or on the unshifted matrix, crossed with exact moments obtained from the Lehmann sum, from the shifted matrix–vector recursion, or by binomial recombination of unshifted powers—the relative residue below order $2K+2$ stays inside $[1.1, 1.8] \times 10^{-15}$ at $K = 0$ and $[1.0, 3.2] \times 10^{-15}$ at $K = 1$, a spread under a factor 3.3, while the absolute residue at the same orders is not stable to two significant figures. We therefore quote relative residues below $2K+2$, and absolute ones only at $2K+2$, where the discrepancy is physical and route independent.

Negative control. Over 300 uniformly drawn coordinate subspaces of the same size $|\mathcal{S}| = 200$, the zeroth moment is wrong by between 1.86×10^{-2} and 4.18×10^{-1} (median 1.67×10^{-1}), i.e. by 3.7% to 84% of $\|\phi\|^2$ (median 33%), with every one of the 300 draws failing; the Krylov-support subspace of identical size is exact to 4.4×10^{-16} . At $|\mathcal{S}| = 280$ the random draws fail by 4.8×10^{-5} to 2.6×10^{-1} (median 1.12×10^{-2}), again without exception, against 5.0×10^{-16} . The discriminating variable is the identity of the determinants, not their number.

Scale. The Krylov ladder at $L = 6$ occupies 0.667, 0.933 and 1.000 of the sector at $K = 0, 1, 2$, so it exhausts the sector at $K = 2$ and the table has exactly three rows. This system can verify Eq. (11) and locate its exact reach; it cannot exhibit the behavior of $\|A - A_{\mathcal{K}_K}\|_1$ at large K , which is the regime Conjecture 1 concerns, and we make no claim there.

Why the open chain. The ladder terminates when the coordinate subspace contains the cyclic subspace generated by ϕ —not when it fills the sector—and at that point the reconstruction is exact, because $\text{span}(\mathcal{S})$ then carries every eigenvector of nonzero Lehmann weight. On the periodic rings of Sec. VII that happens early: at $L = 6$ the ladder runs $200 \rightarrow 296$ and stalls at 296 of 300 determinants, where $A_{\mathcal{S}} = A$ to 1.0×10^{-14} in relative L_1 and the moments are exact past order $2K+1$ at every $K \geq 1$. The ring therefore offers a single row in which the reach of Eq. (11) is visible at all ($K = 0$: relative discrepancy 4.4×10^{-16} through order 1, 4.6×10^{-3} at order 2); the open chain offers two, and is the system on which the method validation of Sec. II is performed. Within the reach they cover, both agree: the identity is exact, sharp at the next order, and decidable from the subspace and the probe alone.

S10.3. The geometric-convergence claim, and why it is left open

Conjecture 1 is easily taken for a consequence of Eq. (11). It is not one: moment exactness gives $2K+2$ linear identities, and $\|A - A_{\mathcal{K}_K}\|_1$ is not a finite linear functional of them. A proof would require (i) a polynomial approximation theorem for $x \mapsto \mathcal{L}_\eta(\omega - x)$, uniform in ω , on a real set containing $\text{spec}(H)$ and $\text{spec}(H_{\mathcal{K}_K})$, and (ii) a tail estimate controlling the complement. For (i) the natural route is a Bernstein-ellipse bound, in which the resolution enters through the poles of the kernel at $x = \omega \pm i\eta$: the largest admissible ellipse has semi-minor axis below η , so its parameter—and with it any geometric rate—approaches 1 as $\eta \rightarrow 0^+$. That is a mechanism and not a derivation: it exhibits neither $C(\eta)$ nor $\rho(\eta)$, and neither appears anywhere in this work or in the deposit.

The prefactor is moreover pinned from below by the two-level family of Sec. IIID. With $H = \begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix}$ and $\phi = e_0$ one has $\langle \phi | H | \phi \rangle = 0$, so $\mathcal{K}_0 = \text{span}\{\phi\}$ carries a single pole at the origin while A carries two at $\pm g$; by adaptive integration over the whole line at $\eta = 0.15$ (`src/moments_table.py`), $\|A - A_{\mathcal{K}_0}\|_1 = 1.5197, 1.8352, 1.9504, 1.9835, 1.9950, 1.9983, 1.9995$ for $g = 1, 3, 10, 30, 100, 300, 1000$ at $\|\phi\|^2 = 1$, approaching the trivial ceiling $2\|\phi\|^2 = 2$ monotonically. Any $C(\eta)$ valid uniformly on this family therefore satisfies $C(\eta) \geq 2\|\phi\|^2$. A geometric bound with a trivial prefactor is not worthless—it would still decay in K —but it cannot be non-vacuous at small K , and small K is where a shot-limited subspace lives.

Conjecture 1 therefore remains open, and we name its two missing ingredients explicitly so that it can be attacked rather than left unremarked. What this work offers in its place is not a rate in K at all but a bound on $\|A - A_{\mathcal{S}}\|_1$ itself, at the η of the figure and on the subspace actually in hand (Theorem 1), measured subspace by subspace, together with a measurement of where that bound is non-vacuous (Sec. V). That is a weaker statement than a rate and a stronger one than a conjecture, and it is the one the deposit can reproduce.

S10.4. The sample-complexity bound

Proposition 1 is stated in Sec. IV C; this is its proof.

Proof of Proposition 1. Let $\mathcal{N}_\varepsilon = \{x : p(x) \geq \varepsilon\}$. Since $\sum_x p(x) = 1$ and each member contributes at least ε , $|\mathcal{N}_\varepsilon| \leq 1/\varepsilon$. For fixed $x \in \mathcal{N}_\varepsilon$, $\Pr[x \notin \mathcal{S}] = (1 - p(x))^T \leq (1 - \varepsilon)^T \leq e^{-\varepsilon T}$ for independent draws from p ; for $T/(K+1)$ draws from each p_k ,

$$\Pr[x \notin \mathcal{S}] = \prod_{k=0}^K (1 - p_k(x))^{T/(K+1)} \leq \exp\left(-\frac{T}{K+1} \sum_{k=0}^K p_k(x)\right) = e^{-Tp(x)} \leq e^{-\varepsilon T}$$

as well. A union bound gives $\Pr[\mathcal{N}_\varepsilon \not\subseteq \mathcal{S}] \leq \varepsilon^{-1} e^{-\varepsilon T}$, which is at most δ for the stated T . $\square$

The cardinality bound $|\mathcal{N}_\varepsilon| \leq 1/\varepsilon$ is deterministic, and it is what removes the circularity of the form $T = O(\varepsilon^{-1} \log(|\mathcal{S}|/\delta))$, in which the random size of the sampler's own output appears inside a bound on the number of draws that produce it. Where $\varepsilon^{-1} \leq |\mathcal{S}|$, the new logarithm is also the smaller one, so nothing is lost.

The scope of ε needs the same care. It thresholds a Born probability of the time-averaged mixture, and Proposition 1 makes no statement about $\|A - A_S\|_1$. Two independent obstructions stand between the two quantities. First, the weight left uncaptured is bounded only by $|\{x : p(x) < \varepsilon\}| \cdot \varepsilon$, which for a delocalized state is $O(1)$: many small probabilities can sum to most of the mixture. Second, and decisively, a subspace holding the *entire* Born support of the probe— $w_S = 1$ exactly, nothing left to capture at any ε —still has relative L_1 error 0.28 at $L = 6$ on the open chain of Table II, and 0.43 at $L = 6$ and 0.35 at $L = 8$ on the periodic rings, all at $\eta = 0.18t$ and all relative to $\|\phi\|^2$ (Sec. III); it is the $K = 0$ row of Table II. Proposition 1 is a statement about the sampler and about nothing else. The spectral error at fixed η is the subject of Theorem 1, and what the composition of the two actually costs is reported as three separate prices in Sec. VII.

S11. THE LEAKAGE CERTIFICATE: HYPOTHESES, PROTOCOL, AND THE ANATOMY OF THE THRESHOLD

Theorem 1 holds under four hypotheses. Each is a statement about the pipeline that produced the reconstruction rather than about the theorem, so each has to be checked against the pipeline. We check all four here, with numbers, for the sweep of Sec. V, and add the two protocol declarations and the decomposition that section quotes.

(H1) *Common origin.* A and A_P must be referred to the same E_0 . A shift common to both cancels exactly—verified to 1.000000 for shifts of 10^{-3} , 10^{-2} and $10^{-1}t$ —which is how the hypothesis is satisfied here, a single E_0 being used for both objects. If A_P is instead referred to $E_0 + \delta$, then

$$\|A - A_P^{(E_0+\delta)}\|_1 \leq \|A - A_P\|_1 + \|\phi_P\|^2 (4/\pi) \arctan(|\delta|/2\eta),$$

an additive term verified to 2×10^{-5} in all 48 cases tested and *which the certificate does not control*: the bound is guaranteed only once that term is added. A ground state obtained by Rayleigh–Ritz on 60% of the N -electron sector at $L = 8$ gives $\delta = 6.7 \times 10^{-4}t$, which adds 0.0024 to a bound of 1.2036—+0.20%—while multiplying the *measured* error by 16; and the dependence is not even monotone, a shift of $10^{-2}t$ lowering the measured error at $L = 6$, fraction 0.85, $\eta = 0.18t$ from 2.7×10^{-2} to 1.2×10^{-2} . The measured errors quoted beside a bound in Sec. V are therefore those of a pipeline handed the exact E_0 , and they bound the error of a pipeline that estimates it from neither side.

No cheap a posteriori control on δ is available: residual (Weinstein) and Temple bounds [5, 6] computed on ground-state subspaces of 20–80% of the N sector at $L = 6$ give arctan terms of 1.76, 1.64, 1.40 and 0.92, all vacuous against the ceiling of 2. In a shot-based pipeline E_0 is itself estimated on whatever subspace the shots populated, and the certificate has never been tested that way.

(H2) *Normalization and support of the L_1 norm.* The bound is on $\int_{\mathbb{R}} |A - A_P| d\omega$ normalized by $\|\phi\|^2 = \int_{\mathbb{R}} A d\omega$. A relative error normalized instead by $\int_W A$ over a finite window W and evaluated by finite quadrature—the convention of the repository, in which Sec. VII reports—is a different quantity, differing by the product of $\int_W |A - A_P| / \int_{\mathbb{R}} |A - A_P|$ and $\|\phi\|^2 / \int_W A$. Both factors are measured here: the second is 1.0100, 1.0309 and 1.0373 at $\eta = 0.05, 0.15$ and $0.18t$ at $L = 6$ and 1.0110, 1.0278 and 1.0335 at $L = 8$, while the first lies between 1.0040 and 1.0115 across the 204 configurations of this study. Every number in Sec. V is in the normalization of Theorem 1, and the two conventions are nowhere interchanged.

(H3) *Exact reference.* A denotes the exact spectral function of the full sector of the probe. If the reference is itself a Lanczos reconstruction of finite depth n_ℓ , the theorem does *not* bound $\|A^{(n_\ell)} - A_P^{(n_\ell)}\|_1$, and a measured error below the reference’s own drift measures the reference rather than the reconstruction. That drift runs from 2.6×10^{-5} ($L = 12$, $\eta = 0.10t$) to 2.8×10^{-7} ($\eta = 0.18t$), and at $L = 10$ the reorthogonalization gap, 1.71×10^{-5} , is of the same order as the depth drift. Every measured error in this work is therefore labeled with the depth of its reference and with that reference’s own drift, and we quote no measured error anywhere in Sec. V within a factor of 20 of its own reference drift: twelve rows at $L \geq 10$ fall under that rule and are left blank rather than rounded, the weakest surviving cell— $L = 12$, $\eta = 0.10t$ —standing at a factor 47. One gap is declared and not bounded. The reconstructions this paper publishes are produced by bare Lanczos at depth 150, while the certificate is evaluated on the reorthogonalized Krylov space; the difference between the two has been measured at a single point at $L = 8$ —relative L_1 of 1.2454×10^{-3} against 1.2464×10^{-3} —and never bounded, and at $L = 12$ it has never been measured at all.

(H4) *Exact probe.* A and A_P must be built from one probe, and it must be the one whose spectral function is wanted: $\phi = \hat{c}_p^\dagger |\Psi_0\rangle$, or its removal, density or spin counterpart, with the exact N -electron ground state Ψ_0 . That holds by construction in every certificate evaluated in this work: A and A_P are formed from the same ϕ , and on the lattices that ϕ is built from a ground state computed on the full N -electron sector—at $L = 14$ the stored checkpoint of Sec. S4, residual 2.7×10^{-9} —while the Rayleigh–Ritz ground state of the (H1) test above supplies E_0 alone and leaves ϕ exact. A pipeline whose Ψ_0 is itself estimated on a sampled subspace forms another vector ϕ ; Theorem 1 then

TABLE S8. **Angle sweep of the fiber exhibit**, $K = 3$, $\delta = 0.01$, $\eta = 0.10$, $|\mathcal{S}| = 4$. Certificate columns as in Table S4; w_S is identical at the two endpoints, and bold marks a certificate below the trivial bound $1 + w_S$.

s	w_S (both endpoints)	$\hat{\Lambda}/\eta$ al.	$\hat{\Lambda}/\eta$ rot.	cert. al.	cert. rot.
0.131	0.982963	0.328	1.610	0.913	1.983 (trivial)
0.262	0.933013	0.317	3.010	1.131	1.933 (trivial)
0.393	0.853553	0.299	4.124	1.312	1.854 (trivial)
0.524	0.750000	0.277	4.849	1.450	1.750 (trivial)
0.654	0.629410	0.252	5.139	1.543	1.629 (trivial)

bounds $\|A[\tilde{\phi}] - A_P[\tilde{\phi}]\|_1$, and nothing the pipeline computes bounds the distance from $A[\tilde{\phi}]$ to $A[\phi]$. The certificate has never been tested with an approximate probe.

Ranking, ties and support. Two properties of the cut are declared with the protocol. Only the reflection-forbidden determinants (4, 18, 48, 200 at $L = 6$ –12) carry exactly zero accumulated Born weight, and they rank last, so the cut is never filled with configurations the device would never sample in the infinite-shot limit; ties at the cut involve two determinants (four at $L = 6$, at fractions 0.80 and 0.85) and move the bound by at most 6.6×10^{-4} relative. Above the support floor $|\text{supp}(\phi)|/D$ of Sec. IIID, on the other hand, the ranking selects among determinants outside $\text{supp}(\phi)$: at fraction 0.85 that is 30.2% of S at $L = 6$, 42.0% at $L = 8$ and $\approx 39.6\%$ at $L = 10$. Those contribute nothing to w_S , and they do change H_S . The depth uncertainty quoted in Sec. V is the full recomputation of FR^* with $L = 12$ capped at $n_\ell = 300$: it moves by 0.0024, 0.0021, 0.0012 and 0.0000 at $\eta = 0.10, 0.15, 0.18$ and $0.25t$.

Reorthogonalization and the certified object. Full reorthogonalization is used at every point of the sweep, without which the projector is not a projector: the orthogonality defect is at most 4.7×10^{-15} , and $P_K \phi = \phi_S$ to 1.1×10^{-14} . At $L = 12$ the certificate is evaluated at $n_\ell = 300$ –500 on up to 658627 determinants, the dense A_S never being formed—which is why what the bound controls is A_K at the declared depth and not A_S (Sec. VA).

Anatomy of the displacement. The percentages quoted in Sec. VE are medians over 15 of the 18 (η , target) cells at $L = 8$; three cells are censored from above, the certificate never reaching the target inside the grid, and are excluded rather than imputed. Pooling $L = 6$ with $L = 8$ shifts them to 52%, 31%, 11% and 4.1%, which is why the $L = 8$ figures are the ones quoted. One caveat travels with them, and it is why they are reported as a diagnosis and not as a quantity: of the rungs of the inequality chain only the last is computable by a user from (H, S, ϕ, η) , so the decomposition is of an object the user never sees.

S12. THE GEMINAL WITNESS: PROOF AND ANGLE SWEEP

Proof of Theorem 3. Each geminal occupies its own four spin-orbitals, so (18) is a product across the K blocks and γ is block diagonal with spatial natural occupations $(n_{b_i}, n_{a_i}) = (2 \cos^2 \theta, 2 \sin^2 \theta)$; hence $N_u = 8K \cos^2 \theta \sin^2 \theta = 2K$ at $\theta = \pi/4$, and $\mathcal{F}_1 = 4K$ by Eq. (17). The Majorana covariance is likewise block diagonal, and at $\theta = \pi/4$ a single maximally-paired geminal is the two-determinant cat, whose covariance vanishes identically ($\Gamma = 0$): every order saturates its maximum, $\mathcal{F}_k = 2n_o = 4$ per geminal for all k , giving $\mathcal{F}_2 = 4K$. This equality of orders is a knife-edge of maximal pairing; away from it the orders separate ($\theta = \pi/6$ gives $\mathcal{F}_1 = 3$, $\mathcal{F}_2 = 3.75$ per pair, both still large on the same family). In the pairing basis each geminal is a two-configuration superposition, so the product carries 2^K determinants. Finally, as a product of two-orbital factors the state is a matrix-product state whose every bond crosses at most one geminal, so $\chi = 2$ in that ordering, and contraction and Born sampling then cost $O(K)$. The values of $\mathcal{F}_1$, $\mathcal{F}_2$ and $|\mathcal{S}|_{\text{pair}}$ are reproduced by exact diagonalization for $K = 1, \dots, 4$. So is the value $\chi = 2$ derived here, the largest Schmidt rank over all cuts of the pairing ordering, for $K = 1, \dots, 4$ and at every interior angle of the sweep ($\chi = 1$ at $\theta = \pi/2$); the Schmidt rank at the central spin-orbital cut alone is 2, 1, 2, 1 for $K = 1, \dots, 4$, because that cut falls between two geminals when K is even. $\square$

Table S8 continues the fiber exhibit of Table S4 into a continuous sweep of the rotation angle at $K = 3$. It stops at $s = 0.654$: at the maximal-pairing angle $s = \pi/4$ the Born distribution of ϕ is exactly flat over 2^K determinants, so the top- $|\mathcal{S}|$ rule does not select a subspace and no contrast measured there is well defined. Across every row the captured weight is identical at the two endpoints to 2.2×10^{-16} , so the entire difference sits in the leakage term, and the certificate at the rotated endpoint is the trivial bound $1 + w_S$ in every row.

S13. ONE-BODY MAGIC CANNOT BOUND THE DETERMINANT SUPPORT

Proposition 3 (Coefficient-insensitivity obstruction). *Let μ be any functional of a fixed finite set of reduced density matrices, continuous in the state. Then μ cannot usefully upper-bound the raw determinant support (the unthresholded Slater rank): for every $\epsilon > 0$ there are states with $|\mu - \mu(|D_0\rangle)| < \epsilon$ (for $\mathcal{F}_k$, $\mu < \epsilon$) whose raw support is every determinant of the particle-number sector, so any bound $|\mathcal{S}|_{\text{raw}} \leq f(\mu)$ valid on the sector takes the trivial value, the sector dimension, at values of μ arbitrarily close to $\mu(|D_0\rangle)$. The count is lower- but not upper-semicontinuous; the CP tensor rank fails upper semicontinuity in the same way and, unlike the count, is not even lower-semicontinuous: its sets of rank at most r are not closed [79] (that reference names the two semicontinuities the other way round, calling matrix rank upper-semicontinuous). Nor can the invariant $\mathcal{F}_1$ usefully lower-bound it: for N electrons on $2n_o$ modes it implies $|\mathcal{S}|_{\text{raw}} \geq N/(N - \mathcal{F}_1/4)$, which never exceeds $2n_o/N$, and at half filling ($N = n_o$), where that is 2, any increasing f with $|\mathcal{S}|_{\text{raw}} \geq f(\mathcal{F}_1)$ satisfies $f \leq 2$.*

Proof. Rank is a coefficient-insensitive count. Take $|\Psi_\epsilon\rangle = |D_0\rangle + \epsilon \sum_{k=1}^K |D_k\rangle$ with the D_k distinct determinants of the sector of $|D_0\rangle$. As $\epsilon \rightarrow 0$ every reduced density matrix converges to that of $|D_0\rangle$, so any continuous $\mu \rightarrow \mu(|D_0\rangle)$ (for $\mathcal{F}_k$, to 0), while the raw support stays $K + 1$ for every K up to the sector dimension minus one. For the second part, every natural orbital lies in the span of the at most $N|\mathcal{S}|_{\text{raw}}$ orbitals occupied in the support, so $\text{rank } \gamma \leq N|\mathcal{S}|_{\text{raw}}$; with $\text{tr } \gamma = N$ and $\text{tr } \gamma^2 = N - \mathcal{F}_1/4$, the Cauchy–Schwarz inequality $(\text{tr } \gamma)^2 \leq \text{rank } \gamma \text{tr } \gamma^2$ gives the bound, and $\text{rank } \gamma \leq 2n_o$ caps it at $2n_o/N$. At half filling with $n_o \geq 2$ the two-determinant cat $(|1 \cdots 10 \cdots 0\rangle + |0 \cdots 01 \cdots 1\rangle)/\sqrt{2}$, whose determinants differ in $N \geq 2$ orbitals, has $\gamma = \frac{1}{2}\mathbb{1}$, hence $\mathcal{F}_1 = 2n_o$ (maximal) and Slater rank 2, so $f(2n_o) \leq 2$; for $n_o = 1$ every state is a single particle on two modes, of raw support at most 2. $\square$

The obstruction is about the unthresholded rank; the operative, coefficient-sensitive support $|\mathcal{S}|_\epsilon$ of Sec. VI collapses to 1 along the same $\epsilon \rightarrow 0$ path, so continuity alone does not obstruct it. What obstructs $|\mathcal{S}|_\epsilon$ itself is the orbital-rotation symmetry—no invariant functional survives it—together with the two equal-weight witnesses below, whose amplitudes are $\Theta(1)$ and therefore threshold-robust. Two explicit witnesses rule out the natural repairs. The two-determinant cat of Prop. 3 is maximal in $\mathcal{F}_1$ yet of Slater rank 2, so at half filling no $\mathcal{F}_1$ -monotone bounds $|\mathcal{S}|$ from below beyond 2; a paired (BCS) Gaussian state has Gaussian rank 1 yet fractional occupations, so no occupation functional bounds it from above. The computable lower bounds on $|\mathcal{S}|$ that are not capped in this way are the basis-dependent orbital-Schmidt (matrix-flattening) ranks [46]—the tensor-network bond dimension—to which any Gaussian-invariant magic measure is blind. That lower bound is exact:

Proposition 4 (The bond dimension lower-bounds the support). *For any state and any bipartition of the spin-orbitals into (L, R), the Schmidt rank χ across the cut (the minimal matrix-product-state bond dimension there) satisfies*

$$\chi \leq |\mathcal{S}|, \quad (\text{S2})$$

where $|\mathcal{S}|$ is the number of fixed-basis Slater determinants of nonzero amplitude. Hence the minimal bond dimension is a rigorous, basis-dependent lower bound on the determinant support. The bound is one-sided: the geminal witness of Thm. 3 has $\chi = 2$ yet $|\mathcal{S}|_{\text{pair}} = 2^K$, so $|\mathcal{S}|$ can exceed χ exponentially already in the pairing basis, which for that family is a natural-orbital basis, while the orbital-invariant one-body magic $\mathcal{F}_1$ cannot usefully upper-bound $|\mathcal{S}|$ and at half filling cannot lower-bound it beyond 2 (Prop. 3).

Proof. Write $|\Psi\rangle = \sum_{D \in \mathcal{S}} c_D |D\rangle$ over its support. Each determinant factorizes across the cut, $|D\rangle = |D_L\rangle \otimes |D_R\rangle$, so the matricization $A_{D_L, D_R} = c_D$ has at most $|\mathcal{S}|$ nonzero entries; its rank—the Schmidt rank χ —is therefore at most $|\mathcal{S}|$. $\square$

Since a matrix-product-state simulation costs $\text{poly}(\chi)$, Prop. 4 says that *in the sampler’s own basis a tensor network is never less compact than the sampled subspace*: this is why we make no advantage claim in one dimension, where χ is small, and why any advantage must live in the higher-entanglement regime where χ itself—and with it the classical tensor-network cost—grows (the chain-to-ladder jump of Fig. S4). Empirically χ is also the operative correlate of the measured cost: across the 74 exact ground states of Fig. S3 the thresholded support $|\mathcal{S}|_\epsilon$ tracks χ (pooled Spearman $\rho = 0.72$; $\rho = 0.90$ within the $n = 38$ molecular suite, where however one-body magic scores a comparable $\rho = 0.86$ —the two resources are separated not there but by the Hubbard sweep, where $\mathcal{F}_1$ anti-orders the support in all six strata while χ orders it at a mean ρ of 0.948; the pooled coefficients for that sweep, $\rho = 0.57$ and -0.11 , are Simpson reversals of the six strata and are not the evidence here, Sec. S5.3). Two caveats belong with that number. A rank correlation is not a statement of tightness, and per the geminal witness the bound can be exponentially loose. And the proposition is proved for the raw support: once amplitudes are thresholded the inequality need not survive—here $|\mathcal{S}|$ is cut at 99% of the weight while χ is cut at $\epsilon = 0.05$, a tighter tolerance—and $|\mathcal{S}|_\epsilon$ falls below χ in 7 of the 38 molecular points. What stands is that χ is a rigorous floor on the raw support and an empirical correlate of the measured one, whereas $\mathcal{F}_1$ orders the support with a sign that is not stable from family to family (Sec. S5.3).

TABLE S9. **The six Hubbard strata.** Each is a five-point sweep in U at fixed (L, N) . The one-sided exact permutation probability of a perfect reversal within one stratum of five distinct values is $1/120 = 0.0083$; ties in χ raise it. “distinct χ ” counts how many of the five χ values differ, which is what caps $\rho(\chi, |\mathcal{S}|)$ below unity in five of the six.

L	N	$\rho(\mathcal{F}_1, \mathcal{S})$	$\rho(\chi, \mathcal{S})$	χ over $U = 1, 2, 4, 8, 16$	distinct χ
6	6	−1.000	+1.000	12, 10, 6, 5, 4	5
6	4	−1.000	+0.894	11, 10, 9, 9, 9	3
8	8	−1.000	+0.949	11, 11, 8, 8, 7	3
8	6	−1.000	+0.975	16, 16, 15, 14, 13	4
10	10	−1.000	+0.975	17, 14, 10, 6, 6	4
10	8	−1.000	+0.894	13, 13, 13, 11, 9	3

S14. THE RESOURCE STATISTICS, STRATIFIED

Section S5.3 reports the correlation evidence within the design that produced it. This appendix gives the design, the per-stratum coefficients and the three tests, so that the decision not to use the two pooled coefficients can be checked rather than taken.

The lattice design. The 30 Hubbard points are six (L, N) strata swept over the same five on-site repulsions $U = 1, 2, 4, 8, 16$; they are not 30 independent observations, and the effective number of independent sweeps is six. Table S9 gives each stratum. The rank correlation of $\mathcal{F}_1$ with the static support $|\mathcal{S}|_{\varepsilon_w}$ is -1 in all six, exactly; that of χ runs from $+0.894$ to $+1.000$ with mean $+0.948$, its departures from unity being ties in χ and not inversions.

Three tests, and what each assumes. The stratified concordance statistics $(n_C - n_D)/(n_C + n_D)$ are -1.000 for $\mathcal{F}_1$, with 0 concordant and 60 discordant within-stratum pairs and no ties, so that the statistic is Kendall’s τ , and $+1.000$ for χ , with 50 concordant, 0 discordant and 10 tied, where it is the Goodman–Kruskal γ , which drops the tied pairs; Kendall’s τ_b , which charges them, is below 1. Because each $\mathcal{F}_1$ stratum is a perfect reversal of five distinct points, the exact one-sided stratified permutation probability is the product of the six per-stratum chances, $(1/120)^6 = 3.35 \times 10^{-13}$, over a relabelling space of 2.99×10^{12} ; the same test applied to χ , which has ties, gives 1.93×10^{-10} . Fisher’s combination of the six per-stratum probabilities is conservative here and returns 6.56×10^{-8} and 1.14×10^{-5} ($\chi^2 = 57.4$ and 44.7 on 12 degrees of freedom); we quote the permutation values and record Fisher’s beside them. All are *one-sided*—a perfect reversal is one of the 120 orderings, not two—and all take a product across strata, so all assume the six strata independent. They are not six independent experiments: they are three lengths crossed with two fillings, on one code path and one U grid. What the numbers establish is that the direction is reproduced six times out of six, not that it generalizes beyond this family.

Why neither pooled coefficient is used. Pooled over the same 30 points the coefficients are $\rho(\chi, |\mathcal{S}|) = +0.569$ and $\rho(\mathcal{F}_1, |\mathcal{S}|) = -0.109$, the second with a nominal two-sided $p = 0.57$ and the first with $p = 0.0010$. Pooling six deterministic five-point sweeps with different offsets destroys the within-stratum structure that carries the signal; it is a Simpson reversal in both cases. It happens to destroy the signal in our favour on χ and against us on $\mathcal{F}_1$, and that is exactly why neither may be kept: a rule that is applied to one coefficient and not to the other is not a rule.

The molecular set. The $n = 38$ molecular points are 19 molecules at two geometries, i.e. dependent pairs. The intraclass correlation of $\log_2 |\mathcal{S}|$ within a molecule is 0.214, a design effect of 1.214 and a design-corrected effective n of 31.3. Ties are pervasive: 29 of the 38 (76%) lie in some tied group of $|\mathcal{S}|_{\varepsilon_w}$ and 11 are pinned at the floor $|\mathcal{S}| = 1$, which caps the attainable Spearman coefficient at $\rho_{\max} = 0.9863$; against that ceiling χ reaches 91.4% and $\mathcal{F}_1$ 87.4%. Under a molecule-level cluster permutation (10^6 resamples) both coefficients are beyond the resolution floor of the test, $p \leq 1.0 \times 10^{-6}$. The comparison the molecular set is asked to settle is not resolvable, and three procedures agree that it is not: $\rho = 0.9011$ for χ against 0.8618 for $\mathcal{F}_1$ differ by 0.0393; a molecule-level cluster bootstrap (2×10^4 resamples) puts a 95% interval on the difference of $[-0.070, +0.189]$, which contains zero; Williams’s test for dependent correlations gives $t = 0.94$ on 35 degrees of freedom, $p = 0.35$, and Steiger’s $z = 0.94$, $p = 0.35$. Both parametric tests assume bivariate normality and untied ranks while 76% of these points sit in a tie, which is why the bootstrap is quoted first; at the cluster-level effective n of 19, Williams gives $p = 0.53$. Pooled over all 74 rows the pair is 0.722 against 0.599 and is likewise unresolved. Recomputed over the 71 *distinct* states—three chain rows duplicate three Hubbard rows—it is 0.746 against 0.592.

One artefact of the deposit, recorded because it changes a digit. The pooled $\rho(\mathcal{F}_1, |\mathcal{S}|)$ stored in the deposit is 0.5991984 and recomputing it from the per-species values gives 0.5992028. The difference is not a bug in either: it is a tie in $\mathcal{F}_1$ created when two rows that are the same state are stored at different precisions. The quantity is quoted to three figures throughout for that reason.

TABLE S10. **Reproducibility data for the nineteen-molecule suite** (cc-pVDZ, valence bond-breaking active spaces). $R_{\text{eq}}, R_{\text{diss}}$ in Å; for the polyatomics ($\text{BeH}_2, \text{H}_2\text{O}, \text{NH}_3, \text{H}_4, \text{H}_6$) the representative varied bond length is shown and full Cartesians are in `src/n19_suite.py` of the deposit. D : dimension of the N -electron (neutral, ground-state) sector, in which the tabulated $|\mathcal{S}|$ lives; $|\mathcal{S}|$: ground-state determinant support ($\geq 99\%$ FCI weight); χ : minimal MPS bond dimension across the central spin-orbital cut. The $|\mathcal{S}|/D$ of Table S12 is a distinct ratio, taken in the $(N\pm 1)$ spectral sector.

Molecule	CAS(n_e, n_o)	R_{eq}	R_{diss}	D	$ \mathcal{S} ^{\text{eq}}$	$ \mathcal{S} ^{\text{diss}}$	χ^{eq}	χ^{diss}
F_2	(1+1,2)	1.412	3.00	4	1	2	2	2
N_2	(3+3,6)	1.098	2.40	400	7	80	7	32
H_2O	(2+2,4)	0.958	2.30	36	1	13	1	13
HF	(1+1,2)	0.917	2.30	4	1	4	1	4
NH_3	(3+3,6)	1.012	2.33	400	1	68	4	42
CO	(3+3,6)	1.128	2.40	400	8	46	14	30
H_6	(3+3,6)	0.900	2.00	400	9	91	8	36
H_4	(2+2,4)	0.900	2.00	36	2	11	4	8
H_2	(1+1,2)	0.741	2.00	4	1	2	1	2
BeH_2	(2+2,4)	1.334	2.90	36	1	6	1	8
C_2	(4+4,6)	1.243	2.60	225	13	6	11	2
NO	(4+3,6)	1.151	2.50	300	5	25	11	7
O_2	(5+3,6)	1.208	2.60	120	11	4	4	2
CN	(4+3,6)	1.172	2.50	300	27	9	17	7
OH	(3+2,4)	0.970	2.40	24	1	5	1	4
BH	(2+2,4)	1.232	2.60	36	5	3	2	2
BeH	(2+1,3)	1.343	2.70	9	1	2	1	2
LiH	(1+1,2)	1.596	3.20	4	1	2	1	4
LiF	(1+1,2)	1.564	3.20	4	1	1	1	1

S15. REPRODUCIBILITY DATA FOR THE MOLECULAR SUITE

Table S10 lists, for every molecule of Table S12, the equilibrium and dissociation bond lengths, the active space, the N -electron (neutral, ground-state) sector dimension D , the ground-state determinant support $|\mathcal{S}|$ (the number of determinants carrying $\geq 99\%$ of the FCI weight), and the minimal matrix-product-state bond dimension χ at both geometries. These are the quantities that enter the $\mathcal{F}_1-|\mathcal{S}|$ and $\chi-|\mathcal{S}|$ correlations of Sec. VI, pooled over the $n = 38$ equilibrium-and-dissociation points (Spearman $\chi-|\mathcal{S}|$ $\rho = 0.90$, $\mathcal{F}_1-|\mathcal{S}|$ $\rho = 0.86$; Pearson $\mathcal{F}_1-\log_2 |\mathcal{S}|$ $r = 0.80$). The $|\mathcal{S}|/D$ reported in Table S12 is a *distinct* quantity—the fraction of the $(N+1)$ sector used in the spectral *reconstruction*, not the ground-state support tabulated here. Every active-space FCI energy is converged to better than 10^{-12} Ha (the largest residual across all 38 points is 1.1×10^{-13} Ha); the full Cartesian geometries and the active-space definitions (frozen core, n_e, n_o) are given in `src/n19_suite.py` of the deposit, and the absolute ground-state energies are not deposited, only these FCI residuals (`data/n19_suite.json`).

S15.1. The molecular suite, and the size of its sectors

The nineteen-molecule suite reconstructs the orbital-projected addition spectral function $A(\omega)$ at dissociation in a valence bond-breaking active space, against a subspace grown by multinomial sampling of the time-evolved probe, seeded with the probe’s largest-amplitude determinant, which is inserted rather than drawn: 8000 draws at each of 27 evolution times, 2.16×10^5 per molecule, at $\eta = 0.10$ Ha, from a single fixed random seed and with no spread reported across seeds. What decides the reach of that test is not the shot budget but the sector it is poured into, and Table S11 gives it.

Twelve of the nineteen reconstructions are exact by construction rather than by approximation, and the dimensions say why: seven sectors have dimension six or less, and in five more the six to twelve sampled determinants already contain $\mathcal{K}(H, \phi)$. The suite therefore cannot show that the reconstruction is efficient across chemistry: on twelve of the nineteen there is nothing for efficiency to mean. Its energy check is an active-space CASCI energy against itself, and the residual the appendix records is 1.1×10^{-13} Ha across all 38 points—a convergence check on a deterministic diagonalization, worth keeping in the reproducibility appendix at its true magnitude, but not a validation against an

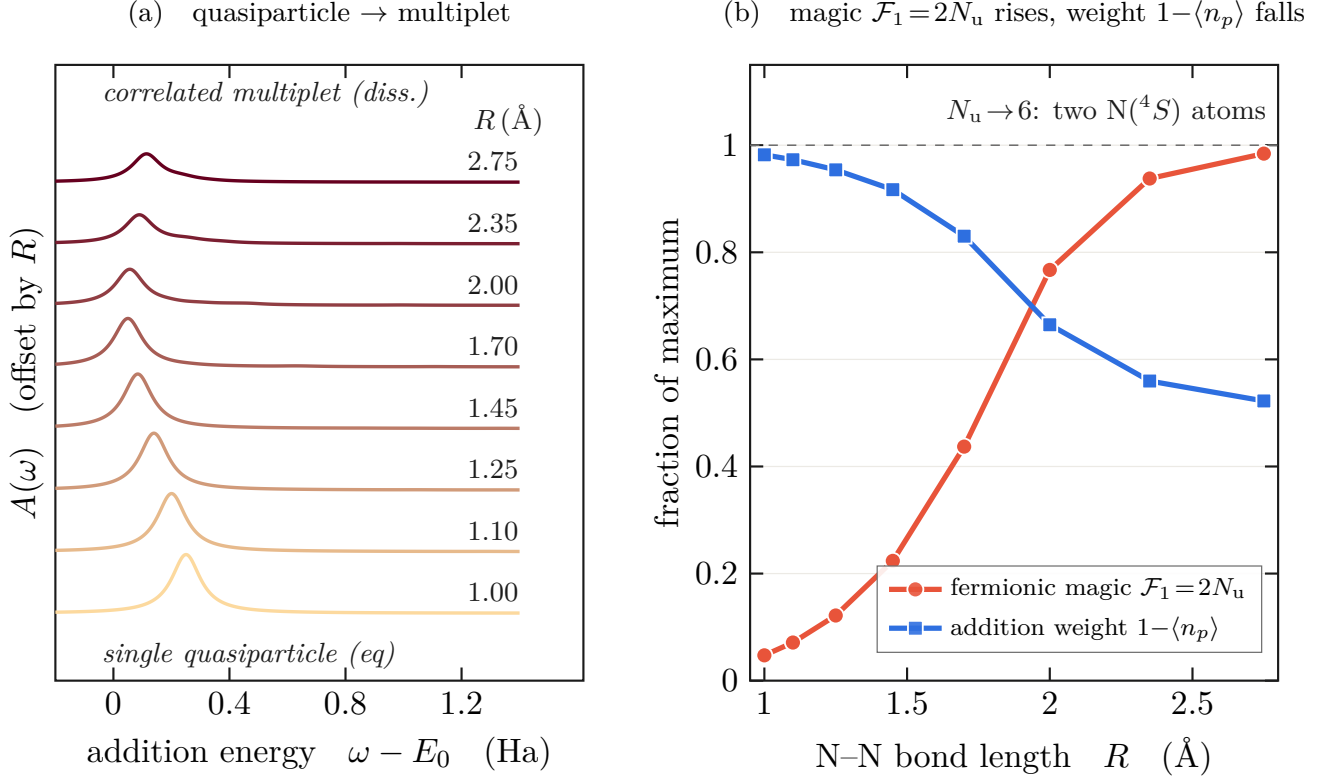


FIG. S6. **Magic and multireference character in N_2 dissociation.** (a) The exact addition spectral function $A(\omega)$ (orbital-projected, CAS(6,6), cc-pVDZ) along the triple-bond stretch, offset vertically, one row per R : the sharp coherent quasiparticle peak near equilibrium ($R = 1.0$ Å, slightly compressed from the $R_{\text{eq}} = 1.098$ Å of Table S10) broadens into a correlated multiplet on dissociation as its spectral weight collapses. (b) Two physically independent quantities along the same coordinate. The fermionic magic obeys the *exact* identity $\mathcal{F}_1 = 2N_u$ [Eq. (17), verified to $< 10^{-13}$], so a single curve carries both: it rises with the orbital-rotation-invariant multireference count—the effective number of unpaired electrons $N_u = \sum_i n_i(2 - n_i)$ from the natural-orbital occupations n_i , approaching its atomic limit of 6 (the two triplet $N(^4S)$ atoms) as the bond fully breaks. The integrated addition weight $\int A^+ d\omega = 1 - \langle \hat{n}_p \rangle$ (from panel a) falls from 0.98 to 0.52 over the same stretch; the two cross as the bond breaks. The active-space CASCI energies are converged to 1.1×10^{-13} Ha, which is a convergence check and not a validation against an independent method (Sec. S15.1).

TABLE S11. **What the nineteen-molecule reconstruction tests.** The suite grouped by whether the reported error is an approximation or an identity. D is the dimension of the $(N+1)$ sector the reconstruction runs in—all nineteen species take the addition branch—and $|\mathcal{S}|$ the absolute subspace size at the end of the protocol. Top row: five sectors of dimension 2, one of 3 and one of 6, where the reported error is identically zero and where in three cases the sampled subspace is the entire two-dimensional sector. Middle row: five further cases whose subspace contains the probe’s Krylov space $\mathcal{K}(H, \phi)$, so what is reported is floating-point round-off (containing the probe’s support alone would not suffice, Sec. IIID). Bottom row: the seven cases in which a truncation costs something. The largest sector in the suite is 300, the dimension of the $L = 6$ Hubbard sector of Sec. IX A, against 731 808 at $L = 12$ on the lattice side of this work.

	count	D	$ \mathcal{S} $	rel- L_1
sector of dimension ≤ 6	7	2–6	1–2	0 exactly
subspace contains $\mathcal{K}(H, \phi)$	5	24–90	6–12	$\leq 1.4 \times 10^{-14}$
genuine truncation	7	20–300	5–149	2.2×10^{-6} – 1.0×10^{-4}

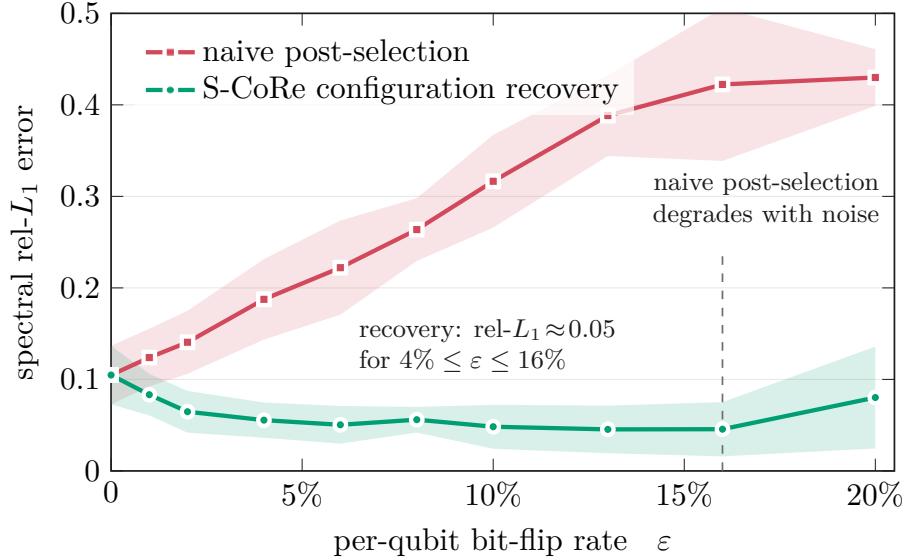


FIG. S7. **Noise robustness (synthetic per-qubit bit-flip channel, $L = 6$).** Spectral relative- L_1 error vs the per-qubit bit-flip rate ε , applied to bitstrings Born-sampled from the exact distribution (this is a controlled classical simulation, not a device run). Naive post-selection degrades steadily, whereas S-CoRe self-consistent configuration recovery, applied to the same corrupted bitstrings, keeps the error near 0.05 for $4\% \leq \varepsilon \leq 16\%$ (periodic Hubbard ring, $U/t = 4$; seed-averaged over 6 seeds at 6000 shots each; bands = ± 1 s.d. across seeds). At $\varepsilon = 0$ the two arms coincide at 0.105, about twice the acceptance threshold 0.05 of the resource scan, and the recovered error at finite ε falls below that noiseless value; that second comparison is not used here (Sec. IX A): it is decided by the size of the recovered subspace, which this sweep does not record.

independent method.

What the suite does test is the bottom row of Table S11, and it is a result. On seven Hamiltonians—O₂, CN, NO, CO, N₂, NH₃ and H₆, in sectors of dimension 20 to 300—the same primitive reproduces the orbital-projected $A(\omega)$ to a relative L_1 between 2.2×10^{-6} and 1.0×10^{-4} from 21% to 50% of the sector. Those seven are independent chemical Hamiltonians rather than one molecule repeated, and their agreement is a consistency check of the primitive outside the Hubbard family. It is not a scaling statement and is not on a common axis with Sec. VII, since the Hamiltonians, the broadening and the error level all differ. Per-species dimensions, subspace sizes and bond dimensions are in Sec. S15, whose column D is the N -electron dimension; the $(N+1)$ dimension of each species is $\binom{n_o}{n_\uparrow+1} \binom{n_o}{n_\downarrow}$ (300 against 400 for N₂).

S16. THE DEQUANTIZATION TEST

The one place where a learned model could remove the quantum sampling from this pipeline is the generation of the dominant determinants, and we tested it on stretched N₂ (cc-pVDZ, $R = 2.5$ Å, CAS(10e,12o), 792 strings per spin, orbitals in the $D_{\infty h}$ symmetry-adapted gauge), where the exact energy is available to score against. A device head is emulated by drawing 1 000 configurations from the exact FCI marginal and corrupting them with a Heron-class noise model—5% depolarizing replacement and asymmetric readout at the median rates of the FakeTorino calibration, 0.0229 for $1 \rightarrow 0$ and 0.0200 for $0 \rightarrow 1$ —after which 24% of the shots carry the wrong particle number. From that one histogram a subspace of $D = 120$ strings per spin is built four ways and scored on the exact energy, over ten seeds paired across the four. A single-pass analogue of the occupation-based configuration recovery of Ref. [64] gives 38.22 ± 2.62 mHa. A cheap classical Epstein–Nesbet perturbative prior [80], which completes the subspace from one application of H to the Hartree–Fock determinant—no additional shots, no additional quantum processing—gives 24.38 ± 4.02 mHa. A generative flow network [81] trained on the recovered histogram fused with that prior gives 22.57 ± 1.60 mHa, and trained on the histogram alone 25.55 ± 1.84 mHa (means and standard deviations over the ten seeds).

The classical step is resolved beyond any reading of the seeds: it lowers the error by 13.85 mHa, a 36% reduction, in all ten seeds, with paired $t_9 = 12.37$; at 500 shots it is larger still, 54% with $t_9 = 36.2$. The generative step is not. On top of the classical prior the fused generator lowers the error by a further 1.80 ± 4.78 mHa, in six of the

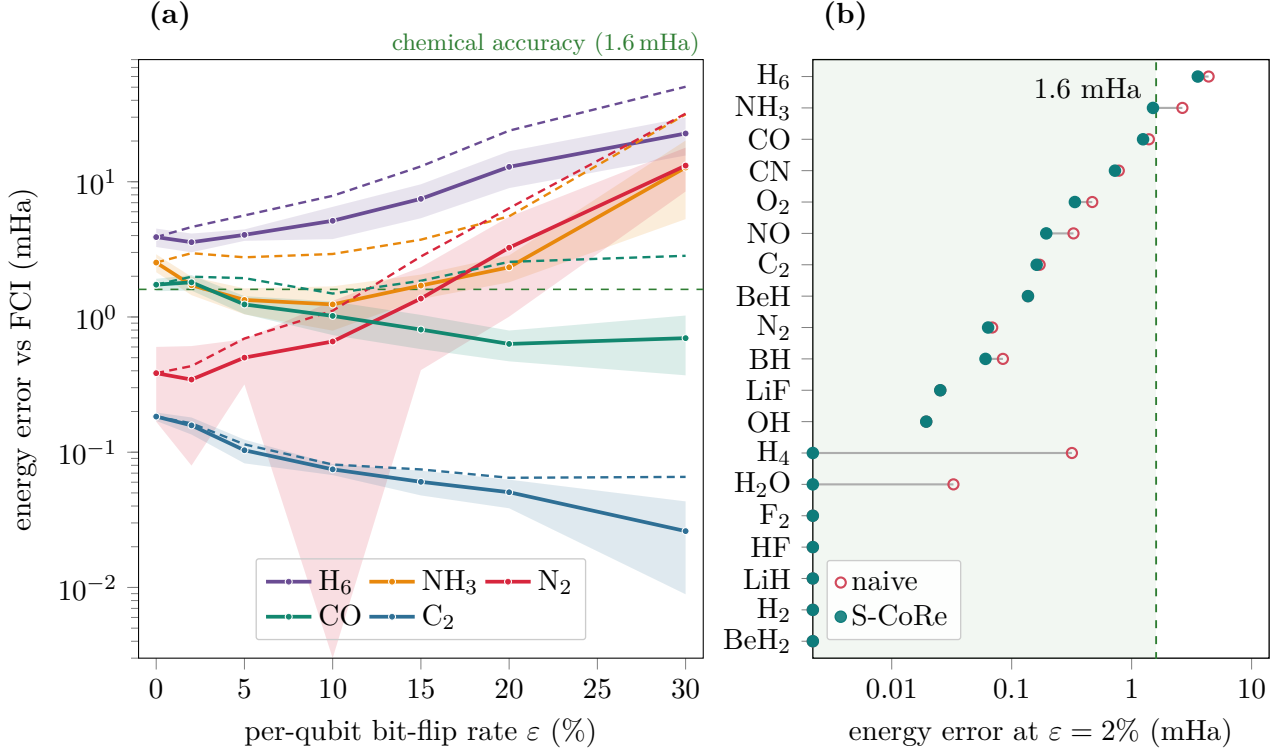


FIG. S8. **SQD ground-state energy under a controlled bit-flip model across the molecular suite.** (a) Energy error vs the per-qubit bit-flip rate ϵ for five representative molecules (H_6 , NH_3 , N_2 , CO , C_2 ; seed-averaged over 8 seeds at 4000 shots each, bands = ± 1 s.d. across seeds, running to the bottom of the panel where the mean minus one s.d. falls below it, as for N_2 at $\epsilon = 10\%$; log scale). Solid = S-CoRe recovery, dashed = naive post-selection. The advantage of recovery over naive post-selection, both arms consuming the same corrupted bitstrings at the same rate with the same seeds, grows with noise. For NH_3 , CO and C_2 the error at finite ϵ also falls below its own $\epsilon = 0$ value, beyond the seed bands (NH_3 $2.53 \rightarrow 1.24$ mHa at $\epsilon = 10\%$); that second comparison is not used here (Sec. IX A): it is decided by the size of the recovered subspace, which this sweep does not record. (b) The full nineteen-molecule suite at a realistic $\epsilon = 2\%$ (log scale, sorted; errors that are exact to round-off, in both arms for F_2 , HF , LiH , H_2 and BeH_2 , where the open circle is hidden behind the filled one, and after recovery for H_4 and H_2O , are drawn at the left edge of the axis): for each molecule the naive post-selection error (open circle) is linked to the S-CoRe value (filled), so the leftward pull is the recovery gain; the shaded band is chemical accuracy (< 1.6 mHa). Eighteen of nineteen land inside it; H_6 , the most multireference case, is the residual challenge. Panels (a) and (b) are independent seed ensembles (panel (a): 8 seeds; panel (b): 5 seeds, 4000 shots), so a given molecule's $\epsilon = 2\%$ value can differ by up to a factor of a few between them (for N_2 , 0.34 vs 0.06 mHa; both far below chemical accuracy).

ten seeds, with paired $t_9 = 1.19$ ($p = 0.26$), and without the prior it is not distinguishable from the classical control (1.17 mHa, $|t_9| = 0.92$). At 500 shots both generative arms are *worse* than the classical control: the fused one by 5.31 mHa ($|t_9| = 3.27$, $p = 0.010$, worse in nine of the ten seeds), the one without the prior by 15.33 mHa ($|t_9| = 10.3$).

The two halves of the ten seeds are the two five-seed runs of the companion study [82], and they show why five seeds do not decide this comparison. Over seeds 0–4 the fused generator is ahead of the classical control by 5.23 ± 3.19 mHa in every seed, with $t_4 = 3.67$, beyond the two-sided critical value 2.776; over seeds 5–9 it is behind, -1.62 ± 3.46 mHa with $t_4 = -1.05$. The companion reports $+5.23 \pm 3.19$ and -1.83 mHa for the same two runs; the first is reproduced here to the fourth decimal and the second to 0.21 mHa. A five-seed t test can therefore clear its critical value and fail to replicate, and a distribution-free test cannot help, since with five seeds the exact two-sided sign test and the signed-rank test both floor at $p = 2/2^5 = 0.0625$; with ten the floor is 0.002, which the classical step reaches. `src/gflow_dequant.py` writes every seed to `data/gflow_runs/`, and `data/gflow.json` holds the paired statistics quoted here.

What the test establishes is the statement we keep. On this instance the dominant determinants are supplied by the sampled histogram (here an emulated one, drawn from the exact ground state and then noise-corrupted, so the test does not measure whether a device state would supply them) and completed by a classical prior, not manufactured by a learned generator: the largest improvement available to an under-sampled head costs no quantum resources at all and should be the default post-processing step, since a demonstration that omits it is measured against a baseline that

TABLE S12. **The nineteen-molecule fermionic-magic suite**, computed in a cc-pVDZ active space; the active-space CASCI energies are converged to 1.1×10^{-13} Ha, which is a convergence check and not a validation against an independent method (Sec. S15.1). The fermionic AntiFlatness $\mathcal{F}_1$ ($k=1$ Majorana antiflatness; $\mathcal{F}_1=0$ iff Gaussian/mean-field, maximum $\mathcal{F}_1^{\max}=2n_o$) is listed near equilibrium and at dissociation, in a valence bond-breaking active space CAS(n_e, n_o). Fermionic magic switches on for covalent bond-breaking, is already large at equilibrium for inherently multireference species (C_2 , O_2), and stays near-Gaussian for ionic and weakly-correlated bonds—so $\mathcal{F}_1$ marks the multireference, strongly-correlated regime rather than bond length *per se*. The last two columns validate the *method*: at the dissociation geometry the reconstruction of the (orbital-projected) spectral function $A(\omega)$ from a subspace drawn in 2.16×10^5 shots matches the exact Lehmann result to relative- L_1 error, on a subspace that is a fraction $|S|/D$ of the full ($N+1$) sector. In twelve of the nineteen the subspace contains the probe’s Krylov space and the agreement is exact by construction; see Table S11. The listed $\mathcal{F}_1 = 4 \text{tr}[\gamma(1-\gamma)]$ are directly computed spin-orbital values (for the five open-shell species they are *not* $2N_u$; see Sec. VI). Per-molecule bond lengths, active spaces, FCI residuals, and the absolute $|S|$, sector dimension D and minimal bond dimension χ are listed in Sec. S15; the absolute FCI energies are not deposited. [†]: near-degenerate ground state.

Molecule	Bonding	CAS(n_e, n_o)	$\mathcal{F}_1^{\text{eq}}$	$\mathcal{F}_1^{\text{diss}}$	$\mathcal{F}_1^{\text{diss}}/\mathcal{F}_1^{\text{max}}$	rel- L_1	$ S /D$
<i>Covalent bond-breaking</i>							
F ₂	covalent single	(1+1,2)	0.068	3.985	1.00	$<10^{-13}$	0.50
N ₂	covalent triple	(3+3,6)	0.847	11.377	0.95	2×10^{-5}	0.21
H ₂ O	multicentre	(2+2,4)	0.007	6.936	0.87	$<10^{-13}$	0.50
HF	polar single	(1+1,2)	0.001	3.460	0.87	$<10^{-13}$	1.00
NH ₃	multicentre	(3+3,6)	0.080	10.289	0.86	1×10^{-5}	0.47
CO	polar triple	(3+3,6)	0.688	8.192	0.68	1×10^{-4}	0.41
H ₆	H-chain	(3+3,6)	0.521	7.735	0.64	2×10^{-5}	0.50
H ₄	H-chain	(2+2,4)	0.252	5.113	0.64	$<10^{-13}$	0.50
H ₂	covalent single	(1+1,2)	0.038	2.511	0.63	$<10^{-13}$	0.50
BeH ₂	multicentre	(2+2,4)	0.039	3.754	0.47	$<10^{-13}$	0.25
<i>Multireference / open-shell</i>							
C ₂ [†]	covalent double	(4+4,6)	4.743	8.001	0.67	$<10^{-13}$	0.10
NO	polar (radical)	(4+3,6)	0.790	7.449	0.62	2×10^{-6}	0.25
O ₂	covalent double (triplet)	(5+3,6)	1.673	6.000	0.50	1×10^{-5}	0.25
CN	polar (radical)	(4+3,6)	1.307	5.938	0.49	1×10^{-5}	0.24
OH [†]	polar (radical)	(3+2,4)	0.013	3.703	0.46	$<10^{-13}$	0.33
<i>Ionic / weakly correlated</i>							
BH	metal hydride	(2+2,4)	0.518	0.573	0.07	$<10^{-13}$	0.25
BeH	metal hydride (radical)	(2+1,3)	0.035	0.386	0.06	$<10^{-13}$	0.67
LiH	metal hydride	(1+1,2)	0.002	0.075	0.02	$<10^{-13}$	1.00
LiF	ionic	(1+1,2)	0.000	0.001	0.00	$<10^{-13}$	1.00

is too weak, and a learned generator adds nothing on top of it that ten seeds resolve.

The class of learned constructions that select or generate the determinants themselves is by now several—neural quantum states built on selected configurations, autoregressive and dual-channel samplers for selected configuration interaction, and neural-network classifiers [83–86]—and the defensible role for a learned model is the one that consumes the device histogram, as physics-informed generative recovery of hardware samples does [87].

A learned model can also schedule device work, as generative models for measurement grouping do [88, 89], alongside the non-learned error-mitigation and validation protocols that already sit at that interface [90, 91] and learners that process device output coherently in quantum memory [92]; this is the boundary the companion study maps from the ground-state side [82].

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